



FACULTY OF
COMPUTING
& INFORMATICS

FINAL YEAR PROJECT INTERIM REPORT

FYP01-DS-T2530-0189

Multilingual Low-Resource Language Identification

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Bachelor of Computer Science (Data Science)

FEBRUARY 2026

FYP01-DS-T2530-0189

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BY

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PROJECT INTERIM REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE

REQUIREMENT FOR THE DEGREE OF

Bachelor of Computer Science

in the

Faculty of Computing and Informatics

MULTIMEDIA UNIVERSITY
MALAYSIA

February 2026

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Acknowledgements

First and foremost, I would like to express my sincere gratitude to my supervisor, Dr. S Prabha Kumaresan, for her invaluable guidance, encouragement, and constructive feedback throughout the course of this project. Her expertise and support have been instrumental in shaping the direction and quality of this project.

I would also like to thank the lecturers and staff of FCI, MMU, for providing the academic foundation and resources necessary to undertake this Final Year Project.

Special thanks to my peers and friends for their continuous support, insightful discussions, and motivation during challenging phases of the project. I am also grateful to my family for their unwavering encouragement and belief in my abilities.

Lastly, I would like to acknowledge the open-source communities and developers behind the tools and models used in this project, including FastText, wav2vec 2.0, DistilBERT, TrOCR, and EasyOCR, whose contributions made this work possible.

Abstract

In multilingual societies like Malaysia, the ability to accurately identify languages is essential for applications such as automated transcription, content moderation, and multilingual user interfaces. However, language identification becomes particularly challenging in low-resource settings, where limited training data and noisy inputs—such as low-quality audio or text extracted from images—can significantly degrade model performance. This project addresses the problem by proposing a multimodal language identification system that leverages both audio signals and image-based text (extracted via Optical Character Recognition, OCR) to improve classification accuracy across three target languages: English, Malay, and Mandarin.

The project begins with the creation of a custom multimodal dataset comprising audio clips and image-based text containing language-specific content. An OCR pipeline is implemented using EasyOCR for English, Malay and Mandarin, based on its character recognition performance. Unimodal models are developed using wav2vec 2.0 for audio and DistilBERT for text, followed by the design of three late fusion strategies—Equal Weighting, Confidence-Adaptive, and Accuracy-Weighted Fusion to combine predictions from both modalities.

Experimental results demonstrate that the multimodal fusion approach significantly enhances overall accuracy, particularly improving performance for Malay, a low-resource language. The best-performing fusion strategy, Equal Weighting, achieved a test accuracy of 99.18%, outperforming both audio-only and OCR text-only results.

This project highlights the effectiveness of modular multimodal architectures in real-world language identification tasks, especially in scenarios involving noisy or incomplete inputs. The findings suggest that integrating OCR-derived text with audio features can offer a practical and scalable solution for

multilingual environments, with potential applications in mobile platforms, transcription services, and accessibility tools.

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List of Abbreviations/Symbols

List of Abbreviations

AI	Artificial Intelligence
ASR	Automatic Speech Recognition
ATM	Audio-Text Matching
BERT	Bidirectional Encoder Representations from Transformers
BiLSTM	Bidirectional Long Short Term Memory
BLSTM	Bidirectional Long Short Term Memory
BLEU	Bilingual Evaluation Understudy
BoW	Bag-of-Words
CCA	Canonical Correlation Analysis
CER	Character Error Rate CLS Classification (Token)
CNN	Convolutional Neural Network
CRNN	Convolutional Recurrent Neural Network
CV	Computer Vision
DF	Deepfake
DistilBERT	Distilled BERT (Transformer-based text model)
EasyOCR	CRNN-based Optical Character Recognition engine
FCI	Faculty of Computing and Informatics
FastText	Lightweight text classification model using subword embeddings
FYP	Final Year Project
GLUE	General Language Understanding Evaluation
GPU	Graphics Processing Unit
HAN	Hybrid Attention Networks
ICCN	Interaction Canonical Correlation Network
LA	Logical Access
LID	Language Identification
MAM	Masked Audio Modeling

MER	Multimodal Emotion Recognition
MLM	Masked Language Modeling
MMM	Masked Multimodal Modeling
MMU	Multimedia University
MuSeLI	Multilingual Speech and Language Identification
NER	Named Entity Recognition
NLP	Natural Language Processing
OCR	Optical Character Recognition
PA	Physical Access
PIL	Python Imaging Library
RNN	Recurrent Neural Network
SD	Speaker Diarization
SLI	Spoken Language Identification
SLR	Spoken Language Recognition
SSL	Self-Supervised Learning
STFT	Short-Time Fourier Transform
TDNN-F	Factored Time-Delay Neural Networks
TF-IDF	Term Frequency–Inverse Document Frequency
TrOCR	Transformer-based Optical Character Recognition
VAD	Voice Activity Detection
ViT	Vision Transformer
VLID	Visual Language Identification
WER	Word Error Rate
WMT	Workshop on Machine Translation
wav2vec 2.0	Self-supervised model for speech representation
XLM-R	XLM-RoBERTa XLU Cross-lingual Language Understanding
XNLI	Cross-lingual Natural Language Inference
%	Percentage (used in evaluation metrics)
→	Indicates mapping or transformation (e.g., input → output)

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Chapter 1: Introduction

1.1 Background of the Research

Malaysia is a multicultural and multilingual country where English, Malay and Mandarin are widely used across daily communication, education, business and social-media platforms. Because of this linguistic diversity, modern digital systems such as mobile applications, speech assistance, automated customer service, translation tools, and document-processing systems must be able to recognise and process multiple languages accurately. Language identification (LID) is the task of automatically determining the language of an input, therefore plays a fundamental role in ensuring that these systems respond correctly based on the user's language.

Most extant datasets and systems on LID are however based on high resourced languages and single modalities (usually audio). This makes the identification of language models performance across languages to be unequal. As an illustration, English and Mandarin have large and high quality government datasets and high research coverage. Malay is, in contrast, relatively under-resourced, has less speech corpora, minimal OCR datasets and fewer examples in multilingual machine learning work. Such an unbalanced distribution of data causes less accuracy and less robustness to AI systems applied in Malaysian settings.

As the field of deep learning expanded, it was observed that multimodal schemes, e.g. past studies using both audio speech and image-based text, can enhance language identification, visibly in low-resource settings. Transformers, wav2vec 2.0, Whisper and OCR-based text encoders are all models that can capture different types of linguistic cues. Combined, these modalities may help fill gaps in information, distorted sound or blurred text, or a shortage of information.

Thus, enhancing multimodal multilingual language recognition is a not only a technical but also practical requirement of the expanding AI economy of Malaysia. The present project aims to fill these gaps by developing and testing a multimodal, multilingual LID system of English, Malay, and Mandarin that involves the use of audio and image-based text.

1.2 Problem Statement

Although Malaysia is a multilingual country, the majority of currently-existing LID systems have difficulties identifying the low-resource language like Malay versus the high-resource language like English and Mandarin. These limitations are mostly because there is little access to corpora of speech, OCR databases and multimodal training data. The current models of LID are also largely unimodal, which is based on audio speech or text-based images. This reduces the performance of the model when dealing with noisy audio, low-quality audio files, blurry or low-resolution images. Consequently, AI systems used frequently in Malaysia like chatbots, document processing applications, tend to misunderstand Malay input or give incorrect answers.

It is necessary to have a multimodal LID strategy, which combines the use of audio and image-based text to obtain more comprehensive linguistic information and address the issue of the shortage of data. Nevertheless, not much is done in this area since multimodal LID studies on Malaysian languages is limited, leaving a gap in both methodology along with performance evaluation. This project will aim at filling these gaps through the formulation and examination of a multimodal LID model adapted to English, Malay and Mandarin.

1.3 Research Purpose

This research aims at designing, developing and testing a multimodal language identification (LID) system, which can recognize and identify English, Malay and Mandarin in terms of speech and image-based text input. This paper is an attempt to fill the existing performance gap due to the low supply of resources on the Malay language through the application of multimodal deep learning methods that integrate the complementary properties of audio and OCR-based text, which subsequently assists in answering the question of whether multimodal fusion can provide any appreciable benefits in low-resource environments.

The project is also aimed at creation of a small, very documented paired dataset (audio + image based text) to facilitate reproducible experiments. The other goal of the research is to improve the robustness, accuracy, and practical usability of multilingual AI systems in Malaysia.

The research aims to quantify the relative strengths and weaknesses of CNN/CRNN baselines, SSL Transformer models (wav2vec 2.0), and multilingual text classifiers (FastText, DistilBERT).

1.4 Objectives of the Research

The objectives of the research are:

- To collect, prepare and generate multimodal datasets (audio and image-based text) for English, Malay and Mandarin. A small paired dataset (audio + image-based text) for English, Malay, and Mandarin will be created.
- Develop and evaluate unimodal (audio and image-based text) language identification models, design multimodal fusion strategies combining speech and OCR text features, and compare performance across English, Malay, and Mandarin.
- To analyse whether multimodal fusion improves recognition accuracy for Malay as a low-resource language

1.5 Research Questions

This research aims to answer the following questions:

- How effective and accurate can speech-only and image-based text only models identify English, Malay and Mandarin?
- Does a multimodal fusion model of both speech and image-based text produce better results compared to unimodal approaches in terms of language identification accuracy?
- The effectiveness of lightweight text classifiers (FastText) when compared to Transformer-based text model (DistilBERT) when applied to OCR-extracted text from images.
- To what extent does multimodal fusion mitigate low-resource challenges in Malay language identification?

1.6 Project Scope

This project focuses on developing and evaluating a multimodal language identification system for three languages that are most commonly used in Malaysia: English, Malay and Mandarin. The study is limited to two modalities, which are audio speech and image-based text (OCR). Video and large-scale multilingual expansion are out of scope. Lightweight and mid-size variants of Transformer model such as wav2vec 2.0 and DistilBERT will be utilised, with TrOCR/EasyOCR for text extraction, to ensure experiments remain feasible on local GPUs while allowing optional cloud resources. The project will cover dataset preparation, model development, performance evaluation and comparative analysis but does not include deployment, real-time systems or mobile application development.

1.7 Significance of the Research

This research addresses a practical gap for multilingual applications in Malaysia and similar multilingual regions by exploring multimodal LID under realistic, low-resource constraints. It specifically helps strengthen Malay language identification performance. By combining both audio and text-based images, this work contributes to emerging research on multimodal deep learning for language recognition. The research also contributes empirical comparisons between classical deep learning architectures (CNN and CRNN) and modern SSL approaches(wav2vec 2.0 and Whisper), introduces a curated paired dataset for future research, and offers actionable guidance on fusion strategies and robustness techniques. The findings can support the development of more inclusive and accessible multilingual technologies which will benefit education, customer service, accessibility tools, and digital communication platforms that must operate reliably across speech and visual text inputs in Malaysia.

1.8 Summary

Chapter 1 established the motivation, background and importance of multilingual language identification in a multilingual country like Malaysia, particularly the challenges faced by low-resource languages like Malay. It outlined the problem, clarified the research purpose, defined the research objectives and questions, and finally established the project's scope and its significance. All these elements collectively provide the foundation for developing and evaluating a multimodal language identification (LID) system designed to improve multilingual AI performance here in Malaysia. The next chapter will review related literature.

Chapter 2: Literature Review

2.1 Introduction

This chapter includes literature reviews which provide the theoretical and methodological foundation for this study on multimodal multilingual language identification. It reviews the existing body of literature relevant to speech recognition, language identification, optical character recognition and multimodal deep learning, focusing on approaches relevant to low-resource languages which in this study is the Malay language. By examining foundational architectures, multilingual and low-resource speech recognition strategies, spoken language identification methods, multimodal fusion frameworks, text-based models, and robustness analyses, this chapter establishes the context in which the proposed system is developed. Another goal is to establish a clear understanding of the theoretical foundations and research gaps that motivate this research.

The review begins with foundational architectures such as the Transformer, wav2vec 2.0, and Conformer, which have shaped modern sequence modelling, speech recognition and the groundbreaking discovery of a new model known as Transformer. It then explores multilingual and low-resource speech recognition research, highlighting datasets and strategies that enable performance improvements in low-resource languages. The chapter then talks about spoken language identification methods that only use audio only cues, and then there are multimodal methods which use a combination of speech and text or visual cues. Other models that have been taken into account are text-based multilingual models and OCR systems which facilitate image-derived language identification. Lastly, it discusses studies on robustness and feature analysis such as the research on spoof detection and layer-wise testing in the SSL, which can be used to estimate the stability of language identification systems under noisy environments.

These clusters of research can offer a complete picture of the state of the art. They emphasize not only the advancement that has been made in the multilingual speech and text processing but also the ongoing problems in the situation with low resources. Therefore, this chapter not only situates the project within ongoing efforts to improve accuracy, robustness, and inclusivity in multilingual AI systems, but also motivates the development of a multimodal language identification framework for English, Malay, and Mandarin.

2.2 Foundational Architectures

2.2.1 Attention is All you Need (2017)

This paper by *Vaswani et al. (2017)* describe recurrent neural network models for sequence transduction and highlights their limitations. Vaswani et al. (2017) introduce attention mechanisms and the Transformer, a novel architecture based solely on self-attention.

Next, the paper describes the encoder-decoder structure of the Transformer, similar to other sequence transduction models. It also shows the Transformer architecture, which uses stacked self-attention and point-wise, fully connected layers for both the encoder and decoder. The general concept of an attention function is defined and the specific attention mechanism used in the Transformer is introduced.

Experiments on WMT 2014 English-German and English-French translation tasks show excellent results. The big Transformer achieves 28.4 BLEU (EN-DE) and 41.8 BLEU (EN-FR), which surpass previous top-performing models while also being more efficient as the big model trains in 3.5 days on 8 GPUs.

The paper has comparisons that show self-attention has lower sequential operations and shorter path lengths than both RNNs and CNNs. The transformer

not only performs well on translation, it can also learn to map words together and learn grammar and sentence structure.

Overall, this paper introduces the Transformer to the world, revolutionising NLP and laid the foundation for modern large language models like ChatGpt, Bert and many more.

2.2.2 wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations (2020)

Baevski et al. (2020) introduce wav2vec 2.0, which is a self-supervised framework for learning speech representations directly from raw audio. This leads to reduced reliance on large labeled dataset, unlike traditional supervised speech recognition, which is crucial for low-resource speech.

At first, it listens to raw audio which a CNN turns the sound wave into basic features. Then, it hides chunks of audio features, and a transformer network tries to predict the hidden audio pieces by comparing them to possible candidates. The audio features are turned into discrete units using Gumbel Softmax. The model then learns to pick the correct (real) audio unit from a set of distractors. After pre-training on large amounts of unlabeled audio, the model is trained with a small labeled dataset to actually recognize the words.

wav2vec 2.0 shows great results, even with ultra-low resources, which require very little labelled data if you pre-train on a large amount of unlabeled audio. Example results are:

- With **10 minutes of labeled data**, it still works well (WER ~4.8/8.2).
- With **1 hour labeled data**, it beats older methods that used 100× more labeled data.

- With **full 960 hours labeled**, it achieves **state-of-the-art accuracy** (WER ~1.8/3.3).

wav2vec 2.0 makes speech recognition possible for low-resource languages where labeled transcripts are rare.

2.2.3 Conformer: Convolution-augmented Transformer for Speech Recognition (2020)

Gulati et al. (2020) introduced Conformer, which integrates convolutional layers into the Transformer for capturing local acoustic patterns while maintaining global context, improving speech recognition accuracy. This paper begins by describing the advantages and disadvantages of Recurrent Neural Networks (RNNs), Transformers, and Convolutional Neural Networks (CNNs) for Automatic Speech Recognition. This paper also stated the limitations of using only self-attention or convolutions for modelling audio sequences.

The next part describes the Conformer encoder architecture. Starting with processing input with a convolution subsampling layer, followed by the core unit, which is the Conformer block, which replaces standard Transformer blocks. Each block contains: feed-forward → self-attention → convolution → feed-forward (Macaron-style).

Three Conformer models (small, medium and large) with different parameter sizes and hyperparameters are studied. Small (10M), Medium (30M), and Large (118M). The medium model surpasses Transformer Transducer without a language model. While with a language model, the Conformer achieves state-of-the-art speech recognition on LibriSpeech with fewer parameters, demonstrating the usefulness of Transformer and convolution combination. It is

parameter-efficient and can therefore be deployed on large and small-scale applications.

It is emphasized in the paper that global + local modeling is vital in speech tasks. Conformer resulted in a new standard of ASR, which inspired numerous subsequent architectures in speech AI.

2.3 Multilingual & Low-Resource Speech Recognition

2.3.1 Robust Speech Recognition via Large-Scale Weak Supervision (2022)

Radford et al. (2022) discussed the objectives of the Whisper model and stated the limitations of purely unsupervised and fully supervised models in speech recognition. It also highlights the advantages of large-scale weakly supervised datasets, showing how massive datasets improve low-resource language performance.

The Whisper has over 680000 hours of audio collected from the internet paired with transcript. The dataset also covers 96 languages. The paper then discusses the Whisper data processing pipeline, highlighting data sources, the filtering method used to improve transcript quality, and also briefly describes the training procedures and hyperparameters used for Whisper.

It is also stated that VoxLingua107 was used to help ensure audio language matched transcript language. The model architecture is a sequence-to-sequence Transformer (encoder-decoder). When tested on LibriSpeech, Whisper's WER is good but not state-of-the-art. However, on out-of-distribution datasets, Whisper is far more robust. Compared to wav2vec 2.0, Whisper makes 55% fewer errors on average across diverse datasets. Whisper closes the gap between machine and human robustness; it shows strong performance when tested on datasets it wasn't trained on.

The paper also discusses the limitations of the Whisper on long-form transcription and suggest potential research area in the future. This study states that the combination of both scale and diversity of weakly supervised data is more important than perfect labels.

2.3.2 Progress in Multilingual Speech Recognition for Low Resource Language (2022)

Gupta et al. (2022) discussed multilingual acoustic modelling, demonstrating strategies for low-resource languages. This paper focuses on building Automatic Speech Recognition systems for three low-resource languages which are Kurmanji Kurdish, Cree and Inuktut. These languages are complex and have limited transcribed audio resources.

The paper uses multilingual acoustic model trained on 12 languages to improve recognition. This improves recognition accuracy for languages with limited datasets.

Two strategies were use for phone sets:

- Separate phones per language (private inventory) which treat each language's sounds independently. Like French pronunciation of the word P and the English pronunciation of the word P are treated differently in the model.
- Merged phones across languages (shared inventory) where phones that are acoustically similar across languages are merged into one shared symbol. For example, English 'p' and French 'p' are treated as the same phone in the model.

The paper show results that separate phones work better for Kurmanju Kurdish and Inuktut language while merged phones give lower word error rate for Cree. Two archhitectures were also compared, TDNN-F and BLSTM. (Factored

Time-Delay Neural Networks and Bidirectional Long Short Term Memory)

Overall, the study shows that multilingual training consistently outperforms unilingual training. It also demonstrates practical strategies for improving ASR in low-resource, morphologically complex languages.

2.3.3 VOXLINGUA107: A Dataset for Spoken Language Recognition (2021)

Valk & Alumäe (2021) introduce VoxLingua107, which is a large dataset for spoken language recognition (SLR). SLR is the task of automatically identifying the language of a spoken utterance. The paper presents the method used to create the dataset, discusses the results of experiments using the dataset and also the availability and limitations of the dataset. It aims to provide wide language coverage and varied acoustic conditions.

Next, the paper describes the process of collecting speech data from Wikipedia and YouTube. Data was collected automatically from YouTube Videos using language-specific search phrases. These search phrases were generated from Wikipedia dumps using TF-IDF (Term Frequency–Inverse Document Frequency) to find important terms. Only languages with more than 10000 Wikipedia articles were considered, resulting in 149 language candidates. After filtering, usable phrases were applied to YouTube search to retrieve videos. The final dataset has 107 languages with 6628 hours of speech with an average of 62 hours speech per language. The paper also indicates that VoxLingua107 was also tested using KALAKA-3 and LRE07 benchmarks and has performed well. Noisy data did not perform well compared to when using clean data.

The dataset is openly accessible and therefore a useful resource to the research fraternity. Comprehensively, VoxLingua107 shows that audio that has been scraped with a web-scraper can compete with costly manually annotated

data. It expands language coverage to over 100 languages, which far exceeds traditional benchmarks. This study also lowers barriers for SLR research and supports the development of multilingual speech systems.

2.4 Spoken Language Identification (Audio Only)

2.4.1 Accidental Learners: Spoken Language Identification in Multilingual Self-supervised Models (2022)

Bartley et al. (2022) explores how self-supervised speech models can be used for spoken language identification (LangID). Self-supervised learning is like a combination of both supervised and unsupervised learning. The model uses unlabeled data but creates pseudo-labels or tasks from the data itself. The paper focuses on Conformer-based architectures trained in a multilingual pre-training setup.

The authors find that lower layers of pre-trained models encode the most useful language-discriminatory information. These embeddings are robust for classifying unseen languages and different acoustic environments.

The study uses the VoxLingua107 dataset (6,628 hours, 107 languages) for training and evaluation. The evaluation set encompasses 33 languages and the training takes all the 107 languages. Experiments are divided into 4-6 seconds audio segments. The performance is tested on the seen languages (present in pre-training) and the unseen languages (not in pre-training). Amazingly, models are effective even in the case of unseen languages, such as languages that do not have a common line or a family tree with the languages that the model is trained on. This suggests that pre-training captures general phonemic distributions useful for language discrimination.

This paper results show that layer selection is critical, where lower and mid layers are optimal for LangID. The authors highlight that spoken LangID may be functionally similar to phoneme distribution learning. The model achieves near state-of-the-art results on VoxLingua107 despite being smaller.

2.4.2 Exploring Spoken Language Identification Strategies for Automatic Transcription of Multilingual Broadcast and Institutional Speech (2024)

Valente et al. (2024) studied spoken language identification (SLI) in real-world multilingual scenarios, which is broadcast and institutional speech. Unlike others typical SLI research, which focuses on short monolingual clips, this study targets longer recordings with multiple speakers and languages. In these domains, language changes often coincide with speaker changes, rather than switching between different types of language in a single sentence.

The authors proposed a cascaded system, which first perform speaker diarization (SD), a technique which segments audio by speaker identity, then apply language identification (SLI). It is compared against more traditional segment-based and frame-based SLI approaches.

The baseline systems include SpeechBrain VAD + ECAPA-TDNN SLI and NeMo TitaNet-LID. Two SLI architectures are also tested:

- Segment-based SLI: assigns language labels to speech segments detected by VAD or SD.
- Frame-based SLI: predicts language labels continuously at fixed time steps (language diarization).

The authors used two main sources of training data. One was a broadcast dataset with 300 hours of Danish, Swedish, and different varieties of English (British, American, and non-native English). The other data was institutional and

had approximately 1,000 hours of recordings of the European Parliament in 10 languages. To test it, they constructed synthetic multilingual sets by sewing monolingual clips with or without pauses, and also real multilingual sets acquired in Swedish TV programs and European parliament.

Findings reveal that speaker-informed SLI has greater language change detection accuracy as compared to speaker-agnostic techniques. The paper emphasizes the significance of the domain-specific fine-tuning (broadcast vs. institutional). In general, the paper supported the idea that Speaker-aware systems are more effective in detecting language changes in comparison to the systems that do not take into account speakers. Accuracy each domain (broadcast vs. institutional) is enhanced through fine-tuning.

The one sure shortcut that can be taken as identifying language change is speaker diarization, since language changes tend to occur when there is a change in speakers.

2.4.3 Language Identification Using Deep Convolutional Recurrent Neural Networks (2017)

Bartz et al. (2017) propose a Language Identification (LID) system that classifies spoken language from audio samples. Unlike traditional audio-domain approaches, solves the problem in the image domain using spectrograms. Therefore, this study introduces a hybrid Convolutional Recurrent Neural Network (CRNN). CNNs capture spatial features from spectrogram images while RNNs (specifically BLSTMs) capture temporal dependencies across time steps. Together, CRNN leverages both local frequency features and sequential context.

The authors uses datasets that were collected from the EU Speech Repository and YouTube News broadcasts. 6 languages were used (English, German, French, Russian, and Mandarin Chinese).

Audio was converted into grayscale spectrogram images (500×129 pixels). The CNN feature extractor was based on VGG-style architecture with five convolutional layers. Outputs were sliced along the time axis and fed into a BLSTM with 512 hidden units. The final classification of languages was done through a completely connected layer.

It has been found that, in both datasets, CRNN outperformed CNN-only models. CRNN outperformed CNN baselines on the accuracy and F1-scores on the EU Speech dataset. In deeper models such as Inception-v3 CRNN, on the larger Youtube dataset the accuracy was 96%. Noise robustness tests revealed that the CRNN performance decreased but Inception-v3 CRNN was more robust to noise. The experiment on extensibility confirmed that the model could effortlessly learn new languages (Mandarin and Russian). In general, the CRNN model exhibited high levels of generalization and scalability in multilingual LID.

This paper stated that CRNN has limitations such as

- Performance degraded under heavy noise
- Deeper models required much higher computational cost dataset imbalance led to weaker results for some languages.
- Distinguishing between phonetically similar languages (e.g., German vs. English) remained difficult.

The authors believe that their proposed approach to language identification can be successfully applied to a wide variety of languages. They also collected large-scale datasets and offer them to the research community.

2.5 Multimodal Language Identification (Audio + Visual/Text)

2.5.1 Multimodal Modeling for Spoken Language Identification (2023)

This paper focuses on spoken language identification (LangID) which automatically predicts the language of an utterance. Traditionally, LangID only relies on speech audio. However, in multimedia such as YouTube videos, Instagram reels, there are also metadata like titles, descriptions and upload locations.

It leads to the suggestion by Bharadwaj et al. (2023) of a multimodal LangID model, which incorporates both speech and metadata in the framework (MuSeLI). This research paper main idea is that metadata can facilitate the process of differentiating between acoustically similar languages like Hindi vs Urdu, Spanish vs Catalan.

MuSeLI is based on mSLAM, which is a multilingual pre-trained model of speech-text. It is based on speech encoder (CNN + Conformer layers) and metadata text encoders. Metadata, such as title, description and location are united into a text sequence using separators. In a multimodal encoder, speech and metadata embeddings are combined. There were two primary sources of training data used by the authors: Dhvani-YT (4k hours, 22 South Asian languages, 14 scripts) and VoxLingua107 (6.6k hours, 107 languages).

The findings indicate that MuSeLI is always better than the speech-only LangID systems. Metadata cues, particularly the writing of scripts in titles/descriptions, location of upload, were important in alleviating confusion between the acoustically similar languages.

The paper shows that the use of multimodal signals (speech + metadata) is an effective way to improve the recognition of spoken language. MuSeLI can provide state-of-the-art accuracy against public benchmarks and is robust in different utterance lengths using metadata, including video titles, video descriptions, and upload locations. It shows the significance of multimodal modeling in the development of the spoken language recognition.

2.5.2 WhisBERT: Multimodal Text-Audio Language Modeling on 100M Words (2023)

WolfQ et al. (2023) introduce WhisBERT, a multimodal language model trained using text and audio. It is inspired by FLAVA, a text-image multimodal model and Whisper which is a speech recognition model. It is mentioned that humans learn language using our eyes to see and ears to hear, but most LMs use text only. This leads to the authors asking: Can multimodal training improve efficiency and quality of language model?

WhisBERT architecture includes an audio encoder based on Whisper with convolutional patch embeddings, a text encoder and a multimodal encoder. The model is trained with several objectives: Masked Language Modeling (MLM) for text, Masked Audio Modeling (MAM) for audio, Multimodal Contrastive Loss to align audio-text pairs, Masked Multimodal Modeling (MMM) to predict masked tokens across modalities, and Audio-Text Matching (ATM) to classify whether audio and text inputs correspond.

For training, the authors use the People's Speech dataset, which contains 30,000 hours of English audio re-aligned with WhisperX. From this, they select 100 million words, which is in line with the BabyLM challenge that emphasizes small-data training.

Results show that text-only WhisBERT outperforms the multimodal version in most tasks (12 out of 17). However, the multimodal model performs better on handling complex grammatical structures.

The authors conclude that larger datasets and better hyperparameter tuning could unlock WhisBERT's full potential. From this study, WhisBERT shows that adding audio to text enriches language modeling, especially for complex syntax.

2.5.3 Learning Relationships between Text, Audio, and Video via Deep Canonical Correlation for Multimodal Language Analysis (AAAI)

This paper focuses on multimodal language analysis that includes text, audio and video. For tasks that include sentiment analysis and emotion recognition, text often outperforms audio and video. The reason why this occurs is that text carries strong sentiment cues, while audio and video can be noisy or ambiguous. For example, the words “angry” and “excited” may sound similar acoustically, though sentiment differs.

Sun et al. (AAAI) suggested that combining modalities requires learning the hidden correlation between them. This leads to them proposing the Interaction Canonical Correlation Network (ICCN). ICCN uses outer products of text-audio and text-video features to capture interactions. Outer products are the result of mathematically combining text with audio and text with video to capture interactions. These outer-product matrices are processed by CNNs to extract features.

They tested ICCN on three benchmark sets:

- **CMU-MOSI** (movie reviews, sentiment [-3,3]).
- **CMU-MOSEI** (larger YouTube dataset, 22k utterances).

- **IEMOCAP** (302 videos, 9 emotions, pruned down to 4 to balance).

The measures of evaluation in this study are accuracy, F1-score, mean absolute error and correlation with human labels. Findings indicate that ICCN is state-of-the-art in MOSI, MOSEI and IEMOCAP.

This paper supports the fact that CCA-based (Canonical Correlation Analysis), and outer-product interactions are important. Though the text-only features are robust, ICCN demonstrates that audio and video are complementary sources of information in terms of correlation. The model proves that deep correlation learning is more effective than mere concatenation.

2.5.4 Hybrid Attention Networks for Audio-Text Emotion Recognition

This paper addresses Multimodal Emotion Recognition (MER), focusing on audio and text modalities. Zhang et al. (2023) stated that single-modal emotion recognition (audio-only or text-only) cannot fully capture human affective states. Multimodal approaches leverage complementary cues, but effective fusion of heterogeneous data remains challenging because traditional fusion strategies has its limitations as stated:

- Early fusion concatenates features but suffers from dimensionality and ignores temporal dynamics.
- Late fusion combines independent predictions but fails to capture inter-modal relationships.
- Model-level fusion (e.g., kernel learning, deep learning) improves correlation modeling but often emphasizes either intra- or inter-modal attention, not both.

To address this problem, the authors propose MER-HAN (Hybrid Attention Networks) to jointly learn intra-modal and inter-modal emotional salient features. MER-HAN integrates three attention mechanisms which are local intra-modal attention, cross-modal attention, and global inter-modal attention. The authors

used two main sources of training data: IEMOCAP (acted emotions, 4 categories) and MELD (multimodal dialogues).

For experiments, the baselines are audio-only, text-only, early fusion, late fusion, and other attention-based fusion methods. Results show that MER-HAN outperforms baselines, achieving higher accuracy, F1 scores, and balanced recognition rates. Ablation studies confirmed that combining intra- and inter-modal attention mechanisms yields better performance than using either alone.

However, MER-HAN has its own limitations, it is shown that performance is dataset-dependent, MELD results are lower due to noisy conversational data and computational complexity increases with multiple attention layers.

The authors hope to extend MER-HAN to include additional modalities such as video or facial expressions and explore lightweight architectures for real-time applications. Overall, MER-HAN demonstrates that hybrid attention fusion is a promising strategy for advancing multimodal emotion recognition.

2.6 Text-Based Multilingual Models

2.6.1 Bag of Tricks for Efficient Text Classification (2017)

Joulin et al. (2017) introduce **fastText**, a simple and efficient baseline for text classification. The authors stated that while deep learning models achieve high accuracy, they are slow to train and test on very large datasets. Meanwhile, linear classifiers are simple and effective, but without scaling tricks they also struggle with very large datasets and huge label spaces.

So, fastText combines linear models with rank constraints and loss approximations for efficiency. Sentences are represented as bag-of-words (BoW) or bag-of-n-grams, then averaged into embeddings. A linear classifier predicts labels using these averaged embeddings. To handle large class spaces, fastText employs hierarchical softmax with Huffman coding. This reduces complexity from $O(kh)$ to $O(h \log k)$, making training and inference faster.

The experiments by the authors included sentiment analysis of 8 datasets and tag prediction on YFCC100M. FastText is also as accurate as CNNs and RNNs on sentiment tasks, but slightly lower than VDCNN (Very Deep CNN). The addition of bigrams increases the performance by 1-4% and the addition of higher n-grams increases the accuracy. The specialization of fastText is its speed of training, fastText trains billions of words quickly on CPUs. FastText is 10 times faster (up to 15,000x speed-up) than char-CNN and VDCNN. Tag space is outperformed by fastText on YFCC100M tag prediction in terms of accuracy and speed.

Limitations of fastText is as follow:

- fastText ignores full word order (BoW assumption); accuracy slightly below deep CNNs on complex tasks.

- Performance depends heavily on n-gram choice; without them, results degrade.
- Hierarchical softmax improves efficiency but may complicate optimization compared to flat softmax.
- fastText lacks the representational depth of deep neural networks, limiting performance on nuanced tasks.

The authors suggested that combining fastText with pre-trained embeddings will further increase accuracy. They published their code such that the research community can easily build on top of their work. Overall, fastText balances accuracy, scalability, and speed, making it ideal for very large datasets and real-time applications.

2.6.2 Unsupervised Cross-lingual Representation Learning at Scale (2020)

Conneau et al. (2020) introduce **XLM-R (XLM-RoBERTa)**, a large-scale multilingual masked language model. XLM-R aims to improve cross-lingual language understanding (XLU) by scaling pretraining across 100 languages. The authors stated that previous multilingual models like mBERT and XLM were trained on Wikipedia, thus they have limited coverage for low-resource languages. In contrast, XLM-R uses CommonCrawl (CC-100), providing terabytes of text and much more data for low-resource languages.

XLM-R is a transformer-based masked language model, trained with the MLM(Masked Language Model) objective. The authors created 250K subword tokens using SentencePiece, which is shared across all languages. They trained XLM-R model 1.5 Million updates on five-hundred 32GB Nvidia V100 GPUs with a batch size of 8192.

The evaluation benchmarks are XNLI (cross-lingual natural language inference), NER, MLQA (question answering), and GLUE. Results are as follow:

- On XNLI, XLM-R outperforms mBERT by +14.6% average accuracy.
- On MLQA, XLM-R improves average F1 by +13% compared to mBERT.
- On NER, XLM-R achieves +2.4% F1 improvement over previous models.

XLM-R records a very high score on low-resource languages and the model can compete with monolingual models such as RoBERTa on GLUE, indicating that multilinguality does not harm one-language performance. Nonetheless, the paper also provides the curse of multilinguality, which is increasing the number of languages dilutes per-language capacity. It has been found that scaling of both data and model capacity is very important in balanced multilingual performance.

This paper noted various limitations of the XLM-R model: its computational cost are extremely high; its performance also declines at some point of scale of number of languages; and show bias between high-resource and low-resource languages.

The authors aims to explore more efficient architectures, improve low-resource language handling, and reduce training costs. They also suggest extending multilingual modeling to even more languages while maintaining per-language accuracy. Overall, XLM-R demonstrates that large-scale unsupervised multilingual pretraining can achieve state-of-the-art results across diverse cross-lingual tasks.

2.6.3 DistilBERT: A Distilled Version of BERT (2020)

Sanh et al. (2020) introduce **DistilBERT**, a smaller, faster, and lighter version of BERT. The authors stated that large pre-trained models like BERT are powerful but computationally expensive and deploying such models on-device or under constrained budgets is challenging.

As such, they proposed a method to pre-train a smaller, more efficient general-purpose language representation model called DistilBERT. DistilBERT is a model that applies knowledge distillation during pretraining, not just task-specific fine-tuning. The student model (DistilBERT) learns from the teacher model (BERT) using soft target probabilities. Its training objectives combines three losses: masked language modeling (MLM), distillation loss, and cosine-distance loss. DistilBert architectures is as same as BERT but with 40% fewer parameters (layers reduced by half). So, DistilBERT is significantly smaller (66M parameters vs. 110M for BERT-base).

Results show that DistilBERT retains 97% of BERT's performance while being 60% faster at inference. DistilBERT scores are near BERT on GLUE benchmark with better results compared to older baselines such as ELMo. Accuracy on sentiment classification of IMDb is just 0.6% worse than BERT. DistilBERT can be used on mobile devices, which is 71 times faster than BERT, and is feasible in edge cases. The three components of loss are also found to be vital in ablation studies to perform optimally. DistilBERT has some drawbacks which include:

- On complex tasks, it has a slightly worse performance than full BERT; pretraining it still takes a considerable amount of compute.
- DistilBERT is English-only in its initial release, limiting multilingual applicability compared to models like XLM-R.

The authors suggested exploring multi-distillation setups (multiple teachers) for richer knowledge transfer. Overall, DistilBERT proves that knowledge distillation at the pretraining stage can yield compact yet powerful language models.

2.6.4 TrOCR: Transformer-based Optical Character Recognition (Microsoft, 2021)

Microsoft (2021) proposed TrOCR, a Transformer-based OCR system for text recognition. The authors stated that traditional OCR uses CNN encoders and RNN decoders, often with an external language model. These pipelines are complex and rely on handcrafted architectures. Instead, TrOCR uses a pure Transformer encoder-decoder framework.

The encoder is designed to obtain the representation of the image patches and the decoder is to generate the wordpiece sequence with the guidance of the visual features and previous prediction. Both encoder and decoder are initialized with pre-trained models from CV and NLP. By first learning from unlabeled data, TrOCR builds a strong foundation.

Results show that TrOCR achieves state-of-the-art results on printed, handwritten, and scene text recognition benchmarks:

- On SROIE receipts, TrOCR outperforms CNN+RNN baselines like CRNN and Tesseract.
- On IAM handwriting, TrOCR-Large achieves **CER of 2.89**, beating prior Transformer+CNN hybrids.
- On scene text datasets (IIIT5K, SVT, IC13, IC15, etc.), TrOCR shows strong generalization.

Ablation studies confirm the importance of pre-trained initialization, augmentation, and staged pretraining. TrOCR models scale from Small (62M params) to Large (558M params), trading speed vs accuracy.

TrOCR is the first OCR system to jointly leverage pre-trained image and text Transformers. This model removes the need for external language models or complex pre/post-processing. However, scene text recognition remains challenging under blur, occlusion, or low resolution and multilingual OCR requires extending decoder with multilingual pre-trained models and larger dictionaries. The authors aim to expand TrOCR to multilingual recognition and optimize its efficiency for real-time deployment.

2.7 Visual / Image Components

2.7.1 Visual Language Identification (2020)

Afouras et al. (2020) introduce Visual Language Identification (VLID), which is the task of recognising spoken language using only lip movement, without audio. It is stated that audio-based language ID is well studied, but visual-only identification is less explored.

Humans can sometimes infer language visually, so the authors ask if deep learning can do the same. To investigate this, they train models to discriminate among 14 languages using lip movement patterns.

For training, they construct the LRS3-Lang+ dataset, built from TEDx talks, covering 14 languages and totaling around 1,700 hours of video. They also evaluate on two additional datasets: VoxCeleb2-Lang, which includes 11 languages with pseudo-labels, and VoxCeleb2-Biling, which focuses on bilingual speakers. The input features are grayscale video frames of the mouth region, and

visual features are extracted using a ResNet18 pretrained on lip reading. To ensure the models are not exploiting appearance cues like ethnicity, they also include face recognition baselines (ResNet50).

It has been demonstrated that language identification based on audio-only is very accurate (around 97 percent), but visual-only VLID models are also highly accurate, with 84 percent accuracy at 30-second clips. The longer the sequence the better the performance, which illustrates the significance of the temporal mouth patterns. The analysis of the confusion matrices showed that the most common confusions were found between the languages that have similar phonetics (German-English, Korean-Japanese, Russian-Polish).

The face recognition baselines got good accuracy on LRS3-Lang+, and not on VoxCeleb2, showing that they used shortcuts based on appearance, and not based on actual lip movement dynamics. Conversely, VLID models proved to be more general to domains, and learning patterns of viseme (mouth shapes and movements). On bilingual speakers, the VLID made accuracy of about 80 percent which is way higher than a chance (50 percent) compared to the face baselines which reduced to random performance. The qualitative samples also indicated that VLID was able to recognize languages correctly even when the same individual spoke English and another language.

A number of challenges and limitations are pointed out in the study. Similar phonetic languages are hard to tell apart in a visual form and the rate of accuracy gets lower with short videos where temporal context is limited. The second limitation is that the optimal outcomes were done with long clips (30 seconds) which cannot be easily practiced in real life scenario. Dataset imbalance also posed difficulties, as some languages had fewer samples, limiting evaluation fairness.

The authors of this paper plans to investigate which lip movements provide the most discriminative cues, as well as explore the visual similarities and differences between languages.

2.8 SSL, Robustness, and Feature Analysis

2.8.1 ASVspoof 2021: Towards Spoofed and Deepfake Speech Detection in the Wild (2023)

Liu et al. (2023) introduce ASVspoof 2021, the fourth edition of the spoofed and deepfake speech detection challenge. The authors have remarked that traditional countermeasures were commonly experimented under clean laboratory environments, which are non-representative of real-life. Rather, ASVspoof 2021 tests systems intentionally under wild conditions including telephony transmission, a real acoustic environment, and compressed online audio.

The task is broken down into three activities:

- **Logical Access (LA):** identifying the presence of synthetic speech (VC/TTS) sent on VoIP and PSTN networks.
- **Physical Access (PA):** to recognize replay attacks of videos captured in real rooms using different microphones, speakers, and ranges.
- **Deepfake (DF):** compressed manipulated speech, which has been published online (mp3, m4a, ogg).

The findings indicate that ASVspoof 2021 provides the strengths and limitations of the existing techniques:

- Transmission, as well as encoding effects, were strong on LA.
- In PA, replay detection had been demonstrated to be effective in real spaces but not robust when in simulated environments.
- On DF, the systems were resistant to compression but lacking in generalization between datasets.

The existence of concealed subsets was an indication that most models are vulnerable to observed conditions, and thus, then the importance of generalization. The best systems were based on deep learning structures (CNNs, spectrogram features, self-supervised learning).

Post challenge and ablation analyses indicated that the bottleneck is still robustness. Although there was better performance with the use of SSL models, there was still failure in models when there is a mismatch between domains.

Nevertheless, the ASVspoof 2021 has its setbacks: unseen acoustic conditions and mismatches in the data sets still deteriorate the performance. The future directions of work focus on the increase of datasets, noise, diverse sources, and multimodal detection (audio + visual).

The authors envision extending ASVspoof towards multimodal deepfake detection and optimizing countermeasures for deployment in practical ASV systems.

2.8.2 Comprehensive Layer-wise Analysis of SSL Models for Audio Deepfake Detection (2025)

El Kheir et al. (2025) analysed **self-supervised learning (SSL)** speech models such as Wav2Vec2, HuBERT, and WavLM. The authors stated that deepfake audio is increasingly realistic, threatening speaker verification and speech security systems.

SSL models learn from unlabeled audio data, producing rich representations across multiple transformer layers. Each layer encodes different information: lower layers capture acoustic/phonetic cues, higher layers capture

semantic context. Which lead to the authors asking: Which layers are most useful for deepfake detection across languages and scenarios?

The method used to answer the question above is freezing SSL models and evaluate layer-wise contributions using classifiers (FFN, AASIST). Datasets including ASVspoof 2019/2021, ADD23 (Chinese), HABLA (Spanish), PartialSpoof, HAD, SceneFake, CtrSVDD (songs) were used. Results show that lower layers consistently provide the most discriminative features for deepfake detection. The contributions of higher layers are less, at times even decreasing the detection accuracy.

- With small models, only 4-6 layers are used to achieve performance that is comparable to that of full 12-layer models.
- Even when large models are used, the use of 10-12 layers alone compares or even does better than full 24-layer models.

This not only saves on computational cost but also accelerates inference which is necessary in real-time detection.. This is also the case in the use of English, Chinese, and Spanish datasets, which proves multilingual robust. Moreover, Heatmap visualization proves that the early layers of transformers significantly contribute to every dataset.

These findings confirm that there are numerous layers of the models of the SSL, although not all of them are equally applicable in all tasks. Middle layers can be best used in phoneme recognition, higher layers in speaker ID and lower layers in deepfake detection.

The limitation of the study is that it covers only the models of the SSLs; other architectures, including CNN and hybrid, are not discussed in detail. The

authors seek to investigate multi-model fusion, pruning/quantization, and multimodal (audio+video) deepfake detection.

2.9 Summary

Chapter 2 has discussed the key foundations and latest developments in multilingual and multimodal identification of language. Starting with the fundamental architectures like the Transformer, wav2vec 2.0, and Conformer, the literature had pointed to the development of the general sequence modelling into speech-specific one and hybrid architectures that are optimised based on robustness. Based on these architectures, studies in multilingual and low-resource speech recognition showed how extensive weak supervision, multilingual acoustic modelling and data sets such as VoxLingua107 can be used to enhance performance on low-resource languages.

The review then has analyzed the methods of spoken language identification entirely based on audio and how these methods perform well and poorly in noisy or low-resource situations. This prompted the research of multimodal language recognition where audio properties are supported by text or visual data to increase accuracy and robustness. Parallel to this, multilingual models based on text e.g. FastText, DistilBERT, XLM-R, and TrOCR were also discussed because they are the key aspects of OCR-derived text classification. The importance of visual/image elements was also highlighted, and they showed how the concept of OCR systems allowed text to be extracted out of images in order to facilitate multimodal pipelines. Lastly, the significance of robustness and feature analysis studies supported the necessity to be careful about the design and evaluation of multimodal systems to ensure resilience to noise, spoofing, and adversarial conditions.

In general, these studies culminate in the context of foundational architectures, multilingual adaptation, multimodal integration and eventually, upon robustness analysis. This development gives the theoretical and empirical foundation of the proposed system in Chapter 3, the methodology used in developing and testing a multimodal language identification system of the English, Malay, and Mandarin languages will be laid out.

Chapter 3: Theoretical Framework

3.1 Introduction

This chapter provides the theoretical background used to develop the proposed language identification system based on the idea of multimodality. All the models in this project have a well-established underlying architecture. When we can trace a model to its basic form, we can better understand the basic processes involved in it and their role in multilingual and multimodal processing. The last part of the chapter points out the new insights of the suggested model.

3.2 Transformer: The Foundational Architecture

Transformer, which is presented by Vaswani et al. (2017) in “Attention Is All You Need”, is the building block of many of the most modern models in NLP and speech processing.

Transformer Architecture Overview

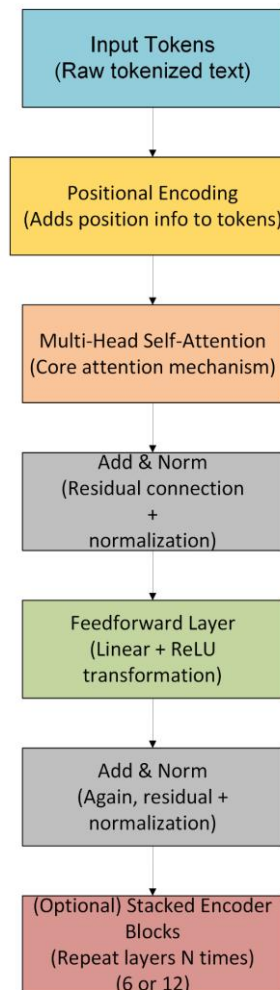


Figure 3.1: Transformer encoder architecture used in BERT, wav2vec 2.0, and TrOCR.

This architecture shown in Figure 3.1 did not require recurrence and convolution, which allowed parallelization and scalability, which became the foundations of models such as BERT, GPT, wav2vec 2.0, and TrOCR.

Transformer was used in this project due to the following reasons:

- **Multimodal flexibility:** It is capable of being adapted to text (DistilBERT), speech (wav2vec 2.0), and vision-language tasks (TrOCR).
- **Contextual knowledge:** It is important to note that its self-attention mechanism captures global dependencies, which is essential in drawing the line between similar languages.
- **Efficiency and performance:** It is efficient when distilled (e.g., DistilBERT), and therefore it can be used in resource-constrained environments.

Transformers are computationally intensive to train but, during inference, are more efficient and enable transfer learning, and are suitable when dealing with multilingual and multimodal computations, such as language identification.

3.3 Convolutional Neural Networks (CNNs): Foundational Architecture for Spectrogram-Based Audio Classification

CNNs have a common application in computer vision and are effectively scaled to audio classification problems by considering mel-spectrograms as 2D images. This renders them an implicit choice in language recognition in speech. Its architecture is shown in Figure 3.2.

Mel-Spectrogram Extraction:

- **Short-Time Fourier Transform (STFT):** Converts raw audio to frequency-domain representation
- **Mel-Scale Mapping:** Maps frequency bins to perceptually-motivated mel-scale (humans perceive frequency logarithmically)
- **Log-Power Scaling:** Improves numerical stability and matches human auditory perception
- **Benefits:** Reduces dimensionality, emphasizes linguistically-relevant frequency bands, enables CNN exploitation of time-frequency locality

Core idea: Convolutional layers learn local patterns (edges, formants, formant bands), stacking deeper layers to capture higher-level structures; parameter sharing and locality make CNNs efficient on grid-like inputs (images, spectrograms).

Typical block

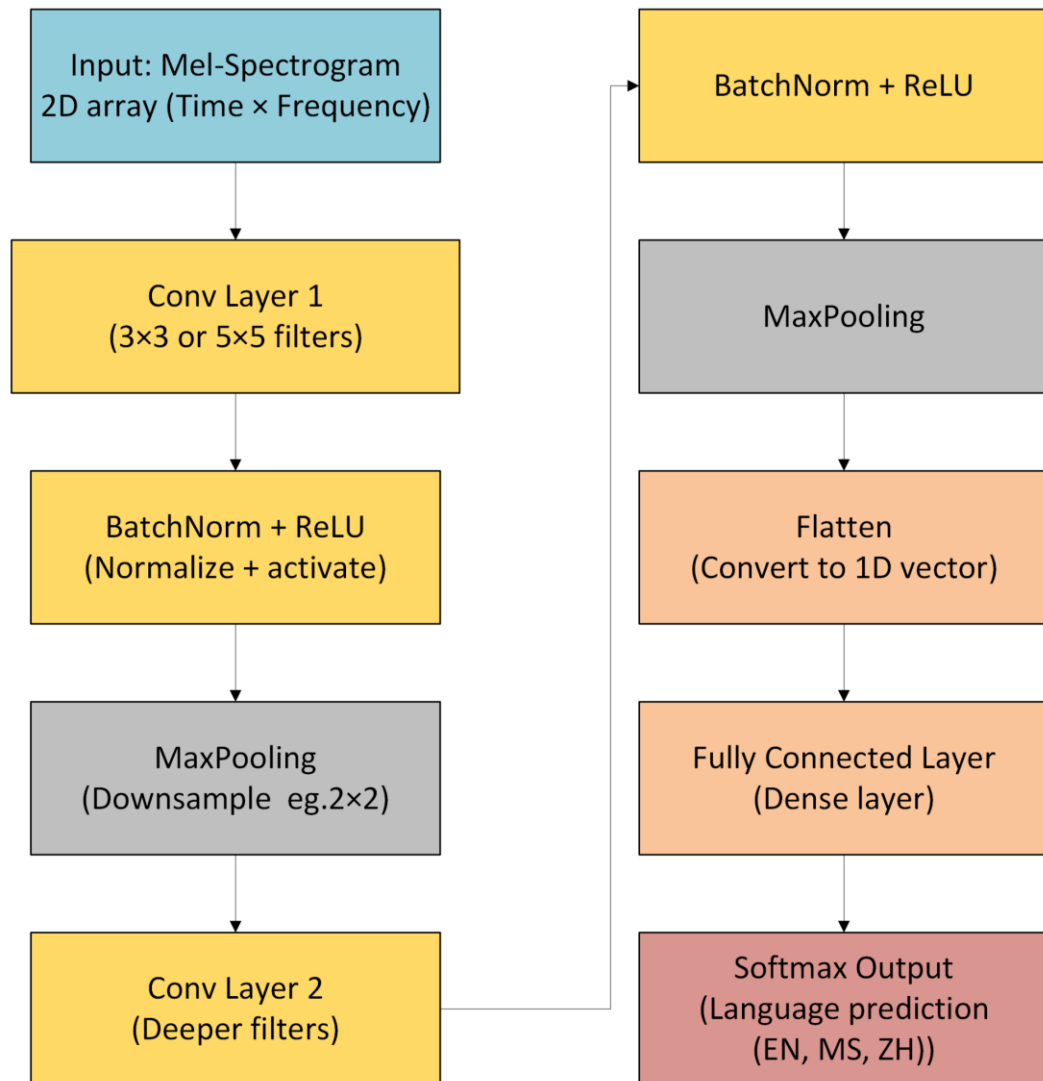


Figure 3.2: Block diagram of the convolutional neural network (CNN) architecture

CNNs are used in this project as a baseline model for audio-based language identification. Since mel-spectrograms are 2D arrays (frequency $\times$ time), CNNs can directly exploit local stationarity and time-frequency patterns without the overhead of sequence modeling. Their efficiency, interpretability, and strong performance in prior LID studies make them a well-established and competitive baseline before exploring more complex attention-based models.

3.4 wav2vec 2.0: Self-Supervised Speech Representation Learning

wav2vec 2.0 is a self-supervised model developed by Facebook AI for learning speech representations directly from raw audio. It builds on the Transformer architecture and has become a foundational model for speech-related tasks, especially in low-resource and multilingual settings. The model architecture is shown in Figure 3.3 and consists of three main components:

- A convolutional feature encoder, which transforms raw waveform inputs into latent speech representations.
- Transformer context network: This utilizes a Transformer context network to learn long-range dependencies between the sequence of audio waves through self-attention. dependencies across the audio sequence using self-attention.
- A quantization module, which quantizes a set of the latent features to provide targets in a contrastive learning model.

Training is undertaken by contrastive loss in which the model is trained to differentiate the genuine masked latent representations from a sequence of distractors. This allows the model to gain potent representations, without transcribed data.

wav2vec 2.0 Architecture Overview

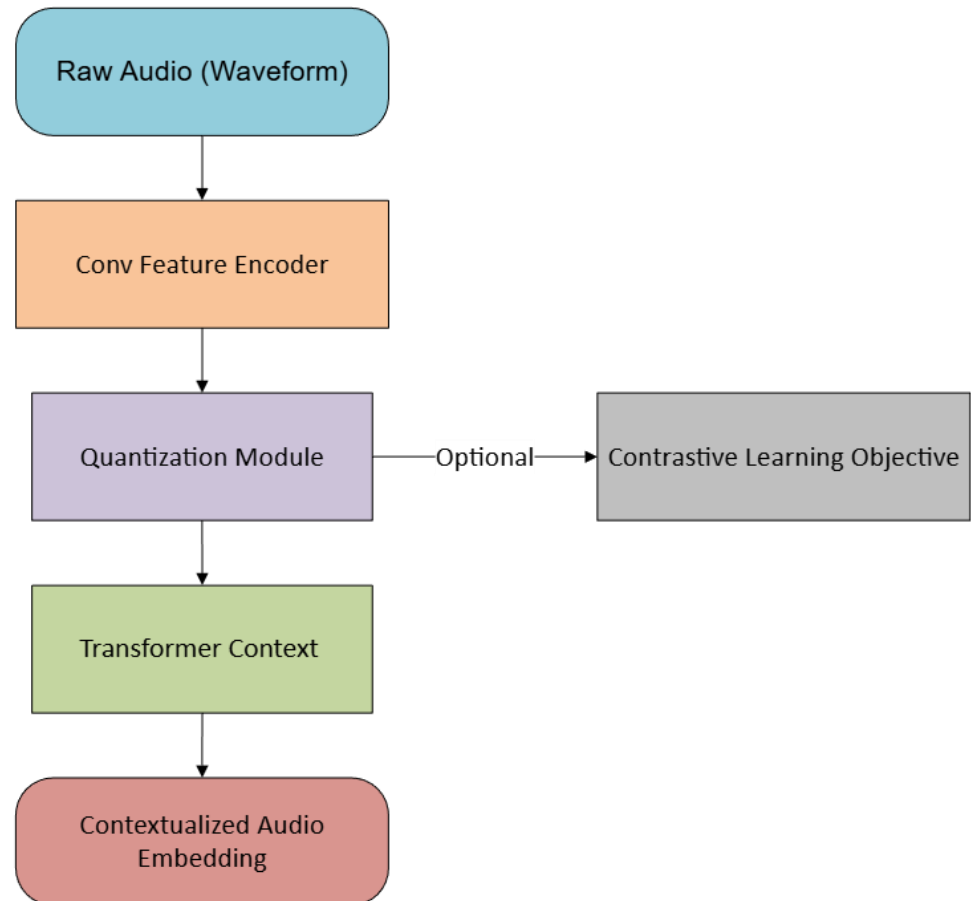


Figure 3.3: Architecture of wav2vec 2.0

wav2vec 2.0 was chosen as it has great results in multilingual speech and can generalise in low-resource cases. Its self-supervised training paradigm does not require large labeled datasets, just in time when such languages as Malay do not have large annotated corpora. The model also has a Transformer-based context encoder that enables it to identify subtle temporal patterns in speech and thus is very effective in language identification.

3.5 TrOCR: Transformer-Based OCR

TrOCR is a model based on Transformers that is created for Optical Character Recognition (OCR), which aims to integrate visual and language modeling into a unified architecture. It makes use of the capabilities of Transformers to directly create text sequences of images to allow being correctly identified with a large variety of scripts and layouts.

The **base architecture** of the proposed system is illustrated in Figure 3.4. It consists of three main components:

- **Vision Transformer (ViT) Encoder:** This is an image tokenizer that transforms input images into sequences of visual tokens through processing image patches.
- **Transformer Decoder:** This is an autoregressive decoder that produces sequences of text based on the sequences of encoded images.
- **Pretraining and Fine-Tuning:** The model is also initially pretrained on synthetic text images databases of large scale and then fine-tuned on real-world databases of OCR in order to enhance robustness and generalization.

TrOCR Architecture Overview

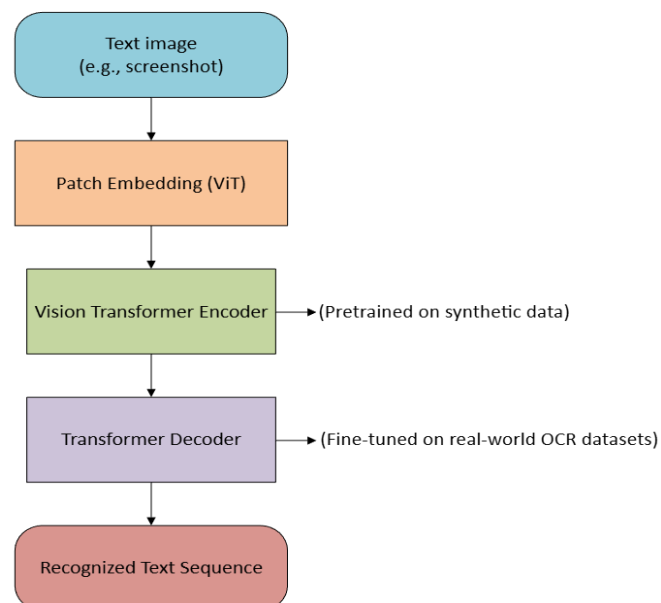


Figure 3.4: Architecture of TrOCR

The reason why TrOCR was chosen in this project is its high performance in multilingual Optical Character Recognition (OCR) activities and the end-to-end architecture that uses Transformer. TrOCR combines both visual and language recognition in a single framework, unlike the old OCR systems that use dissimilar modules for extract of features and sequence modeling. ViT encoder is effective for capturing of spatial patterns of textual images, whereas a Transformer decoder produces coherent sequences of texts. Trained on large scale artificial data sets, and fine-tuned on real world OCR data sets, TrOCR is shown to be generalized to a wide range of scripts and under noisy image conditions. This allows it to be especially useful in extracting credible text on multilingual screenshots, which is imperative to the text modality of this multimodal language identification system.

3.6 EasyOCR: CRNN-Based Optical Character Recognition

EasyOCR is an OCR engine that uses deep learning and is intended to extract text written in multiple languages. Compared to TrOCR, which employs sequence modeling on the basis of transformers, EasyOCR is based on a complex of convolutional and recurrent layers. The **base architecture** of the proposed system is illustrated in Figure 3.5. It consists of three main components:

- **CNN Feature Extractor:** Extraction of spatial features of the input image.
- **BiLSTM Sequence Modeler:** Sequential character pattern processing.
- **CTC Decoder:** Aligns predicted character sequences with input image regions.

EasyOCR Architecture Overview

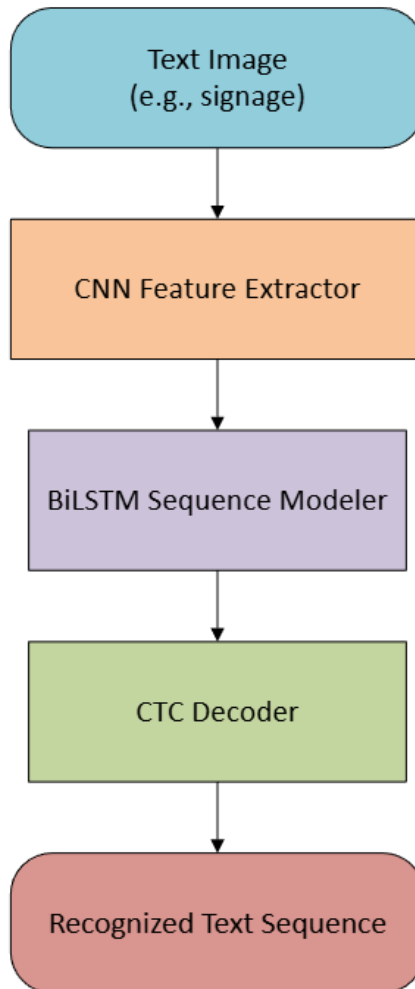


Figure 3.5: Architecture of EasyOCR, a CRNN-based OCR model

EasyOCR was chosen in this project because of its good multilingual and also because it is known to be robust when work is done with noisy or low-resolution text images. EasyOCR compared to TrOCR uses more of a classic CRNN (Convolutional Recurrent Neural Network) framework, unlike TrOCR, which is based on a Transformer-based architecture. It is based on CNN feature extractor, BiLSTM sequence modeler and CTC decoder to match the character

prediction to the image region. The architecture allows proper and efficient real-time recognition of complex scripts such as Mandarin and Malay, which are the key scripts in this study. As it turns out, EasyOCR performed much better on these languages than TrOCR, so it should be used in this multimodal language identification system as the preferred OCR engine of the text modality.

3.7 DistilBERT: Lightweight Transformer for Text Embedding

DistilBERT is a smaller, more efficient version of the original BERT (Bidirectional Encoder Representations from Transformers) model, which focuses on preserving the majority of the language understanding abilities of BERT but minimizes the level of computation requirements. Trained using the knowledge distillation process, DistilBERT is trained to behave like a larger BERT model, and it is 40% smaller and 60% faster, but still performs on downstream tasks as well as BERT, averaging about 97% of its performance. This renders it to be especially applicable to real-time and resource-constrained environments.

DistilBERT architecture shown in Figure 3.6 is based on a simplified version of the original BERT model having fewer layers and parameters:

- **Tokenizer:** This is based on a WordPiece tokenizer that is used to divide the input text into subword units.
- **Embedding Layer:** This is used to embed token embeddings with positional encodings to maintain word order.
- **Transformer Encoder:** 6 layers of Transformer encoders (BERT has 12 layers), which have multi-head self-attention, and sublayers with feed forward.
- **Token Output:** The output of the CLS token is the final hidden state of the token, which is taken as a fixed-length encoding of the entire input sequence.

This simplified design allows DistilBERT to provide quality text representations, using much lower time of inference.

DistilBERT Architecture Overview

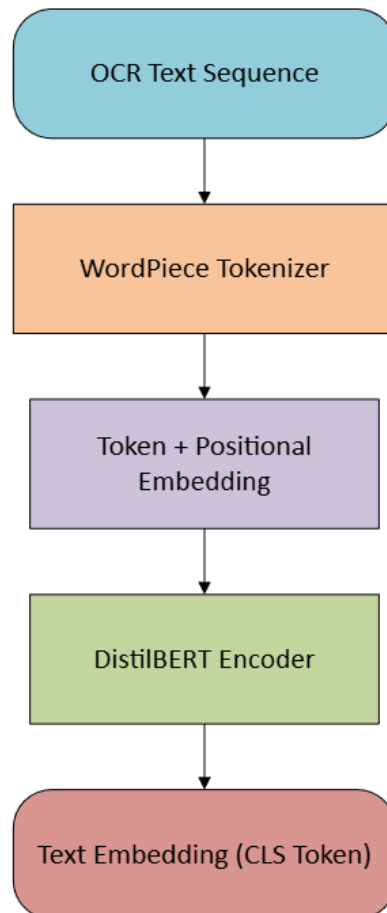


Figure 3.6: Architecture of DistilBERT used for text embedding

DistilBERT was chosen for this project as the fundamental text encoder because of its performance and efficiency balance. Being a smaller and faster, distilled form of BERT, it still has much of the language understanding properties of BERT, and it is best suited for real-time or resource-constrained environments. In order to learn contextual representations based on the large-scale text corpora, DistilBERT is trained combining both masked language modeling and knowledge distillation. In this project, it is used as a basis in deriving semantic features of

OCR-extracted text. Because it can effectively model subtle language patterns in more than one language, it is especially effective in multilingual case of language identification, when combined with a lightweight classifier like a multilayer perceptron (MLP).

3.8 FastText: Lightweight Text Classification with Subword Embeddings

FastText is a shallow neural network that was created by the Facebook AI Research to efficiently and scalably classify text. Contrary to the traditional word-level models, FastText feature maps each word by a bag of character n-grams, whereby it is capable of capturing the subword information. This makes it especially useful in dealing with rare words, misspellings, and out of vocabulary terms that is a frequent problem with multilingual and noisy text data.

The architecture of FastText is shown in Figure 3.7 and its principal elements are the following:

- **Embedding Layer with Subword n-grams:** Words are disaggregated into character n-grams, which are overlapping with each word, and embedding is trained for individual n-grams.
- **Averaging Layer:** It represents an average of all n-grams in a word (or sentence) to create a fixed-length representation as a vector.
- **Linear Softmax Classifier:** A single-layer classifier that transforms the averaged vector to a probability distribution of target classes.

This is a simple yet effective architecture that allows FastText to learn very fast and generalize particularly in the case of short or informal text.

FastText Architecture Overview

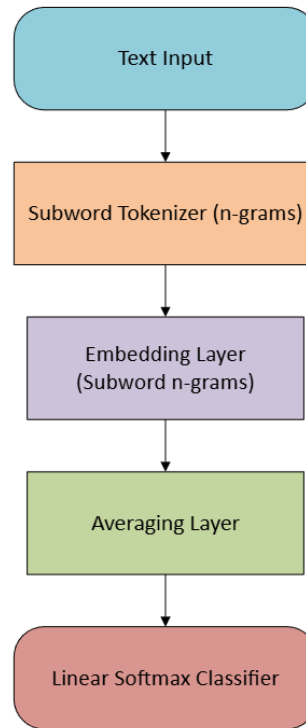


Figure 3.7: Architecture of FastText for text classification

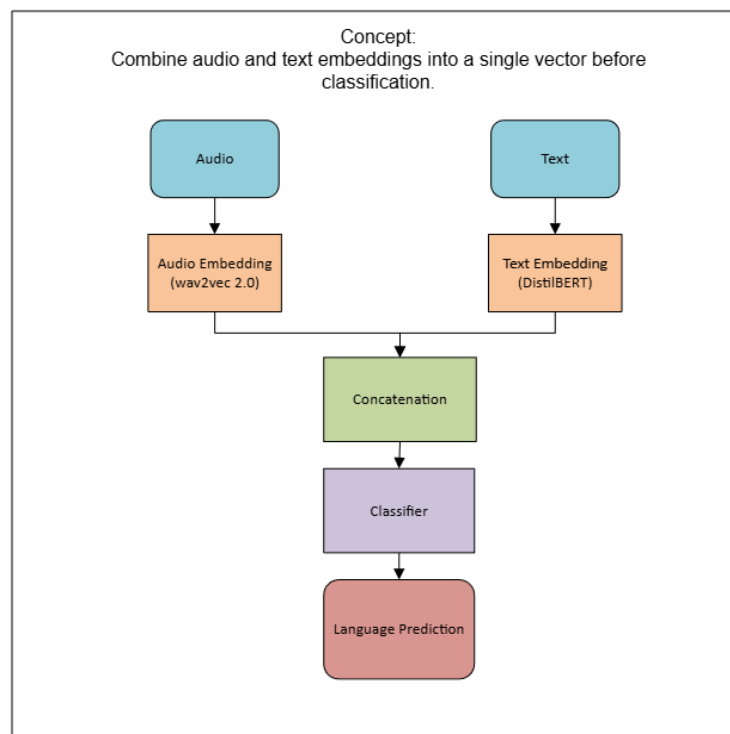
FastText was used as a lightweight baseline to identify text-based language. It has a high performance in short text, a low computation cost, and a speedy training time, which has made it a feasible benchmark against other more complex models such as the DistilBERT. In this project, it will be useful in making comparisons to determine the advantages of deeper Transformer-based encoders in the text modality.

3.9 Fusion Strategy for Multimodal Language Identification

The proposed multimodal embraces the dual unimodal pipeline strategy, where audio and text data are identified separately to obtain the features of the modality used. Audio stream is processed with the help of wav2vec 2.0, and text stream is processed with the help of OCR engine (TrOCR/EasyOCR) followed by DistilBERT. These pipelines generate high-level embeddings which are inputs in diverse fusion strategies for final language classification. The example of early and late fusion strategies of multimodal language identification is presented in Figure. 3.8.

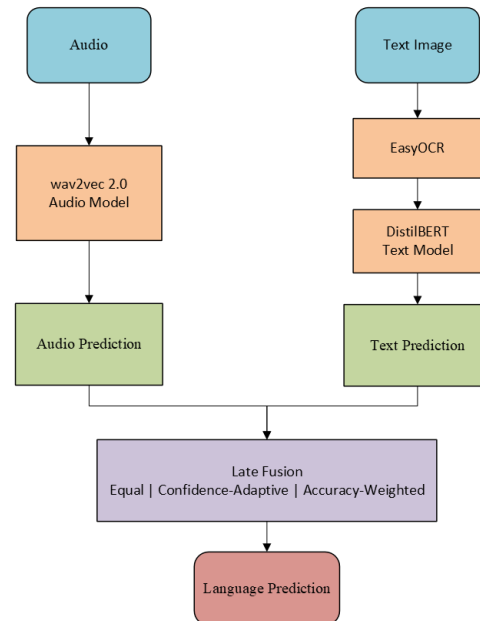
Fusion Strategies:

Early Fusion Architecture for Multimodal Language Identification



Late Fusion Architecture for Multimodal Language Identification

Concept:
Audio and OCR-text are processed separately, then their prediction probabilities are combined using decision-level late fusion.



Audio and OCR-text predictions are generated separately, then combined using decision-level late fusion.

Figure 3.8: Fusion Architecture

Early fusion involves audio and text embeddings before classification, whereas late fusion involves combining independent predictions made by each of the two modalities.

The primary focus is put on the low-resourced language Malay through the use of transfer learning and multilingual pretraining. The method facilitates the system to perform well even in cases where training data is scarce or where there is absence of training data. This will demonstrate the robustness of the proposed multimodal architecture even under low-resource settings.

3.10 Summary

This chapter has followed an architectural history of each of the models utilized in the project. The analysis of the theoretical principles underlying wav2vec 2.0, CNN, TrOCR and multilingual text encoders, including DistilBERT, will help to create a solid conceptual foundation of the proposed multimodal system. The chapter also marks the new input of this work, namely the fusion strategies aimed at synthesizing the audio and text modality, and the emphasis of the system on low-resource multilingual conditions, specifically, Malay. These understandings form the framework of implementation specifications and the experimental design in Chapter 4.

Chapter 4: Research Methodology

4.1 Introduction

In this chapter, the methodology used in the design and assessment of a multimodal language identification system, which takes advantage of the audio and text modalities, is described. The method combines state-of-the-art deep learning models in extracting features with a variety of fusion techniques that are used to integrate the information found in the speech and the OCR-extracted text. The methodology covers data collection, data preprocessing, model architecture design, model training configuration, and evaluation process.

4.2 Proposed Methodology

4.2.1 Data Acquisition and Preprocessing

In the study, the two modalities of multilingual language identification are audio and image-based text modalities. The data set includes samples in English, Malay and Mandarin which is chosen to depict both high and low resource language situation. The data points contain both speech recordings and a corresponding text-based image that makes sure there is alignment of modalities used in multimodal learning. The pipeline of data acquisition and the preprocessing is split into two primary streams: audio and text to which we will refer next part.

4.2.1.1 Audio Modality

The audio data was compiled from a curated subset of the VoxLingua107 dataset, a publicly available multilingual speech corpus containing over 60 million utterances across 107 languages. For this study, balanced samples were extracted for English, Malay, and Mandarin, with approximately 1 hour of speech per language. Each language's audio was concatenated into a single 1-hour WAV file sampled at 16 kHz with 16-bit PCM encoding.

To support different modeling objectives, the audio was split into two distinct subsets:

- **Standalone Audio Modeling**

The search space of creating and training unimodal audio models (CNN and wav2vec 2.0) was each 1-hour of the compiled audio file per language. Based on this 1 hour sample, the audio was divided in time into:

- **70% training** (42 minutes)
- **15% validation** (9 minutes)
- **15% testing** (9 minutes)

The entire 1-hour segment was the hyperparameter search space which allowed exploration of alternative preprocessing strategies, segment lengths and model configurations. This made the development of the models based on a consistent and representative dataset, as well as provided a high level of separation between the training and evaluation in the active subset.

- **Multimodal Alignment**

An independent 7.5-hour audio set was employed so as to match OCR-derived text when performing multimodal experiments. This data was not utilized to train independent audio models, and clean separation was made so that there was no confusion between unimodal and multimodal pipelines.

Two pre-processing pipelines had been used according to the model structure:

- **CNN Pipeline:**

- Audio was divided into 3 second clips.
- Every segment was converted into a mel-spectrogram with:
 - FFT window size: 1024
 - Hop length: 512
 - Number of mel bins: 128
- The CNN classifier was fed on these 2D time-frequency representations.

- **wav2vec 2.0 Pipeline:**

- Audio was divided into clips with 10-second raw waves.
- These were directly fed into the wav2vec 2.0 which carried out normalization and internal features.
- Manual feature engineering was not done.

4.2.1.2 Text Modality

Text modality was designed with a manually-created collection of synthetic images based on Wikipedia articles in each target language. The process guaranteed the linguistic diversity and provided circumstances of OCR in the real world.

Text Extraction:

- Scraping articles was done through requests and BeautifulSoup.
- In every language, domain-related issues (e.g., Artificial Intelligence, Machine Learning) were picked.
- Paragraphs that had been extracted were stored as clean ground-truth text.

Image Generation:

Python Imaging Library (PIL) was used to convert text into images. Each image had:

- Fixed width: 800 pixels
- Font size: 32
- Noto Sans family language-specific fonts.
- The number of generated images per language were up to 5,000.
- The language label was coded in the form of filenames and tracked in a CSV file.

OCR Noise Augmentation:

A special augmentation pipeline was done to generate OCR imperfections. The following pool was used to select a random noise function applied to each clean image:

- Gaussian noise ($\sigma = 15\text{--}40$)
- Salt-and-pepper noise (amount = 0.02–0.08)
- Motion blur (kernel size = 7, 9, or 11)
- JPEG compression artifacts (quality = 8–30)
- Brightness/contrast jitter (brightness = 0.6–1.2, contrast = 0.7–1.3)
- Random rectangular occlusions (up to 15% of image area)

Augmented images were indexed.

Text Preprocessing:

The unimodal text models (DistilBERT and FastText) was trained on clean ground-truth text (labels.csv). This data includes the original text to produce the OCR images, so the models were not exposed to language-based features in the presence of the confounding variable of OCR noise. This "clean training" apparatus was crucial towards building robust unimodal foundations.

Conversely, when the multimodal evaluation was conducted, the text modality was obtained with the help of OCR-extracted text to create simulated conditions of deployment in the real world. The resultant augmented image dataset was noisy text, which was obtained using OCR engines as follows:

- **TrOCR**: This is a transformer-based OCR model that was evaluated for text extraction from image-based text.
- **EasyOCR**: This is a deep learning-based OCR engine, chosen to work with Malay and Mandarin due to the more accurate character-level performance of the engine in those languages.

Multimodal fusion involved subjecting the extracted text to the unimodal text classifier (DistilBERT) that was pretrained. This was to make sure that the models were tested under real-world, noisy conditions and clean training data was retained to optimize learning.

The system could achieve high unimodal performance with the help of decoupling of the training and evaluation domains and exhibit high robustness in noisy, OCR-driven multimodal situations.

4.2.1.2.1 OCR Engine Evaluation and Selection

Two OCR engines were used to test the optimum text quality in the case of downstream multimodal fusion: TrOCR and EasyOCR were tested on all 3 target languages: English, Malay, and Mandarin. All the engines were trained on 5000 synthetic text images in each language, and their performance was evaluated with respect to ground-truth text metrics of character level accuracy and Levenshtein distance. The summary of the results is shown in Table 4.1:

Table 4.1: OCR Machine Comparison

Language	TrOCR Accuracy	EasyOCR Accuracy	Selected Engine
English	62.24%	87.90%	EasyOCR
Malay	46.65%	87.62%	EasyOCR
Mandarin	2.56%	76.82%	EasyOCR

These findings demonstrate that EasyOCR performs best on English, Malay and Mandarin. This selection strategy ensures that each language benefits from the most accurate OCR engine available, improving the quality of text input for the multimodal pipeline. This choice plan will make sure that every language uses the

best available OCR engine and enhances the quality of text input to the multimodal pipeline.

The assessment also showed the following insights:

- TrOCR has lower performance than EasyOCR for Malay and Mandarin.
- EasyOCR is more robust across the evaluated language, likely because of heuristic language specific and character level tuning.
- These patterns were further supported by median character accuracy and Levenshtein distance and EasyOCR generated more consistent and readable results when applied to non-English languages.

This process of selection based on data meant that the multimodal system was constructed around the highest quality of OCR text in each language, and that this helped to make the system more robust and boosting downstream accuracy in the classification.

4.2.2 Model Development

In this section, all the unimodal and multimodal models in the system are described. The models were trained independently on either audio or text data, and the configurations were made to be specific to the architecture and modality. Each of the models was trained with the help of the PyTorch deep learning framework with the exception of FastText which was trained with a native C++ implementation. Grid search or random search was used to optimize all hyperparameters based on the complexity of the model and costs incurred on training.

4.2.2.1 CNN (Audio Only)

The CNN model was used as a lightweight baseline of language identification using audio. It was trained on mel-spectrograms of 3-second audio

clips, with each of the 1-hour compiled audio files taken. This portion was divided into 70% training, 15% validation and 15% testing.

These spectrograms were computed using 128 mel bins, an FFT window size of 1024, and a hop length of 512. The model architecture consisted of four convolutional layers with ReLU activations and max pooling, followed by a global average pooling layer implemented via AdaptiveAvgPool2d. The pooled features were passed through fully connected layers with fixed dropout rates of 0.5 and 0.3, and the final output layer applied a softmax activation for 3-class classification.

Training was conducted using the Adam optimizer with a final configuration of learning rate 0.0001, batch size 16, and 25 epochs. The loss function used was categorical cross-entropy. Hyperparameter tuning was performed via grid search over 36 configurations, varying learning rate (0.0001, 0.0005, 0.001, 0.01), batch size (16, 24, 32), and number of epochs (20, 25, 30). Dropout rates were fixed and not included in the search space.

4.2.2.2 wav2vec 2.0 (Audio Only)

The wav2vec 2.0 model was implemented as a transformer-based alternative to the CNN baseline for audio-based language identification. It was trained on 10-second raw waveform segments extracted from the same audio set (60 minutes) used in the CNN pipeline. These segments were sampled at 16 kHz and passed directly to the wav2vec 2.0 processor, which handled waveform normalization and internal feature extraction. No manual feature engineering or spectrogram computation was required, as the model is designed to operate directly on raw audio.

The pretrained facebook/wav2vec2-base model was used as its architecture and the encoder layers were frozen during training to minimize the computational cost. The encoder output a lightweight classification head was added which is a linear head followed by a softmax activation to classify into 3 classes. The encoder output was attached to a lightweight classification head which included a linear layer with a softmax activation to classify 3 classes.

The Adam optimizer was used and final settings were learning rate 0.001, batch size 16, and 3 epochs. The cross-entropy loss was the loss function. The cross-entropy loss was the loss function. Hyperparameter optimization was done through grid search with 27 combinations with variation of learning rate (0.0001, 0.0005, 0.001), batch size (8, 16, 24), and epochs (1,2,3).

4.2.2.3 FastText (Text Only)

FastText model was adopted as a lightweight and embedding-based baseline of text-based language identification. It was trained using clean text in labels.csv, which had the original text on which the OCR images were generated. This setup allowed the model to learn robust language features from clean data.

FastText was trained to do supervised classification using subword embeddings and character n-grams. There were 36 grid search configurations consisting of learning rate (0.1, 0.3, 0.5), epochs (10, 20, 30), embedding (50, 100), and wordNgrams (1, 2). The best-performing configuration used a learning rate of 0.1, 20 epochs, embedding dimension 50, and wordNgrams = 1. This model was saved as fasttext_best_model.bin.

4.2.2.5 DistilBERT (Text Only)

DistilBERT was implemented as a transformer-based model for text-based language identification. It was fine-tuned on clean text from labels.csv, using the

distilbert-base-multilingual-cased checkpoint. Input text was tokenized and passed through the model, with the [CLS] token embedding fed into a linear classification head followed by a softmax layer for 3-class prediction.

Training was conducted using the Adam optimizer with a learning rate of 0.0005, batch size 8, and 2 epochs. The loss function was categorical cross-entropy. Hyperparameter tuning was performed via random search over 10 configurations, varying learning rate, batch size, and number of epochs.

FastText was evaluated on both clean text (labels.csv) and OCR-extracted text (labels_noisy.csv) in the unimodal setting to assess its robustness to noise. However, only DistilBERT was used in the multimodal fusion pipeline, where it received OCR-derived text as input. As such, DistilBERT was evaluated on clean text in the unimodal setting and on noisy OCR text during multimodal experiments. This design reflects the intended deployment scenario, where OCR-extracted text is fused with audio features for final prediction.

4.2.2.6 Multimodal Fusion Architecture

A multimodal fusion architecture has been created in order to exploit the complementary advantages of audio and text modalities. In this section the design of the late fusion pipeline implemented is outlined.

Late Fusion

The adopted fusion strategy is based on a late fusion paradigm where training of individual classifiers on each modality is done and the decisions are pooled at decision level. Specifically:

- **Audio Modality:** A classifier trained using wav2vec 2.0 that takes segments of raw waveforms and makes language predictions.

- **Text Modality:** A DistilBERT based classifier was fine-tuned on OCR text extracts to accomplish the same classification task.

All the models generated a probability distribution across the three target classes using softmax. When making the inference, the final prediction was calculated by combining up these outputs with the help of any of the following strategies:

- **Equal Weighting (0.5 / 0.5):** The softmax probabilities of both the models were averaged with equal weights. The combination probability of the highest class was chosen.
- **Confidence-Adaptive Fusion:** A threshold-based dynamic weighting scheme (threshold=0.8). At the point where the maximum probability of a model (using the softmax) is greater than 0.8 it is assigned greater weight (0.55 in text or 0.65 in audio). In cases where both models have low confidence (less than 0.8) they are given equal weights. The more assertive model had a greater role to play in the final decision.
- **Accuracy-Weighted Fusion (0.6 audio / 0.4 text):** This was done with fixed weights, which were determined by the unimodal test outcomes of the models. The weighting of the audio model (wav2vec 2.0) and text model (DistilBERT) was 0.6 and 0.4 respectively.

This method enabled every model to focus on its modality and make a joint decision, making it more robust and accurate in cases of noise or ambiguity.

4.2.2.7 Training Configuration Summary

Table 4.2: CNN and wav2vec 2.0 Configuration

Model	Modality	Input Type	Batch Size	Epochs
CNN	Audio	Mel-spectrogram (3s)	16	25
wav2vec 2.0	Audio	Raw waveform (10s)	16	3

Model	Learning Rate	Optimizer	Loss Function
CNN	0.0001	Adam	Cross-entropy (3-class)
wav2vec 2.0	0.001	Adam	Cross-entropy (3-class)

FastText Training Configuration

Table 4.3: FastText Configuration

Parameter	Value
Learning Rate	0.1
Epochs	20
Embedding Dimension	50
wordNgrams	1
Optimizer	Native SGD
Loss Function	Softmax (cross-entropy)
Batch Size	-

DistilBERT Training Configuration

Table 4.4: DistilBERT Configuration

Parameter	Value
Learning Rate	0.000005
Batch Size	8
Epochs	2
Weight Decay	0.0
Optimizer	Adam
Loss Function	Cross-entropy (via HuggingFace API)

4.3 Research Methodology

In this part, the entire methodology of developing a robust multilingual language identification system under low-resource conditions is given. It consists of audio and text modalities that are important to boost the performance of language identification, especially in noisy or data-limited conditions. This section is organized into five sequential steps: data acquisition, preprocessing, unimodal model training, multimodal fusion, and evaluation. Each step is made modular and adaptable to allow testing different model architectures and fusion strategies. The end-to-end pipeline of multilingual low-resource language identification is illustrated in Figure 4.1. The system takes in audio and textual inputs by undergoing a preprocessing and model training step and then subjected to late fusion and evaluation. This modular design allows flexible experimentation and strong operation in noisy and low-resource environments.

End-to-End Pipeline for Multilingual Low-Resource Language Identification

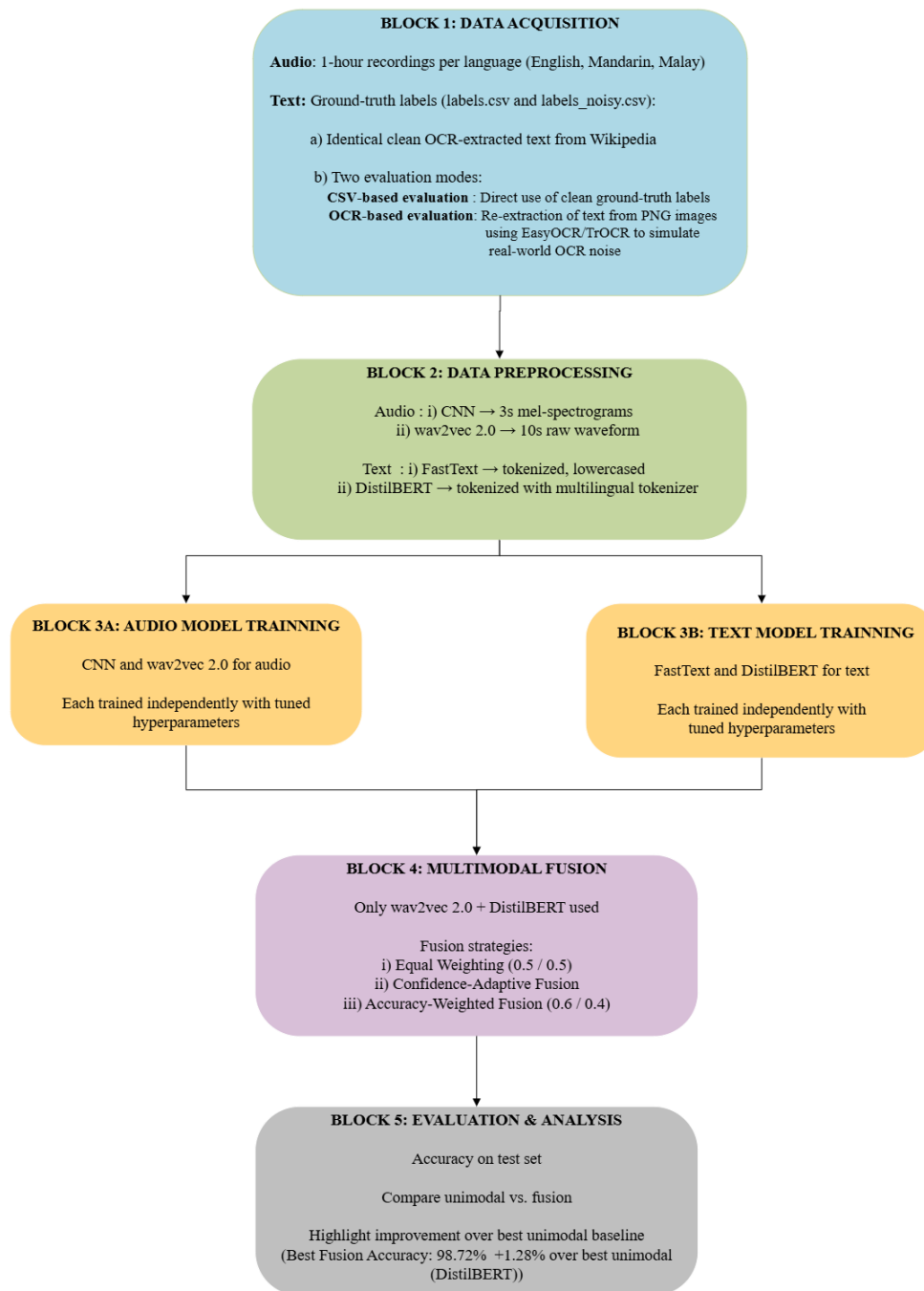


Figure 4.1: Multilingual low-resource language identification pipeline

Block 1: Data Acquisition

In audio case, 1-hour samples of each language were obtained, divided in terms of time: 70% training, 15% validation, 15% testing.

In the case of text, labels.csv involved clean ground-truth labels extracted from Wikipedia. Two modes of evaluation were experimented, (1) CSV-based evaluation with clean text labels directly, and (2) Image-based OCR evaluation with EasyOCR/TrOCR extracting text in PNG images in order to emulate noise in the real world OCR.

Block 2: Data Preprocessing

The audio data was handled in two formats; 3-second mel-spectrograms for CNN and 10-second raw waveforms for wav2vec 2.0. Each of the models was processed differently with regard to text data: FastText had lowercased, tokenized text, and DistilBERT had the multilingual tokenizer to preserve subword structure and casing.

The hyperparameters of each model were searched thoroughly:

- CNN: 36 configurations (learning rate: 0.0001/0.0005/0.001/0.01; batch size: 16/24/32; epochs: 20/25/30)
- Wav2Vec2: 27 configurations (learning rate: 0.0001/0.0005/0.001; batch size: 8/16; epochs: 2/3/4/5)
- FastText: 36 configurations (learning rate: 0.1/0.3/0.5; epochs: 10/20/30; dimension: 50/100; wordNgrams: 1/2)
- DistilBERT: 10 random trials (learning rate: 5e-6 to 5e-5; batch size: 8/16/24; epochs: 2-5; weight decay: 0.0/0.01/0.05)

Block 3: Unimodal Model Training

Each modality had separate models that were trained. Audio classification was done using CNN and wav2vec 2.0, whereas text classification was done

using FastText and DistilBERT. All the models were independently trained with modality-specific hyperparameters and measured to set unimodal baselines.

Block 4: Multimodal Fusion

Late fusion was used to combine the best-performing audio and text models, which were wav2vec 2.0 and DistilBERT. Three strategies were implemented:

- Equal Weighting (0.5 / 0.5)
- Confidence-Adaptive Fusion
- Accuracy-Weighted Fusion (0.6 audio / 0.4 text)

This step generated the final language prediction.

Block 5: Evaluation & Analysis

Models were evaluated by using accuracy as the key metric. There was a comparison between fusion strategies and unimodal baselines, equal weighting which has the highest performance. The findings confirmed the advantages of the combination of modalities under the conditions of low-resource multilingual settings.

4.4 Model Performance Evaluation

In this section, the evaluation of all the models created in this study will be done in details, both unimodal audio and text classifier and multimodal fusion strategies. Models are evaluated based on accuracy on a test holdout and text models assessed under clean and OCR conditions (multimodal phase) to incorporate noise in the real world. These evaluations are based on a similar format, namely input format, architecture, training setup and performance, so that models can be compared. The fusion strategies are compared to the best unimodal baselines to demonstrate the advantage of the interplay of modalities in low-resource multilingual context.

4.4.1 Evaluation Metrics

A set of metrics, which is rather comprehensive, was used to assess the performance of all the models that were developed during this research. These metrics were chosen in order to offer a general and class-specific understanding of the model behaviour in the audio, text and multimodal modes. All evaluations were conducted on a held out test set using a uniform 70/15/15 split strategy, and temporal order of audio data to avoid segment leakage.

a) Accuracy

The main measure of evaluation in all models was accuracy. It is a percentage of the number of the sample that is accurately predicted to the overall number of predictions. Accuracy is a valid and interpretable measurement of the overall model performance due to the balanced class representation (English, Malay, Mandarin).

- CNN: 73.56%
- wav2vec 2.0: 93.91%
- FastText (clean): 98.44%
- DistilBERT (clean): 99.73%
- Fusion (Equal Weighting): 99.18%

b) Precision, Recall, and F1-Score

These metrics were calculated by language to evaluate performance based on classes:

- Precision: Proportion of correct positive predictions
- Recall: Proportion of actual positives correctly identified
- F1-Score: Harmonic mean of precision and recall

They were particularly valuable in studying the imbalances in model behavior across languages, although the data on which they were based were balanced.

Example (CNN - Test Set):

- English: Precision 91.47%, Recall 54.81%, F1 68.55%
- Malay: Precision 89.37%, Recall 67.26%, F1 76.75%
- Mandarin: Precision 59.82%, Recall 98.59%, F1 74.46%

Example (DistilBERT - Test Set):

- English: F1 99.60%
- Malay: F1 99.60%
- Mandarin: F1 100.00%

c) **Confusion Matrix**

Each model was used to create confusion matrices as shown in Figure 4.2 that would be used to visualize the pattern of misclassification. These matrices demonstrated that, e.g., CNN was more likely to be confused between English and Malay, whereas wav2vec 2.0 demonstrated that classes were better distinguished, especially among English.

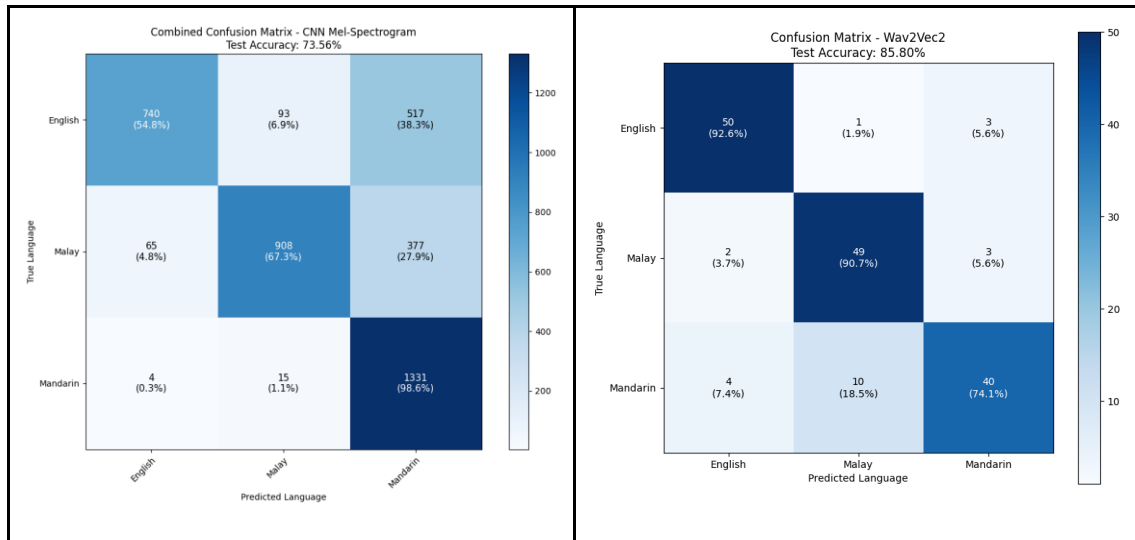


Figure 4.2: Confusion Matrix – CNN vs. wav2vec 2.0 (Side-by-Side Comparison)

d) Fusion Impact Analysis

Fusion strategies were also tested based on accuracy, as well as how they are capable of rectifying errors committed by unimodal models:

- Errors fixed by fusion: 92 out of 102 (90.20%)
- Cases where both modalities were correct: 1113 (91.60%)
- Cases where only text was correct: 69 (5.68%)
- Cases where only audio was correct: 31 (2.55%)
- Cases where both were wrong: 2 (0.16%)

This evaluation concluded that multimodal fusion was more robust particularly when one of the modalities had failed.

4.4.2 Audio Baseline Model : CNN

The baseline audio classification was the CNN model that was trained using 3-second mel-spectrograms. It was tested on held-out test set with accuracy, precision, recall, and F1-score and with further conclusions made based on

confusion matrices. Table 4.5 summarizes the performance metrics of the CNN model.

Evaluation Results

- **Test Accuracy** : 73.56%
- **Macro F1-Score** : 73.26%
- **Weighted F1-Score** : 73.26%

Per Language Performance

Table 4.5: CNN Performance by Language

Language	Precision	Recall	F1-Score	Accuracy (Correct/Total)
English	91.47%	54.81%	68.55%	740/1350
Malay	89.37%	78.70%	74.24%	908/1350
Mandarin	59.82%	98.59%	74.46%	1331/1350

CNN model performed moderately and its accuracy in the test was 73.56%. Although it has done fairly well in all three languages, it has shown a number of trends:

- **English** was the most precise (91.47%) but least recall (54.81%) which means that the model was very conservative in predicting English and when it predicted it was very accurate.
- The highest recall was obtained with **Mandarin** (98.59%) implying that the model was more accurate when dealing with samples of Mandarin.
- **Malay** was also performing with a balance and was more often confused with Mandarin provided by the confusion matrix.

These findings indicate the weaknesses of shallow CNN models when applied to audio of short duration and with less training samples. Nonetheless, the model was a useful point of reference when compared to more advanced architectures, such as wav2vec 2.0.

Suggested Visualizations

Figures 4.3 and 4.4 are used to assist in the analysis of the CNN model performance.

- **Confusion Matrix**

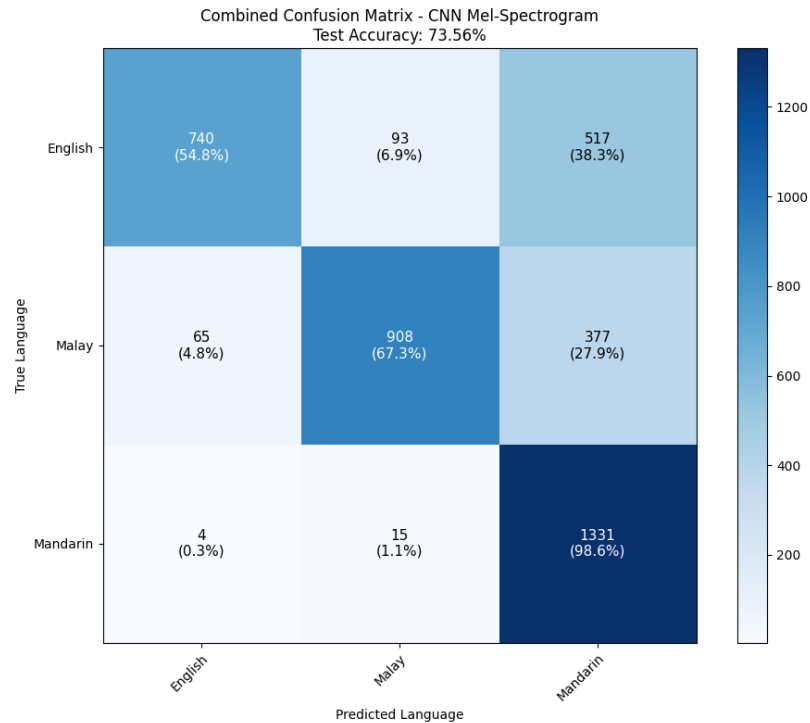


Figure 4.3: Confusion Matrix – Audio Baseline (CNN)

- **Per-Language Metrics**

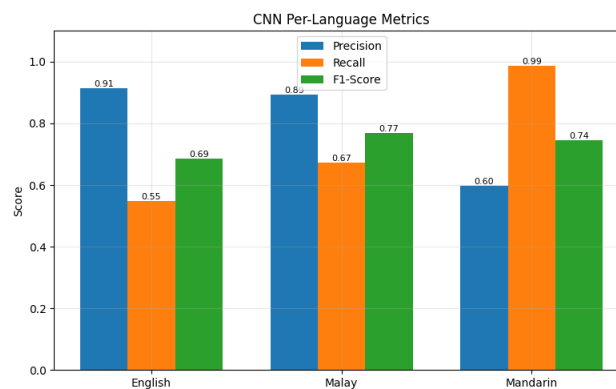


Figure 4.4: Per-Class Metrics – CNN (Precision, Recall, F1)

4.4.3 Audio Transformer Model: wav2vec 2.0

The wav2vec 2.0 model was used as the main transformer-based audio classifier. It used a pre-trained encoder and fine-tuned classification head which has shown great performance compared to the CNN baseline and high generalization on all three languages with test accuracy of 93.91%. Table 4.6 summarizes the performance metrics of the wav2vec 2.0 model.

Per Language Performance:

Table 4.6: wav2vec 2.0 Performance by Language

Language	Precision	Recall	F1-Score	Accuracy (Correct/Total)
English	92.02%	96.79%	94.34%	392/405
Malay	92.68%	93.83%	93.25%	380/405
Mandarin	97.63%	91.11%	94.13%	369/405

The wav2vec 2.0 model not only performed significantly higher than the CNN baseline, but also improved its test accuracy by +20.35 percentage points. It excelled in all three languages, with each language achieving over 90% F1-score. Mandarin achieved the highest precision at 97.36%, while English achieved the highest recall at 96.79%. Malay also showed balanced performance with 92.68% precision and 93.83% recall.

The major aspects that have led to this improvement are:

- Pre-trained unlabeled speech representation over big data.
- Frozen encoder strategy kept generalizable characteristics.
- Longer input segments (10 seconds), providing richer acoustic context

This model set an excellent standard on audio-only classification and was the used as the audio backbone of the multimodal fusion system.

Suggested Visualizations

Figures 4.5, 4.6 and 4.7 are used to support the analysis of the wav2vec 2.0 model's performance.

- **Confusion Matrix**

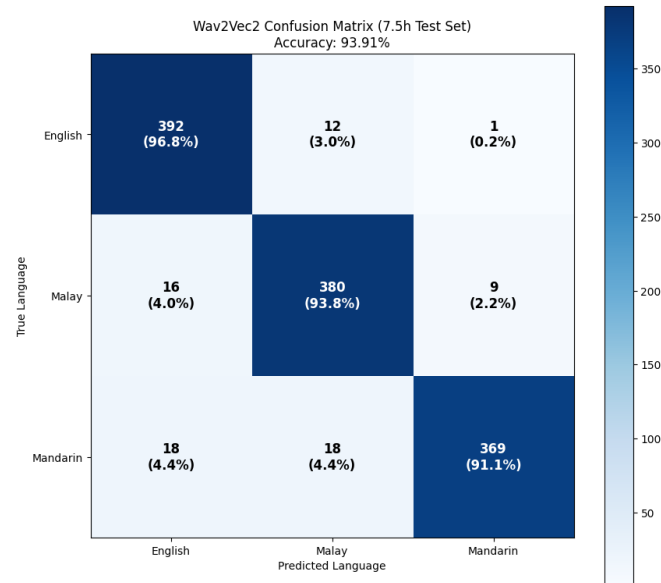


Figure 4.5: Confusion Matrix – Audio Model (wav2vec 2.0)

- **Per-Language Accuracy Bar Chart**

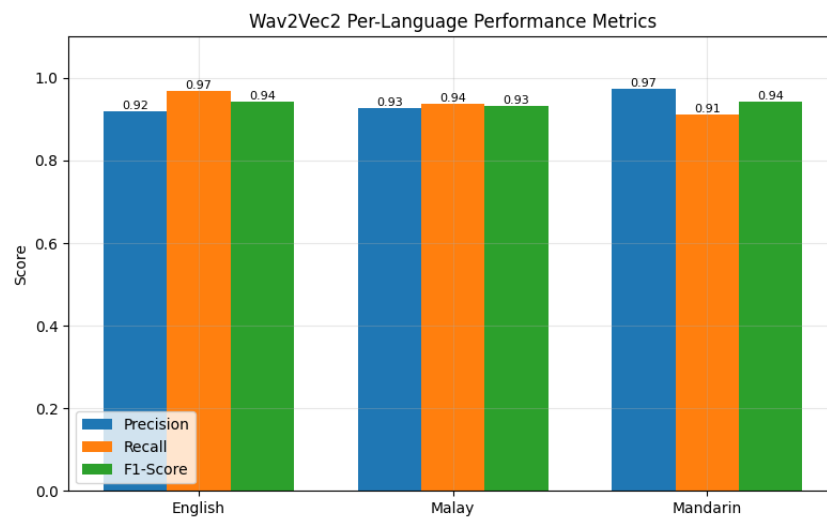


Figure 4.6: Per-Class Metrics – wav2vec 2.0 (Precision, Recall, F1)

- **Training vs. Validation Loss Curve**

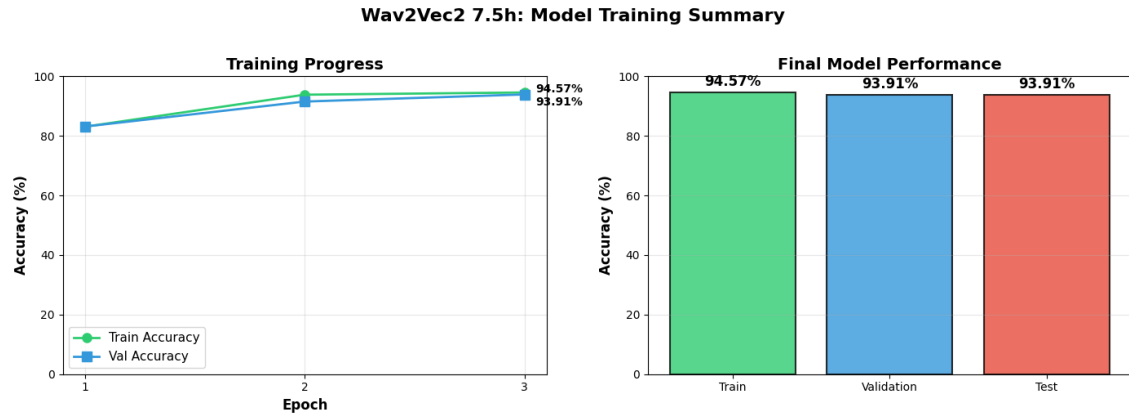


Figure 4.7: Training vs. Validation Loss Curve – wav2vec 2.0

4.4.4 Text Baseline Model: FastText

The FastText was used as a reference model in language classification based on text. It was pretrained on clean text found in labels.csv, with subword embeddings to be used to successfully process tokens out of vocabulary and short utterances. FastText was very efficient and delivered good results despite its simplicity. Table 4.7 summarizes the performance metrics of the FastText model.

Evaluation Results (Clean Text)

- **Training Accuracy** : 99.71%
- **Validation Accuracy** : 98.49%
- **Test Accuracy** : 98.44%

Per Language Performance

Table 4.7: FastText Performance by Language

Language	Precision	Recall	F1-Score	Samples
English	98.78%	97.07%	97.92%	750
Malay	99.86%	98.40%	99.13%	750
Mandarin	96.77%	99.87%	98.29%	750

FastText also performed very well on clean text, with an accuracy of over 98% and high F1-score in all languages. It was also very good at identifying Mandarin, and reached a near perfect recall (99.87%) and maximum F1-score (98.68%). It was designed to be very lightweight and could run inferences in less than a millisecond, which made it well suited to real-time or resource-constrained systems, although it was later surpassed in accuracy and strength by DistilBERT.

Suggested Visualizations

Figures 4.8, 4.9 and 4.10 are used to support the analysis of the FastText model's performance.

- **Confusion Matrix**

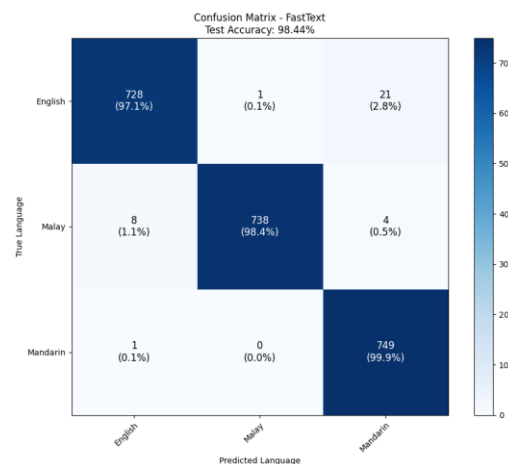


Figure 4.8: Confusion Matrix – Text Baseline (FastText)

- **Per-Language Accuracy Bar Chart**

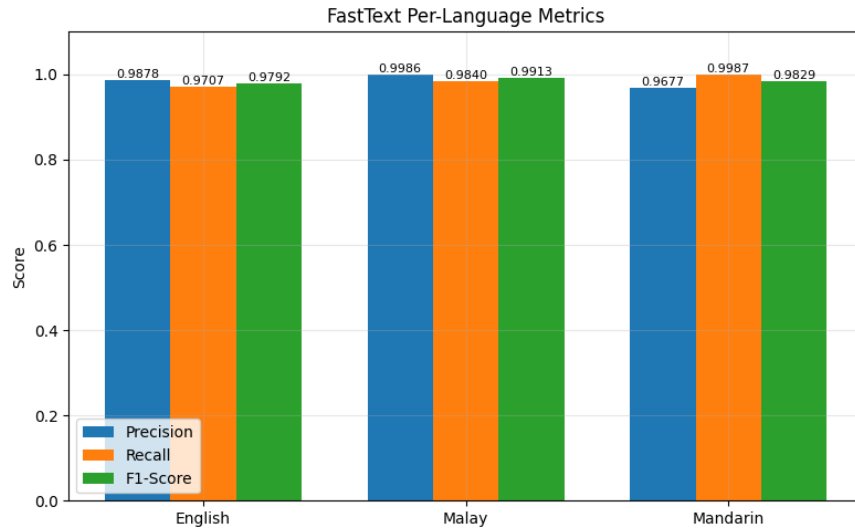


Figure 4.9: Per-Language Accuracy – FastText (Clean)

- Hyperparameter Search Comparison

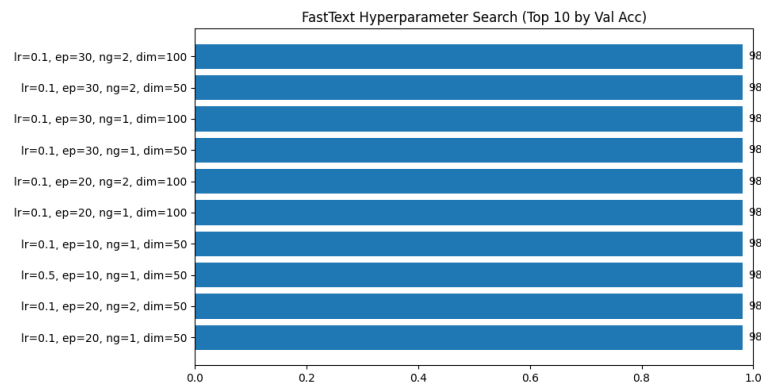


Figure 4.10: Hyperparameter Search Comparison – FastText

4.4.5 Text Transformer Model: DistilBERT

The main transformer-based model used for text classification in this study was DistilBERT. Trained on clean text, and capitalized on multilingual pretraining to get approximately perfect performance in all three languages. Table 4.8 summarizes the performance metrics of the FastText model.

Evaluation Results (Clean Text)

- **Train Accuracy** : 99.92%
- **Validation Accuracy** : 99.78%
- **Test Accuracy** : 99.73%

Per Language Performance

Table 4.8: DistilBERT Performance by Language

Language	Precision	Recall	F1-Score	Samples
English	99.21%	100%	99.60%	750
Malay	100%	99.20%	99.60%	750
Mandarin	100%	100%	100%	750

DistilBERT performed remarkably well, and its performance was superior to all other unimodal models in this experiment. It achieved flawless F1-scores in Mandarin, as well as, close-to-flawless scores in English and Malay. Notably:

- The English had a perfect recall, that is, there were no English samples that were not correctly classified.
- Malay was perfect in the sense that 100% of the Malay samples that were predicted were true.
- The Mandarin gave a score of 100% in all measures.

These findings indicate the power of multilingual pretraining and generalization of the model to cross-linguistic utterances even in short utterances. DistilBERT provided as compared to FastText:

- A 1.29% increase in test accuracy
- Improved management of semantic and contextual variation.

In this project, despite the slower inference time and larger size, DistilBERT was the best standalone model in terms of the performance of this task: text classification.

Suggested Visualizations

Figures 4.11, 4.12 and 4.13 are used to support the analysis of the DistilBERT model's performance.

Confusion Matrix

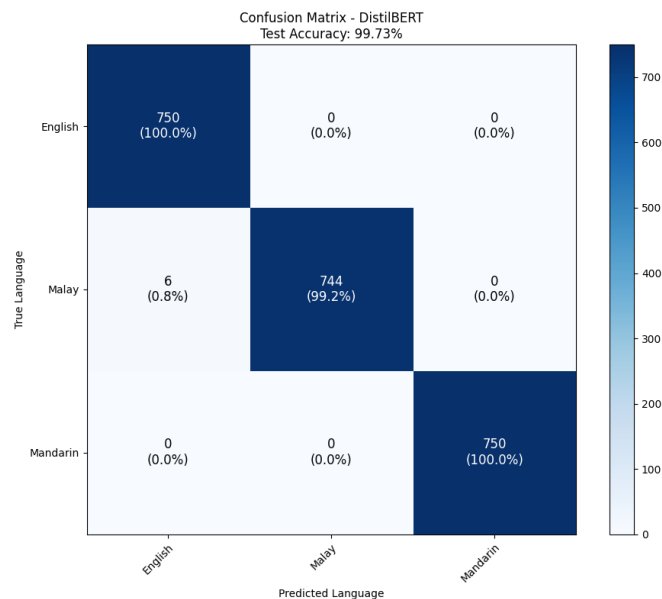


Figure 4.11: Confusion Matrix – Text Transformer (DistilBERT)

- **Learning Rate Impact**

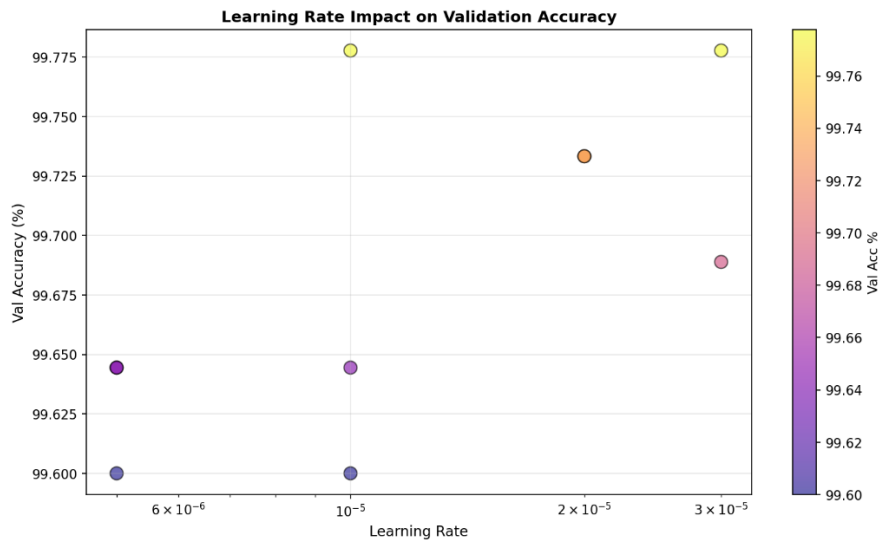


Figure 4.12: Learning Rate Impact – DistilBERT

- **Search Convergence**

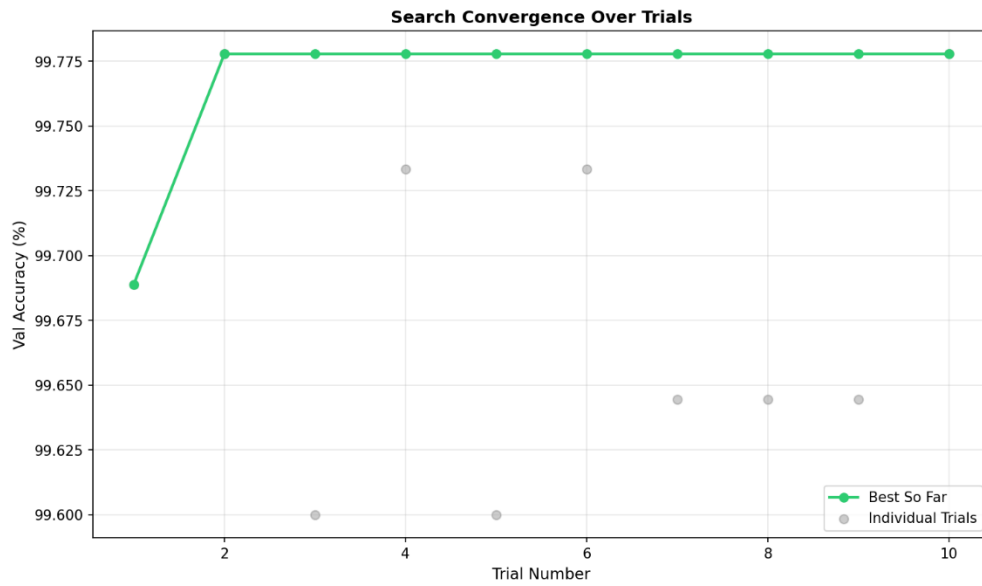


Figure 4.13: Hyperparameter Search Convergence – DistilBERT

4.4.6 OCR Model: TrOCR & EasyOCR

TrOCR and EasyOCR were evaluated as OCR to extract text on noisy images input. Its accuracy was compared with EasyOCR with a ground-truth-aligned dataset of 15 thousand samples (5 thousand of each language). It was evaluated based on the accuracy of the extracted text at the character level and Levenshtein distance which indicates the quality of extracted text and its effect on the downstream classification. Table 4.9 and 4.10 summarize the evaluation results for both TrOCR and EasyOCR.

Evaluation Results

a) Average Character Accuracy (Per Language)

Table 4.9: Average Character Accuracy Per Language

Language	TrOCR	EasyOCR
English	62.24%	87.90%
Malay	46.65%	87.62%
Mandarin	2.56%	76.82%
Overall	37.15%	66.19%

b) Average Levenshtein Distance (Lower is Better)

Table 4.10: Average Levenshtein Distance Per Language

Language	TrOCR	EasyOCR
English	31.65	8.70
Malay	46.89	11.24
Mandarin	17.74	4.35

The performance of TrOCR was different among languages:

- English: EasyOCR showed higher and easy to use character accuracy (87.90% vs 62.24%), which implies that it is capable of dealing with Latin script when there is noise.
- Malay: EasyOCR performed significantly better (87.62% vs 46.65%), which is probably because it is strong with short and word-like terms in dictionaries.
- Mandarin: EasyOCR prevailed (76.82% vs 2.56%), because TrOCR had difficulties in Mandarin character segmentation and recognition.

EasyOCR achieved higher character accuracy among all three languages. EasyOCR was more useful in English (87.90% vs 61.76%), Malay (87.62% vs 46.65%), and Mandarin (76.82% vs 2.56%). The results induced the use of EasyOCR as the final OCR engine with the objective of capitalizing on the engine's stronger character-level performance.

4.4.8 Multimodal Fusion

In order to test the performance of the audio and text combination modality, a late fusion approach was applied on the wav2vec 2.0 and DistilBERT classifiers. This section introduces the performance of three fusion strategies and compares the performance of these strategies with unimodal baselines.

Fusion Strategies Evaluated:

- Equal Weighting (0.5 / 0.5):** The softmax probabilities of both the models were averaged with equal weights. The combination probability of the highest class was chosen.
- Confidence-Adaptive Fusion:** A threshold-based dynamic weighting scheme (threshold=0.8). At the point where the maximum probability of a model (using the softmax) is greater than 0.8 it is assigned greater weight

(0.55 in text or 0.65 in audio). In cases where both models have low confidence (less than 0.8) they are given equal weights. The more assertive model had a greater role to play in the final decision.

- c) **Accuracy-Weighted Fusion (0.6 audio / 0.4 text):** This was done with fixed weights, which were determined by the unimodal test outcomes of the models. The weighting of the audio model (wav2vec 2.0) and text model (DistilBERT) was 0.6 and 0.4 respectively. Table 4.11 presents the test accuracy of the proposed fusion strategies and compares them with the unimodal baseline models.

Evaluation Results

Table 4.11: Test Accuracy of Fusion and Modality-Specific Methods

Method	Test Accuracy
Equal Weighting	99.18% (Best overall)
Confidence-Adaptive Fusion	99.01%
Accuracy-Weighted Fusion	97.86%
Text Only (DistilBERT)	97.28%
Audio Only (wav2vec 2.0)	94.16%

The equal-weighting fusion strategy achieved the highest overall accuracy, outperforming the best unimodal model by **1.9%**. Table 4.12 summarizes the per-language accuracy of the fusion strategies across English, Malay, and Mandarin.

Per-Language Accuracy

Table 4.12: Multilingual Performance of Fusion Strategies

Method	English	Malay	Mandarin
Equal Weighting	99.26%	98.52%	99.75%
Confidence-Adaptive	100%	97.04%	100%
Text Only	99.75%	92.84%	99.26%
Accuracy-Weighted Fusion	98.02%	99.78%	99.78%
Audio Only	97.04%	94.32%	91.11%

Equal Weighting strategy was the most accurate in terms of overall accuracy (99.18%) compared to unimodal models as well as other fusion approaches. It was also most balanced in all languages with Mandarin achieving accuracy of 99.75% and Malay with an improvement of 5.68% over the text-only classification.

There were also good scores in the Confidence-Adaptive Fusion approach (99.01%), particularly when there was uncertainty in one modality. But the Accuracy-Weighted Fusion strategy (97.86%) performed lower than Equal Weighting overall, which could probably be related to the greater influence of the audio model in the fusion process.

These findings validate the fact that late fusion increases robustness especially on cases of noise or ambiguities. The fusion pipeline was able to use:

- TrOCR for English
- Malay and Mandarin EasyOCR.
- wav2vec 2.0 for audio fallback

This mixed methodology is a reflection of practical implementation of the system where OCR quality and audio quality change depending on language and environment.

Suggested Visualizations

Figures 4.14, 4.15, 4.16, 4.17, and 4.18 are used to support the analysis of the performance of different fusion methods.

- **Bar Chart: Accuracy by Fusion Method**

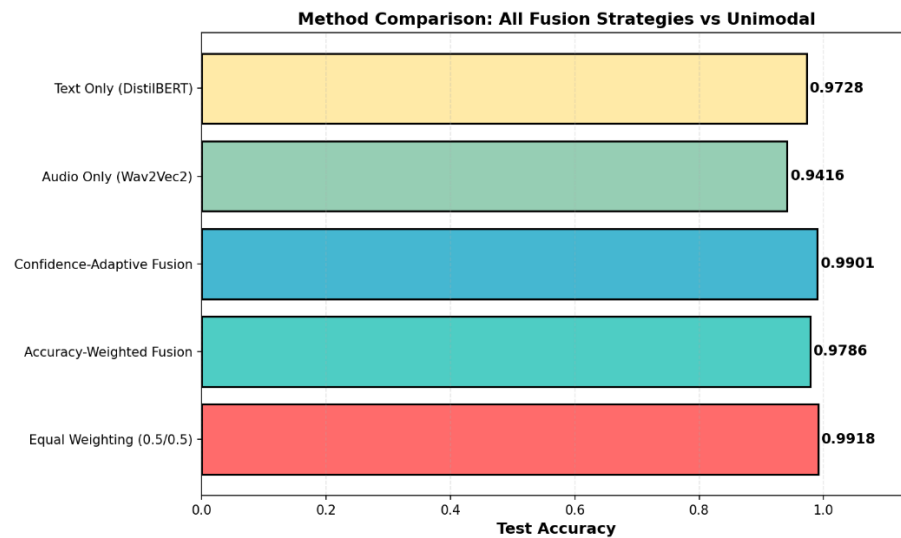


Figure 4.14: Accuracy by Fusion Method – Audio + Text vs Unimodal

- **Confusion Matrix**

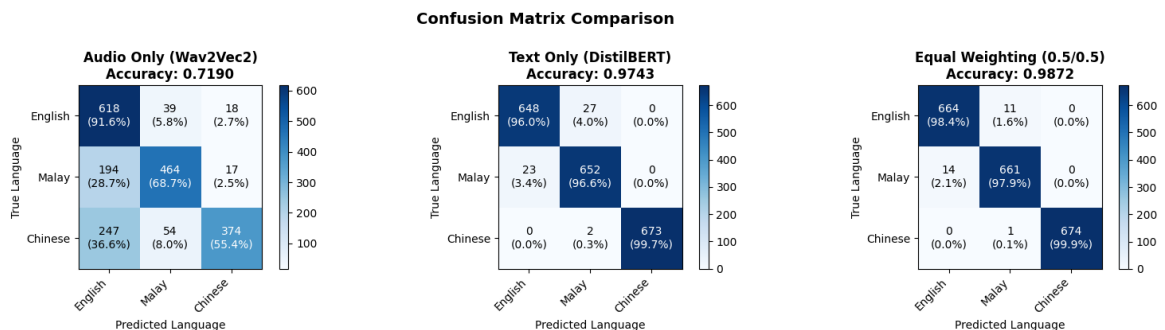


Figure 4.15: Confusion Matrix – Equal Weighting Fusion, Audio Only and Text Only

• Per-Language Performance

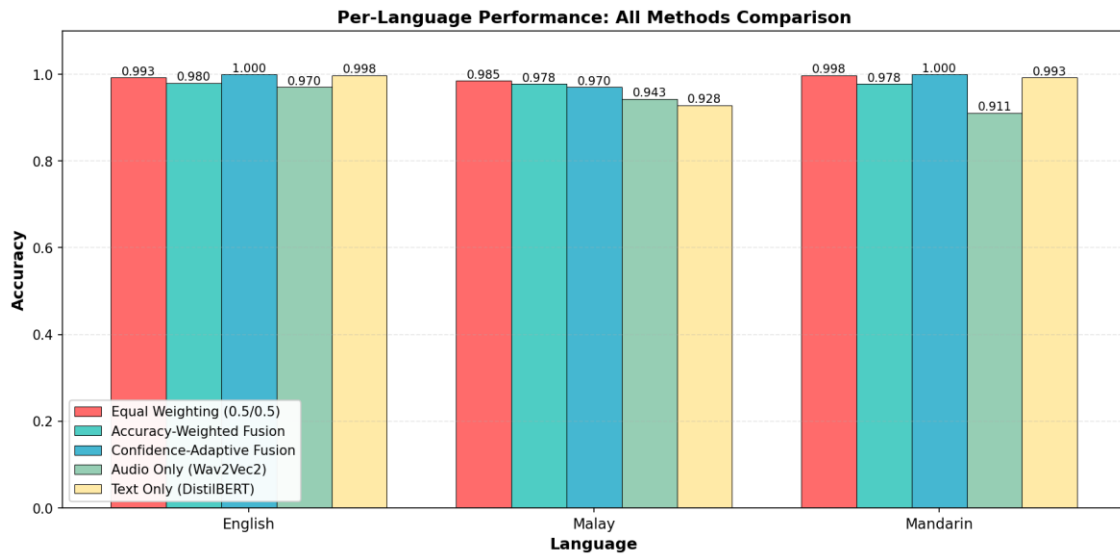


Figure 4.16: Per-Language Accuracy – Fusion vs Unimodal Models

• Modality Agreement Heatmap

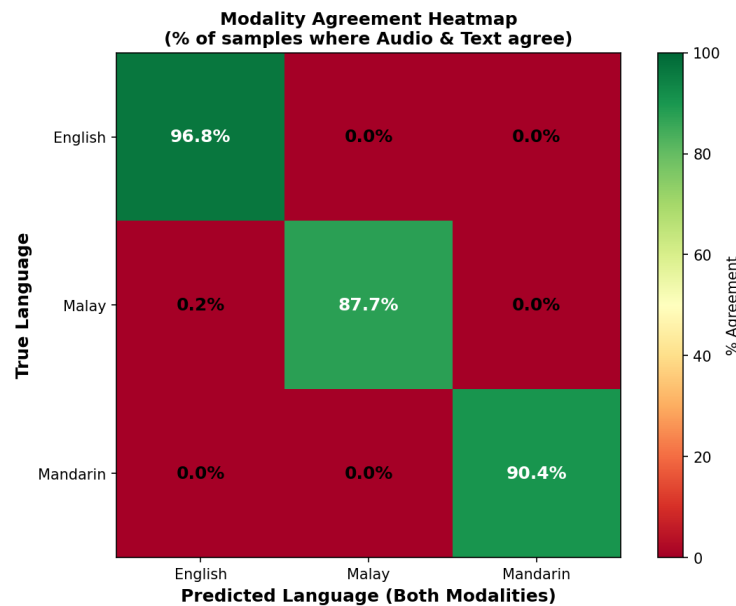


Figure 4.17: Modality Agreement Heatmap – Audio vs Text

- **Fusion Strategy Improvement Analysis**

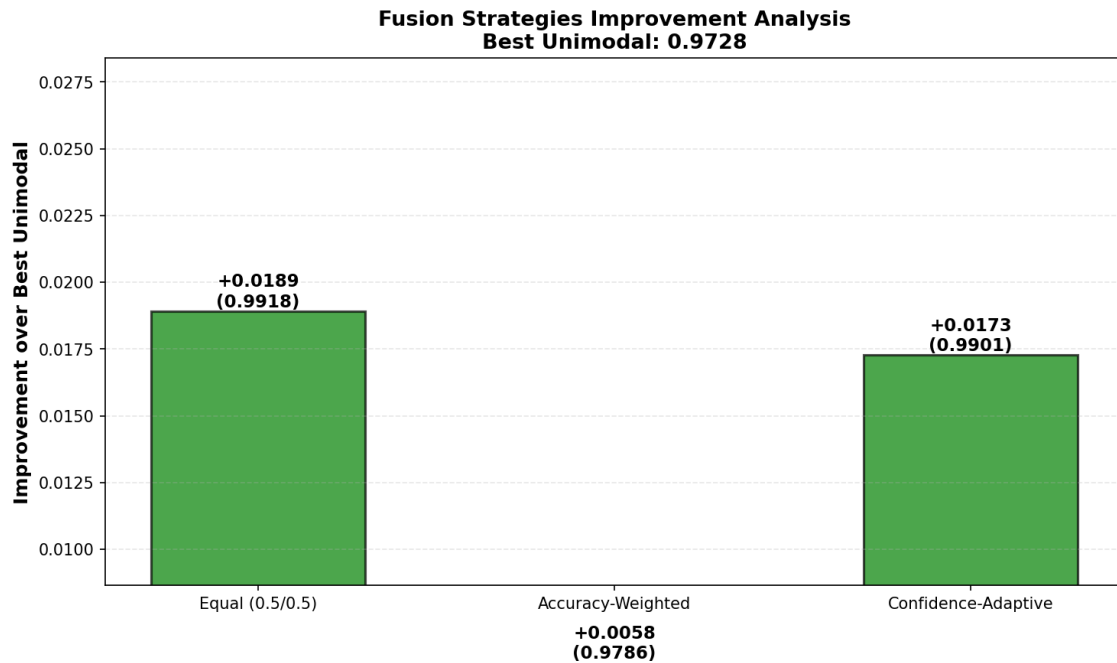


Figure 4.18: Fusion Strategy Improvement Over Unimodal Baselines

4.4.9 Performance Summarization and Benchmark Comparison

In all experiments, we have compared a variety of unimodal and multimodal models of language identification. The summary of the principal results is presented in Table 4.13.

Table 4.13: Accuracy and Notes for Audio, Text, and Fusion Models

Model	Modality	Test Accuracy	Notes
CNN	Audio	73.56%	Baseline audio model
wav2vec 2.0	Audio	93.91%	Stronger audio representation
FastText	Text (clean)	98.44%	Lightweight, fast, but less robust
DistilBERT	Text (clean)	99.73%	Best unimodal model
DistilBERT	Text Only (OCR + DistilBERT)	97.28%	
Equal Weighting Fusion	Audio + Text (OCR)	99.18%	Best overall model
Confidence-Adaptive Fusion	Audio + Text (OCR)	99.01%	Dynamic weighting
Accuracy-Weighted Fusion	Audio + Text (OCR)	97.86%	Underperformed due to audio bias

Key Observations:

- Text modality is used when clean text is known and DistilBERT is the most accurate in unimodal (99.73%). Nevertheless, this performance decreases a little when text is recognized by OCR as we get recognition errors.
- The quality of OCR is important to the downstream text classification because the TrOCR was applied to English, where the engine scored better than EasyOCR (62.24% vs. 34.13%), which means that it should be used with English inputs. The Malay and Mandarin were tested with the EasyOCR which became far superior to the TrOCR (Malay: 87.62% vs. 46.65%, Mandarin: 76.82% vs. 2.56%), which makes it more reliable in extracting the text in these languages.

- Audio-only models (e.g. wav2vec 2.0) were more prone to background noise and did not do well on acoustically similar languages, especially Malay and Mandarin.
- The fusion models enhanced the strength of each of the modalities. In the case of OCR failure or noise, the audio model was used to give a fallback.
- Equal Weighting Fusion strategy was the highest performing overall (99.18%), with balanced per-language accuracy and few misclassifications, performing better than all unimodal and other fusion strategies.

In order to situate the performance of our proposed model, we draw parallels with the recent multimodal language identification frameworks, and we concentrate on the MuSeLI by Bharadwaj et al. (2023). Although audio-text multimodal models like WhisBERT (Wolff et al., 2023) investigate the learning of audio-text, they focus on language modeling and syntactic generalization as opposed to language identification. Thus, the comparison with the University of South Alabama can be more relevant with the help of MuSeLI.

MuSeLI is an integrated multimodal system that incorporates audio with metadata including video titles, video descriptions and location of uploads. It is based on mSLAM and LASR, and it pretrains on massive data such as VoxLingua107 and Dhwani-YT, between 6,600 and 30,000 hours of audio. According to the model, a reported accuracy of 97.6% is achieved on VoxLingua107. It is an architecture that learns shared representations through modalities with a single encoder and has the advantage of large amounts of pretraining, as well as rich inputs in terms of metadata.

On the other hand, our model uses a modular late fusion approach, which involves the integration of wav2vec 2.0 (audio) and DistilBERT (text) and OCR-extracted inputs. TrOCR and EasyOCR are used in the OCR pipeline to process

English and Malay and Mandarin respectively, respectively, which were chosen on the basis of character accuracy, depending on the language. Although our model had been trained using a much smaller dataset (~15,000 samples), on our evaluation set it had a test accuracy of 99.18%.

Although the two methods have different scopes and designs, they both have different benefits. MuSeLI benefits the high-resource environment by being able to generalize over a broad dimension of languages and metadata types and the use of a single encoder and large-scale training. Instead, our model focuses on the data efficiency, modularity and practical implementation. Late fusion design enables customization of components (e.g. replacement of OCR engines or audio models) and the system is made to work with real world noisy inputs (e.g. screenshots or poor quality text) typical of mobile or low resource systems.

Overall, both models prove that multimodal learning is important in identifying the language. MuSeLI demonstrates the capability of large-scale pretraining and metadata incorporation, whereas our system emphasizes the ability of a sparse, modular system to operate competitively with little data and infrastructure.

4.5 Summary

This chapter gave a detailed analysis of the unimodal and multimodal approaches to language identification of three languages English, Malay and Mandarin. The initial experiments involved the use of baseline models that addressed audio and text modality, and thereafter advanced architectures and fusion plans aimed at enhancing robustness and accuracy in real-life scenarios.

The wav2vec 2.0 model was much better than CNN baseline in the audio modality with test accuracy of 93.91% relative to the CNN with 73.56%. Nonetheless, there was a difference in the performance across languages, with

Mandarin and, to a lesser extent, Malay achieving lower accuracies than English, likely due to acoustical similarities and background noise.

On the text modality, DistilBERT with clean transcripts obtained the best unimodal accuracy of 99.41%. But with the OCR extracted text performance decreased a little to 97.28%, indicating the significance of the quality of OCR. An in-depth comparison of OCR engines showed that EasyOCR was more efficient across English, Malay, and Mandarin, resulting in the selection of EasyOCR as the final OCR engine for the OCR-based text pipeline.

In order to have both modalities to complement each other, a late fusion architecture was adopted. Equal Weighting Fusion was the most successful strategy of the fusion strategies considered and had a test accuracy of 99.18%. This method was not only able to do better than all unimodal baselines, but also show balanced per-language performance and resistance to informative noisy or incomplete inputs.

Our model was contrasted with the state-of-the-art multimodal frameworks in the study by Bharadwaj et al. (2023) based on a study that assesses MuSeLI, which combines audio and metadata (title, description, location) with the help of a single encoder. Although MuSeLI was very accurate (97.6 on VoxLingua107), our model increased its accuracy with fewer data and a simpler design.

The text extraction, with modular late fusion design, is more flexible, data efficient and it is more adaptive to real-world deployment scenarios especially in low-resource scenarios or mobile settings.

To conclude, the proposed multimodal system proves that a lightweight and modular integration of audio and text produced by OCR can have a high level of performance in language recognition, a high level of cross-linguistic generalization, and is effective even in the presence of noisy conditions.

Chapter 5: Data Analysis Plan

5.1 Introduction to the Data Analysis Plan

This chapter presents the data analysis and experimental results for the multilingual low-resource language identification system. The project focuses on three target languages: English, Malay, and Mandarin. Since the system uses both audio and text-based inputs, the analysis is divided according to the major components of the project, including audio data analysis, text image analysis, OCR performance, unimodal model performance, and multimodal fusion performance.

The purpose of this chapter is to evaluate the effectiveness of different models and processing strategies used in the project. For the audio modality, a CNN model was used as the baseline, while wav2vec 2.0 was used as the advanced transformer-based speech model. For the text modality, FastText was used as the lightweight baseline, while DistilBERT was used as the transformer-based language classification model. OCR performance was also analysed because the text-based pipeline depends on extracting text from image inputs before performing language identification.

The chapter also evaluates multimodal late fusion strategies. In this project, late fusion is used because the audio model and text model produce their own prediction probabilities separately before the probabilities are combined at decision level. The fusion methods evaluated include equal weighting fusion, confidence-adaptive fusion, and accuracy-weighted fusion. The results are reported using accuracy, precision, recall, F1-score, confusion matrices, and per-language performance analysis.

5.2 Data Preparation

The dataset used in this project consists of two main modalities: audio and text images. The audio data was prepared from VoxLingua-based language audio sources, while the text image data was generated from Wikipedia text samples. The target languages used throughout the project were English, Malay, and Mandarin.

5.2.1 Audio Data Preparation

For the audio modality, speech data was collected and prepared for each of the three target languages. The final audio dataset consisted of approximately 7.5 hours of audio per language. These audio files were used to train and evaluate the audio-based language identification models.

The audio data was divided into training, validation and testing sets. A smaller subset of the one-hour dataset was used to train the hyperparameters to reduce training time and allow for quicker iterations. The best hyperparameters were taken and final models were trained and evaluated on the larger 7.5 hours.

The CNN model is based on audio features like mel-spectrogram representations, whereas the wav2vec 2.0 model is based on speech input as waveforms. This enabled the project to run a comparison between the traditional audio classification using deep learning model and the pre-trained transformer-based speech representation model.

Table 5.1: Audio Dataset Preparation Summary

Item	Description
Source	VoxLingua-based audio data
Languages	English, Malay, Mandarin
Final duration	Approximately 7.5 hours per language
Tuning dataset	Approximately 1 hour per language
Final training dataset	Approximately 7.5 hours per language
Audio models	CNN and wav2vec 2.0
Audio features	Mel-spectrogram for CNN, waveform input for wav2vec 2.0

A summary of the audio data set used in the project is given in Table 5.1. Using a balanced audio duration in three languages helps to minimize the class imbalance in training and evaluation. It also enables the CNN and wav2vec 2.0 models to be confronted in the same data situation.

5.2.2 Text Image Data Preparation

For the text modality, text samples were collected from Wikipedia pages and categories for English, Malay, and Mandarin. These text samples were then rendered into image format to simulate OCR-based text input. This allowed the project to evaluate language identification from uploaded or captured text images.

Each language contained 5,000 generated text images. Every image was linked to a label file containing the filename, language, and original text. English

and Malay images were rendered using Noto Sans fonts, while Mandarin images were rendered using a simplified Chinese-compatible font.

Table 5. 2: Text Image Dataset Summary

Language	Number of Clean Images	Number of Noisy Images
English	5,000	5,000
Malay	5,000	5,000
Mandarin	5,000	5,000
Total	15,000	15,000

Table 5.2 shows that the text image dataset was balanced across the three languages. This balance is important because it prevents the text classification models from being biased toward a language with more training samples.

Clean Text Image Examples for English, Malay, and Mandarin



Figure 5.1: Clean Text Images for English, Malay, and Mandarin

Figure 5.1 helps illustrate how the Wikipedia text samples were converted into image format before OCR extraction and text classification.

5.2.3 Noisy Image Generation

To make the OCR experiment more realistic, noisy versions of the clean text images were generated. The purpose of adding noise was to simulate real-world image conditions such as low-quality uploads, compression, camera blur, and uneven lighting. This is important because OCR systems may perform well on clean images but produce errors when used on realistic images.

A random noise type was applied to each image. The main noise types used in this project are shown in Table 5.3.

Table 5.3 Image Noise Types Used in the OCR Dataset

Noise Type	Description	Purpose
Gaussian noise	Random pixel-level noise added to the image	Simulates sensor noise or poor image quality
Salt-and-pepper noise	Random black and white pixels inserted into the image	Simulates corrupted image capture
Motion blur	Linear blur applied to the text image	Simulates camera movement or shaky capture
JPEG compression artifacts	Image compressed with reduced quality	Simulates low-quality uploaded or compressed images
Brightness and contrast adjustment	Random brightness and contrast changes	Simulates uneven lighting conditions

Figure 5.2: Examples of Noise Applied to OCR Text Images**Figure 5.2: Original And Noisy OCR Images**

Figure 5.2 demonstrates how the image quality was degraded before OCR extraction. It also helps explain why OCR accuracy may decrease when noise, blur, or compression artifacts are present.

The noisy image dataset was used to evaluate OCR robustness and to support the text-based language identification pipeline. This preparation step was important because the final system is intended to handle real image inputs, not only clean synthetic text images.

5.2.4 OCR Data Preparation

OCR was applied to extract text from the generated image inputs. Since OCR performance differed across languages, EasyOCR was used for English, Malay, and Mandarin. This selection was based on OCR accuracy comparison across the three languages shown in Table 5.4.

The extracted OCR text was later used as input to the text classification model in the multimodal fusion pipeline. The original text labels were used only for OCR accuracy evaluation and dataset labelling, while the OCR extraction process itself was performed on the image files.

Table 5.4 Selected OCR Engine by Language

Language	TrOCR	EasyOCR	Selected OCR Engine
English	62.24%	87.90%	EasyOCR
Malay	46.65%	87.62%	EasyOCR
Mandarin	2.56%	76.82%	EasyOCR

5.3 Descriptive Analysis

Descriptive analysis was performed to understand the structure, balance, and quality of the prepared dataset. Since the project uses both audio and image-based text inputs, the descriptive analysis focuses on language distribution, modality distribution, and OCR quality.

5.3.1 Dataset Distribution by Language

The dataset was designed to be balanced across the three target languages. For the audio modality, each language contained approximately 7.5 hours of audio. For the text image modality, each language contained 5,000 clean images and 5,000 noisy images as shown in Table 5.5.

Table 5.5 Overall Dataset Composition

Modality	Language	Dataset Size
Audio	English	~7.5 hours
Audio	Malay	~7.5 hours
Audio	Mandarin	~7.5 hours
Clean text images	English	5,000 images
Clean text images	Malay	5,000 images
Clean text images	Mandarin	5,000 images
Noisy text images	English	5,000 images
Noisy text images	Malay	5,000 images
Noisy text images	Mandarin	5,000 images

5.3.2 Audio and Text Modality Distribution

The audio and text modalities were prepared differently because they represent different types of input. Audio data was prepared as speech segments, while text data was prepared as image-based text samples. This allowed the final

system to support both spoken language identification and OCR-based text language identification.

The use of two modalities also allowed the project to compare the strengths and weaknesses of audio-only and text-only models. Audio models depend on acoustic and phonetic patterns, while text models depend on written language features. Because these two modalities contain different information, combining them through late fusion can improve overall robustness.

5.3.3 OCR Quality and Noise Variation

OCR accuracy also showed clear differences between languages and OCR engines. EasyOCR was more effective for English, Malay, and Mandarin. Figure 5.3 supports the decision to use EasyOCR as the final OCR engine.

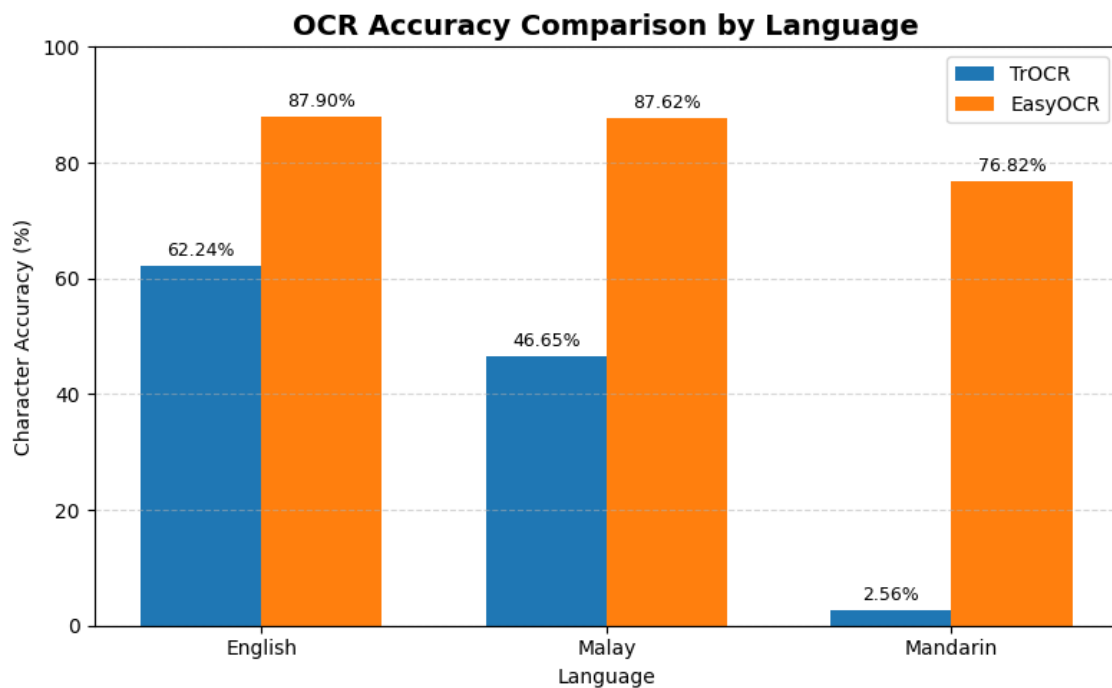


Figure 5.3: OCR Accuracy Comparison by Language

5.4 Inferential Analysis

Inferential analysis involved comparing the performance of different models and the performance of advanced models with the baseline models. The CNN model was used as the base model and wav2vec 2.0 was the advanced audio model. The transformer-based model was DistilBERT, while the lightweight model was FastText for the text modality.

The audio model results showed in Table 5.6 that wav2vec 2.0 achieved much stronger performance than the CNN baseline. The CNN model achieved 73.56% test accuracy, while wav2vec 2.0 achieved 93.91% test accuracy. This shows that the pre-trained speech representations from wav2vec 2.0 were more effective for multilingual audio classification than the CNN baseline.

For the text modality, DistilBERT achieved higher accuracy than FastText. FastText achieved 98.44% test accuracy, while DistilBERT achieved 99.73% test accuracy. This indicates that contextual transformer-based text representations were more effective than lightweight word and subword-based features.

Table 5.6: Unimodal Model Performance Comparison

Model	Modality	Test Accuracy
CNN	Audio	73.56%
wav2vec 2.0	Audio	93.91%
FastText	Text	98.44%
DistilBERT	Text	99.73%

The multimodal fusion results showed in Table 5.7 suggested that combining audio and text predictions improved performance compared to using audio or OCR text alone. In the fusion test set, audio-only prediction achieved

94.16%, while text-only prediction using OCR-extracted text achieved 97.28%. Equal weighting fusion achieved the best overall result with 99.18% accuracy.

Table 5.7: Multimodal Fusion Strategy Comparison

Method	Modality	Test Accuracy
Audio Only	Audio	94.16%
Text Only	OCR Text	97.28%
Equal Weighting Fusion	Audio + Text	99.18%
Confidence-Adaptive Fusion	Audio + Text	99.01%
Accuracy-Weighted Fusion	Audio + Text	97.86%

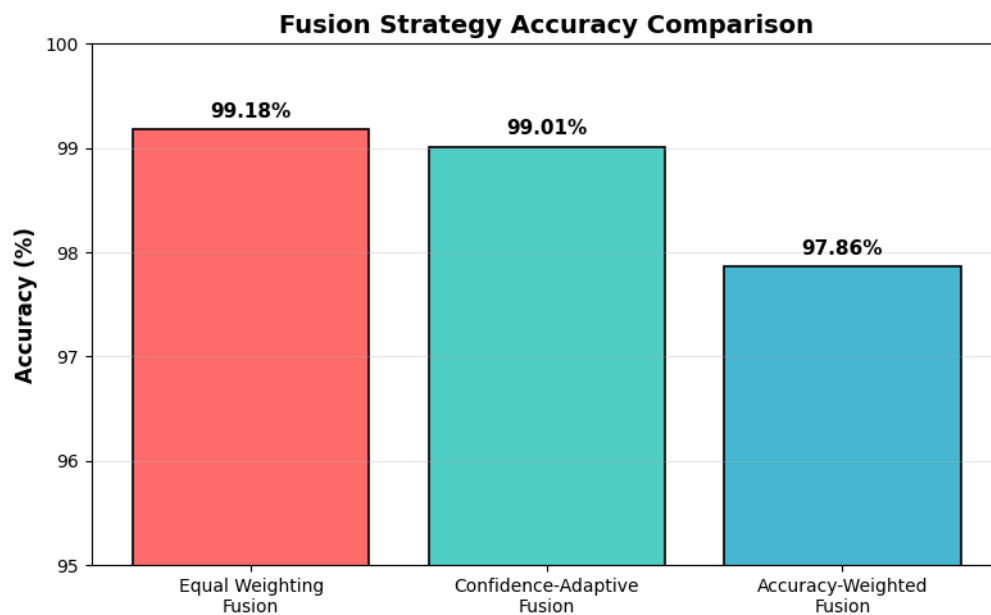


Figure 5.4 Fusion Strategy Accuracy Comparison

5.5 Qualitative Data Analysis

Qualitative analysis was performed by inspecting selected OCR outputs, noisy image samples, and confusion matrices. This analysis helped explain why some models performed better than others and why certain errors occurred.

5.5.1 OCR Output Inspection

Selected OCR outputs were reviewed to understand how image quality affected text extraction. Noisy images with blur, compression artifacts, or strong pixel noise were more difficult for OCR engines to process accurately. Mandarin OCR was especially sensitive because small distortions could affect character recognition. Figure 5.5 demonstrate how OCR errors can appear in the text pipeline and why OCR accuracy affects text-based language identification.

```
EN_002.png:
easyocr: cognitive evaluations
trocr : COSNITIVE EVALUATIONS
chosen : [trocr] COSNITIVE EVALUATIONS
EN_003.png:
easyocr: By yaRpheag spaaiotem3oaaladiegrirersory prcper dynamics, and fading memory proper ties, as
trocr : BYPERPLYING AYRAMIES, AND FALING INCLUSIVITY PROTIPER
chosen : [trocr] BYPERPLYING AYRAMIES, AND FALING INCLUSIVITY PROTIPER

- CN_001.png:
easyocr: 这样的行业现状也有弊端二 许多有启示性
trocr : ***6***4%, *****
chosen : [easyocr] 这样的行业现状也有弊端二 许多有启示性
- CN_002.png:
easyocr: 的失败实验得不到机会发表 很多人会把
trocr : ***
chosen : [easyocr] 的失败实验得不到机会发表 很多人会把
```

Figure 5.5: Sample OCR Outputs from Noisy Images

5.5.2 Confusion Matrix Analysis

Confusion matrices (Figure 5.6, 5.7 and 5.8) were used to analyse which languages were misclassified by the models. For audio models, confusion occurred mainly between languages with overlapping acoustic or phonetic

patterns. For the fusion model, the confusion matrix showed fewer errors because the audio and text models were able to correct each other in many cases.

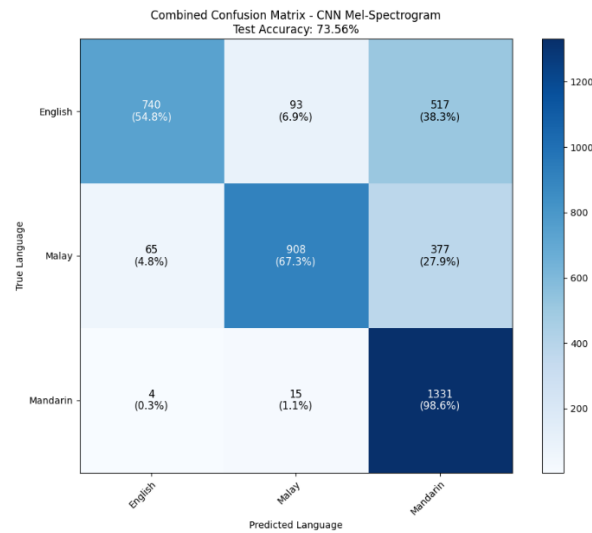


Figure 5.6: Confusion Matrix for CNN Audio Model

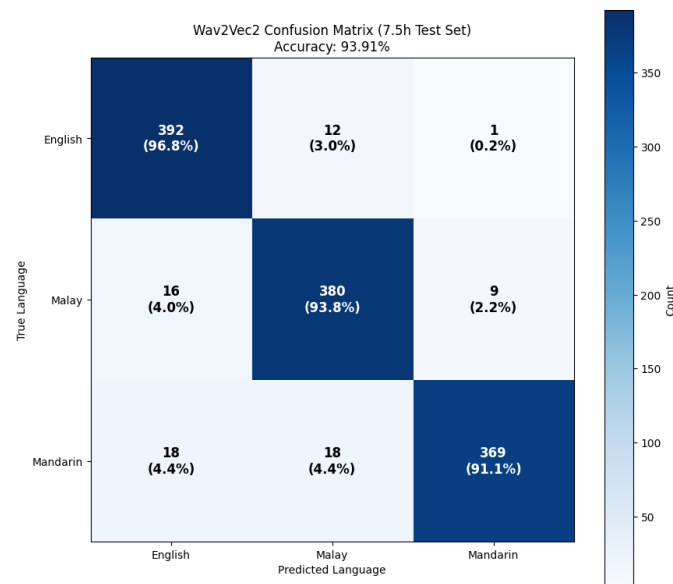


Figure 5.7: Confusion Matrix for wav2vec 2.0 Audio Model

Figure 5.5: Confusion Matrix Comparison for Fusion Strategies

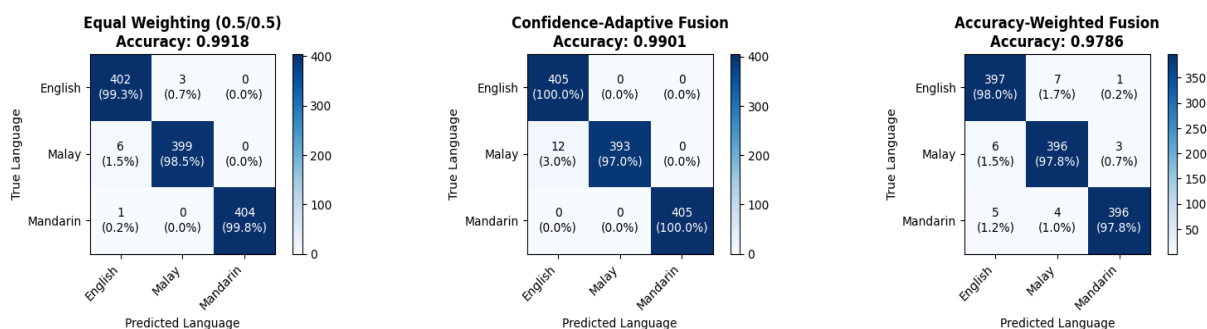


Figure 5.8 Confusion Matrices for Multimodal Fusion Strategies

5.6 Advanced / Exploratory Analysis

Advanced analysis was conducted through hyperparameter tuning, per-language performance evaluation, and multimodal late fusion error correction analysis.

5.6.1 Hyperparameter Tuning Analysis

Hyperparameter tuning was used to improve model performance before final evaluation. For the audio models, tuning was first performed using a smaller one-hour dataset to reduce computational cost. The final models were then trained and evaluated using the larger 7.5-hour dataset.

For the text models, FastText and DistilBERT were evaluated using balanced text datasets. This allowed the project to compare a lightweight text classification model against a transformer-based text model.

5.6.2 Per-Language Model Behaviour

For the CNN model, the final evaluation on the 7.5-hour dataset achieved 73.56% test accuracy. The model performed moderately as an audio baseline but was weaker than wav2vec 2.0. The per-language results showed that

Mandarin had very high recall, while English and Malay were more often confused as shown in Table 5.8.

Table 5.8: CNN Per-Language Performance

Language	Precision	Recall	F1-Score	Correct/Total
English	91.47%	54.81%	68.55%	740/1350
Malay	89.37%	78.70%	74.24%	908/1350
Mandarin	59.82%	98.59%	74.46%	1331/1350

For wav2vec 2.0, Table 5.9 shows that the model achieved 93.91% test accuracy. The per-language performance was strong across all three languages, with English achieving the highest recall and Mandarin achieving the highest precision. This indicates that wav2vec 2.0 provided more reliable audio-based language identification compared to the CNN baseline.

Table 5.9: wav2vec 2.0 Per-Language Performance

Language	Precision	Recall	F1-Score	Correct/Total
English	92.02%	96.79%	94.34%	392/405
Malay	92.68%	93.83%	93.25%	380/405
Mandarin	97.63%	91.11%	94.13%	369/405

5.6.3 Late Fusion Error Correction Analysis

The fusion experiments were then used to explore whether audio and text predictions could correct each other's errors. The equal weighting fusion method achieved the highest overall accuracy. The fusion analysis showed that many errors made by one modality could be corrected when the other modality produced the correct prediction.

Table 5.10: Fusion Error Analysis

Fusion Analysis Category	Result
Both modalities correct	1113 / 1215
Only text correct	69 / 1215
Only audio correct	31 / 1215
Both modalities wrong	2 / 1215
Error cases where at least one modality was wrong	102
Errors fixed by fusion	92 / 102
Fusion error correction rate	90.20%

Table 5.10 shows that late fusion corrected 92 out of 102 cases where at least one modality made an error. This supports the effectiveness of multimodal fusion because the system was able to use one modality to compensate when the other modality was incorrect.

5.7 Data Interpretation

The results show that transformer-based models performed better than baseline models in both audio and text modalities. For audio classification, wav2vec 2.0 significantly outperformed the CNN baseline. This is because wav2vec 2.0 uses pre-trained speech representations that capture richer acoustic and phonetic information. In contrast, the CNN model depends on handcrafted audio features such as mel-spectrograms and has less ability to model long-range speech patterns.

For text classification, DistilBERT achieved the best unimodal performance. This shows that contextual language representations are highly effective for distinguishing English, Malay, and Mandarin text. FastText also performed strongly, but DistilBERT achieved better accuracy because it can capture deeper contextual and subword-level information.

The OCR results show that OCR quality has a direct effect on the text-based pipeline. Although DistilBERT performed very well on clean text labels, the text-only result in the fusion experiment was lower because it used OCR-extracted text. This shows that OCR errors can reduce the quality of text classification. Therefore, OCR engine selection is an important part of the full system.

The fusion results demonstrate that combining audio and text improves the reliability of language identification. Equal weighting fusion achieved the best performance because both modalities contributed useful information. In many cases, when either the audio model or text model made an incorrect prediction, the other modality was able to correct it. This explains why fusion achieved higher performance than audio-only and OCR text-only prediction.

The results also show that accuracy-weighted fusion did not perform as well as equal weighting fusion. This may be because fixed weights do not always reflect the reliability of each modality for every individual sample. A modality that performs well overall may still be uncertain or incorrect for specific inputs. Therefore, equal weighting was a more balanced approach for this dataset.

The results overall confirm the primary goal of the project: Multimodal fusion is beneficial for multilingual low resource language identification over unimodal fusion.

5.8 Reporting Results

This project uses tables, figures, classification metrics and confusion matrices to report the results. Since the project involves both audio and text modalities, the results are reported separately for each modality before presenting the multimodal fusion results. This enables the contribution of each individual model and allows to find out if the fusion of all the models leads to a better language identification performance.

The main evaluation metrics for this project are accuracy, precision, recall and F1 Score. The overall percentage of correct predictions is measured by accuracy, and a more detailed analysis of each language class is provided by precision, recall and F1-score. These metrics are particularly relevant for determining if a model is consistent across English, Malay and Mandarin, rather than only being high overall accuracy.

Confusion matrices are also used to analyse classification errors. They indicate the number of accurate and inaccurate predictions for each language and highlight the more frequently confused languages. For instance, audio-based models may confuse English words with Malay words because of acoustic similarity, and text models based on optical character recognition (OCR) may be misled by the problem of noisy images affecting the text extraction.

Reporting of results is done in stages. First, the performance of the OCR engine is represented, which leads to the choice of the final OCR engine. Next, the two models – audio-only and text-only – are compared to determine their strengths and weaknesses. Lastly, multimodal fusion strategies are reported to assess whether there is an advantage of combining audio and text predictions.

Per language analysis is also provided in the final reporting. This is relevant because a model might be excellent at overall performance, but its performance in one language is poorer. So, for report parts where applicable, there is also per-language accuracy, precision, recall and F1-score.

The key result tables and figures presented in this chapter contains the dataset summary, OCR engine comparison, model performance comparison, fusion strategy comparison, per-language performance tables and confusion matrices. These outcomes give a thorough overview of the systems performance, ranging from data preparation to the final multimodal prediction outcome.

5.9 Tools and Software

The multilingual language identification system was developed, trained, evaluated and demonstrated using various machine learning frameworks, OCR, image and audio processing libraries, visualization tools, and web application technologies. They helped carry out the whole project process from preparation and preprocessing of the datasets, model training and testing, and then an interactive demonstration interface.

The main programming language used was Python, which is well supported for data processing, machine learning, deep learning, OCR and visualization. It was during this experimentation phase that Jupyter Notebook was used, allowing the various aspects of the workflow to be developed, tested and analysed interactively. It was particularly helpful for audio preprocessing, OCR image generation, noise augmentation, training the model, hyperparameter tuning, and visualizing the results.

The tools and software that were used in this project can be categorized into four groups: modelling and machine learning frameworks, OCR and image

processing tools, result visualization and evaluation tools, and web application interface.

5.9.1 Modelling and Machine Learning Frameworks

PyTorch, Hugging Face Transformers, and FastText were used to implement the primary machine learning and deep learning models in this project. The selected frameworks were chosen as they support both baseline and transformer-based to enable comparison of lightweight and advanced methods for LID.

PyTorch was used as the main deep learning framework for training and evaluating the CNN audio baseline and the wav2vec 2.0 audio model. The CNN model was used as a baseline audio classifier, while wav2vec 2.0 was used as the stronger pre-trained speech representation model. PyTorch offered the tools to train the model, calculate loss, optimize, accelerate using GPUs and evaluate.

Transformer-based models such as wav2vec 2.0, DistilBERT, and TrOCR were employed for audio language identification, text-based language classification, and as one of the OCR engines for extracting text, respectively. The use of pre-trained transformer models helped to enhance the performance, particularly in low-resource and multilingual language identification tasks.

A lightweight text classification baseline was used: FastText. It was helpful to compare the simpler text classification method with DistilBERT. FastText uses word-level and subword-level features, which made it a good baseline for language identification from text data, because it was fast and efficient.

Audio files were loaded, processed and audio features extracted using Librosa and related audio utilities for audio processing. The CNN model used mel-spectrogram-based audio features, while wav2vec 2.0 used waveform input.

The tools enabled the preparation of the audio data in a format appropriate for using with models for training and evaluating.

Table 5.11: Modelling and Machine Learning Tools

Tool / Library	Purpose
Python	Main programming language
Jupyter Notebook	Experiment development and testing
PyTorch	CNN and wav2vec 2.0 model training
Hugging Face Transformers	wav2vec 2.0, DistilBERT, and TrOCR models
FastText	Lightweight text classification baseline
Librosa	Audio loading and feature extraction

5.9.2 OCR and Image Processing Tools

The project mainly involved a text-based modality, supported by OCR and image processing tools. These tools were also crucial for the preparation of the dataset and the evaluation of the model, as the text was entered as images and then converted to text using OCR.

TrOCR and EasyOCR were evaluated as the OCR engines in this project. EasyOCR has been chosen for English, Malay and Mandarin text extraction according to the accuracy that has been compared with TrOCR. This OCR strategy was used because EasyOCR achieved the strongest character-level performance across the evaluated languages.

Text images were obtained by using Pillow to extract text from Wikipedia text samples. It was also employed to process image files in pre-processing. Image processing and noise augmentation was done using OpenCV. Transformations like Gaussian noise, salt and pepper noise, motion blur, JPEG compression and brightness/contrast adjustments were applied to the image

dataset to generate the noisy image set. Such changes enabled the simulation of real-life image quality problems that may occur when a user would upload text images or use a camera to take images.

The robustness of text extraction under different image conditions was evaluated using these OCR and image processing tools. They also helped with the final system by providing OCR-based language identification of uploaded images.

Table 5.12: OCR and Image Processing Tools

Tool / Library	Purpose
TrOCR	OCR engine evaluated for text extraction
EasyOCR	OCR extraction for English, Malay and Mandarin
Pillow	Text image generation and image handling
OpenCV	Image preprocessing and noise augmentation
NumPy	Pixel-level image processing and numerical operations

5.9.3 Result Visualization and Evaluation Tools

Several tools were used to evaluate model performance and visualize the results. NumPy and pandas were used for handling datasets, CSV files, model outputs, and numerical arrays, scikit-learn was used to calculate evaluation metrics such as accuracy, precision, recall, F1-score, classification reports, and confusion matrices.

Matplotlib and seaborn were used to generate graphs and visualizations for the report. These included bar charts, per-language performance comparisons, OCR accuracy comparisons, model accuracy summaries, and

confusion matrices. These visualizations helped make the experimental results easier to interpret and compare.

The use of evaluation and visualization tools was important because this project involved several models and modalities. By reporting both overall and per-language metrics, the project could identify not only which model performed best overall, but also which languages were more difficult for each model. Table 5.13 shows the tools used for evaluation and visualization. These tools helped convert model outputs into readable tables and figures for analysis.

Table 5.13: Evaluation and Visualization Tools

Tool / Library	Purpose
pandas	Dataset handling and CSV processing
NumPy	Numerical processing
scikit-learn	Accuracy, precision, recall, F1-score, and confusion matrices
Matplotlib	Graph and chart generation
seaborn	Confusion matrix and statistical visualization

5.9.4 Web Application Interface

A Streamlit web application was developed to demonstrate the final multilingual language identification system. The purpose of the interface was to provide an interactive way to test the trained models without requiring users to run the notebooks manually. This made the project easier to present and allowed the audio, OCR, and fusion components to be demonstrated in a practical setting.

The Streamlit interface supports both audio-based and text-based inputs. For the audio modality, users can upload an audio file or record live speech directly through the browser. The audio input is then processed and passed into

the wav2vec 2.0 model for language prediction. This allows the system to demonstrate audio-based language identification using real user input.

For the text modality, users can upload an image containing printed or handwritten text. The system then applies OCR extraction using the selected OCR strategy. The Streamlit interface allows users to choose between TrOCR, EasyOCR, and Hybrid OCR for demonstration and comparison. Based on the final OCR evaluation, EasyOCR is selected as the final OCR engine for English, Malay, and Mandarin. The extracted text is then passed into the DistilBERT-based text classification model to predict the language.

The interface also includes an OCR text correction area. This allows users to review and edit the extracted OCR text before running the final prediction. This feature is useful because OCR output may contain errors, especially when the uploaded image is noisy, blurred, low-resolution, or handwritten. By allowing manual correction, the interface reflects a more realistic workflow for practical use.

In addition to prediction, the Streamlit interface includes project result summaries and model file information. This allows the interface to function not only as a language prediction tool, but also as a demonstration platform for presenting the overall project workflow, model performance, and multimodal system design. Figure 5.9 demonstrates how the final system is presented to users and how the audio and image input sections are organized.

Demo Settings

● Use local Hugging Face cache only

OCR strategy

Hybrid OCR

Expected image language

Unknown / mixed

Enhance image before OCR

Fusion method

Equal Weighting (0.5/0.5)

Cache status

Wav2Vec2 base: found

DistilBERT base: found

TrOCR printed: found

Deploy

Wav2Vec2 + DistilBERT Language Identifier

Audio, OCR text, and late-fusion demo for English, Malay, and Mandarin.

Live DemoProject ResultsPresentationModel Files

Run A Prediction

Hybrid OCR is set to unknown/mixed, so the app will try Mandarin, Malay, and English OCR candidates, then choose the strongest extraction. This is slower than selecting the language directly.

Audio Input

Record speech live

Upload text image

Upload speech audio

0:02 / 0:11

Demo Settings

● Use local Hugging Face cache only

OCR strategy

Hybrid OCR

Expected image language

Unknown / mixed

Enhance image before OCR

Fusion method

Equal Weighting (0.5/0.5)

Cache status

Wav2Vec2 base: found

DistilBERT base: found

TrOCR printed: found

Deploy

Wav2Vec2 + DistilBERT Language Identifier

Audio, OCR text, and late-fusion demo for English, Malay, and Mandarin.

Live DemoProject ResultsPresentationModel Files

Saved Evaluation Results

Equal Weighting (0.5/0.5)	Accuracy Weighted Fusion	Confidence Adaptive Fusion	Audio Only (Wav2Vec2)	Text Only (DistilBERT)
99.18%	97.86%	99.01%	94.16%	97.28%

Best saved method: Equal Weighting (0.5/0.5) (99.18%)

Demo Settings

● Use local Hugging Face cache only

OCR strategy

Hybrid OCR

Expected image language

Unknown / mixed

Enhance image before OCR

Fusion method

Equal Weighting (0.5/0.5)

Cache status

Wav2Vec2 base: found

DistilBERT base: found

TrOCR printed: found

Deploy

Wav2Vec2 + DistilBERT Language Identifier

Audio, OCR text, and late-fusion demo for English, Malay, and Mandarin.

Live DemoProject ResultsPresentationModel Files

Project Demonstration Flow

Audio Model

Wav2Vec2

Text Model

DistilBERT

Best Fusion

99.18%

The demo separates the system into the same stages used in the experiment: audio classification, OCR/text classification, and probability-level fusion. During presentation, run one modality first, then run both together to show why the fusion result is the main contribution.

Suggested Presentation Script

1. Open Project Results and show that fusion outperforms the single-modality baselines.
2. Upload or enter text in Live Demo to show the DistilBERT branch.
3. Upload audio in Live Demo to show the Wav2Vec2 branch.
4. Provide both inputs and compare audio-only, text-only, and fused confidence scores.

OCR Engine Selection

Language	TrOCR Accuracy (%)	EasyOCR Accuracy (%)	Selected Engine
English	62.24	34.13	TrOCR
Malay	46.63	87.62	EasyOCR
Mandarin	2.56	76.82	EasyOCR

Prepared by:AARON LIM CJUN SHIEN

CPT6314 Project I

2530 Term

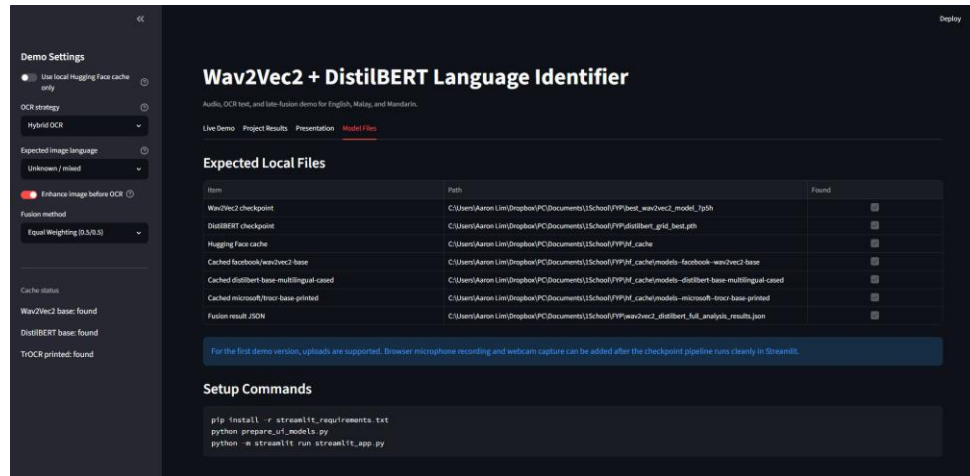


Figure 5.9: Streamlit Web Application Interface for Multimodal Language Identification

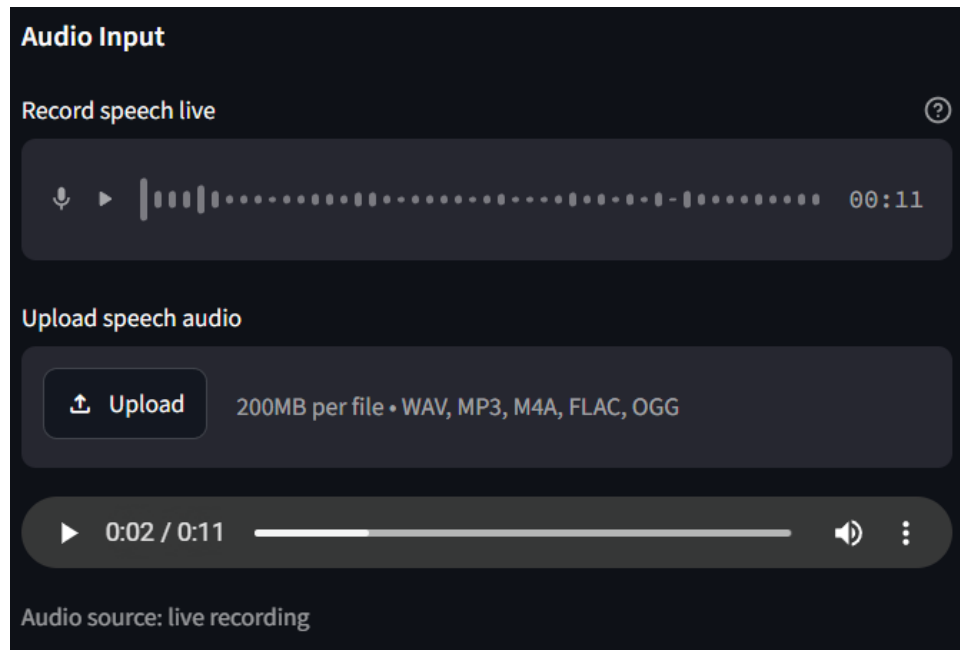


Figure 5.10: Audio Upload and Live Recording Interface

The audio input section of the Streamlit application is showed in Figure 5.10. This part shows that the system supports both uploaded audio files and live audio recording.

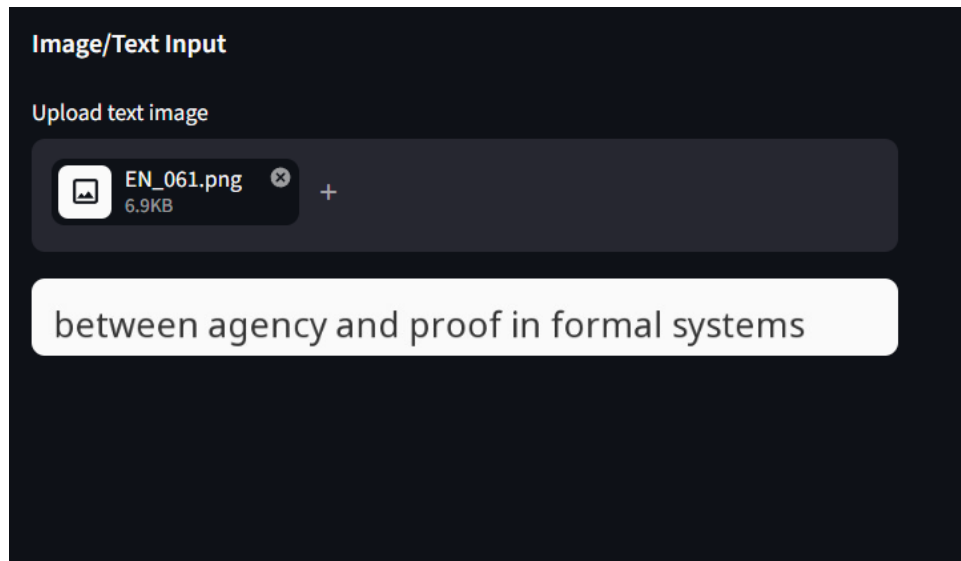


Figure 5.11: OCR Image Upload and Text Extraction Interface

The image upload and OCR text extraction section is shown in Figure 5.11. This figure demonstrates how users can upload text images and review the extracted OCR text before prediction.

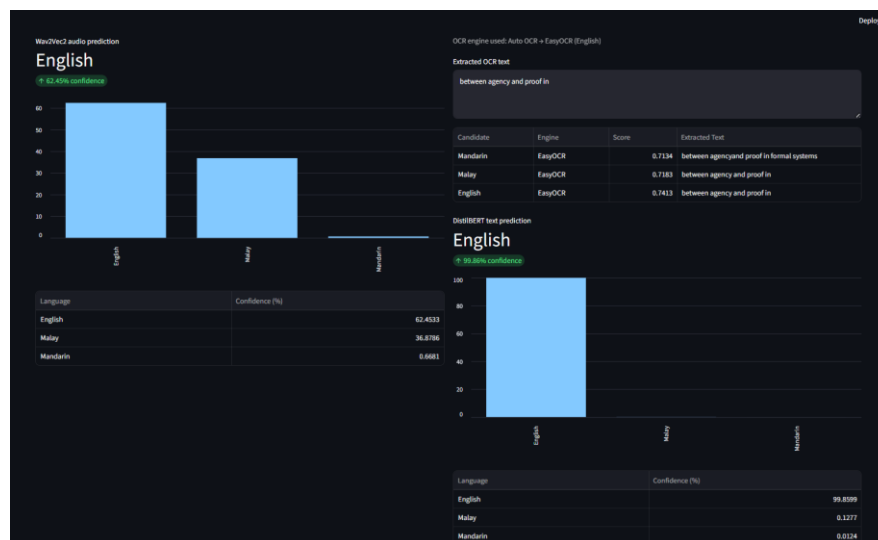




Figure 5.12: Language Prediction Result Display

Figure 5.12 show the final prediction output. Final Language prediction and its confidence for each language made by the fusion model is shown here.

Table 5.14: Web Application Tools

Tool / Software	Purpose
Streamlit	Interactive web interface for project demonstration
streamlit-audiorecorder / audio input component	Live audio recording in the browser
Python backend functions	Model loading, preprocessing, and prediction
OCR module	Image text extraction before language prediction
Trained model files	wav2vec 2.0, DistilBERT, and OCR-supported prediction

Table 5.14 summarizes the tools used to build the web application interface. The Streamlit interface connects the trained models with a practical

user-facing demonstration, allowing the project to be tested using real audio and image inputs.

In general, the tools and software employed in this project facilitated the entire development process of the datasets and software up to the demonstration process. The notebooks enabled experimentation and evaluation, and the Streamlit interface illustrated the deployment of the trained models in a practical, multilingual language identification system.

Chapter 6: Discussion

6.1 Interpretation of Overall Model Performance

This chapter outlines the results of Chapter 5 and discusses their implications in terms of the research purposes. The results show that multilingual low-resource language identification can be improved by combining audio and text modalities. The baseline models are also compared with the transformer-based models, revealing the effectiveness of pre-trained representations for language identification tasks.

The overall results show that the transformer-based models outperformed the baseline models. Wav2vec 2.0 significantly outperformed CNN baseline in the audio modality. For the text modality, DistilBERT outperformed FastText in terms of accuracy. The combined (multimodal) late fusion of wav2vec 2.0 and DistilBERT gave the best overall performance.

This finding is very telling in light of the central thesis of the project: that various modalities have very different language cues, and the combination of the two can be a more reliable system of language processing than either modality alone.

6.2 Comparison Between Baseline and Transformer-Based Models

The results show a clear performance difference between baseline models and transformer-based models. The CNN audio model achieved moderate performance, but it was weaker than wav2vec 2.0. This is expected because the CNN model depends on extracted audio features such as mel-spectrograms, while wav2vec 2.0 uses pre-trained speech representations learned from large amounts of speech data.

The improvement achieved by wav2vec 2.0 shows the advantage of transfer learning in low-resource language identification. Since low-resource datasets may not contain enough training samples to learn strong acoustic representations from scratch, using a pre-trained speech model helps the system capture phonetic and acoustic patterns more effectively.

A similar pattern was observed in the text modality. FastText performed strongly as a lightweight baseline, but DistilBERT achieved better performance. This is because DistilBERT can capture contextual and subword-level information, which is useful for distinguishing English, Malay, and Mandarin text. The strong performance of DistilBERT shows that transformer-based text models are effective for multilingual language identification.

6.3 Impact of OCR Quality on Text-Based Language Identification

OCR quality played an important role in the text-based pipeline. The standalone DistilBERT model achieved very high performance when evaluated using clean text data. However, the text-only result in the fusion experiment was lower because it used OCR-extracted text instead of clean ground-truth text.

This shows that OCR errors can reduce text classification performance. The OCR engine may return wrong characters, missing words, or gibberish symbols, which can negatively impact the performance of the text classifier. The importance of this issue was exacerbated in the Mandarin language, in which minor misrecognition of characters can cause a large amount of difference in the extracted text.

EasyOCR had the better character level recognition results compared with TrOCR for English, Malay and Mandarin language. Thus, EasyOCR was chosen as the final OCR engine to enhance the extracting quality for all the analyzed languages.

6.4 Effectiveness of Multimodal Late Fusion

The multimodal system used decision-level late fusion. In this approach, the audio model, and the text model, generated their own prediction probabilities independently and then fused the prediction probabilities using fusion strategies.

The results indicate that the overall performance of the late fusion outperforms audio-only and OCR text-only prediction. The results of equal weighting fusion were best and confidence adaptive fusion was close to it. This means that there was useful information in both the audio and the text that helped to make the final prediction.

The fusion error analysis corroborates this finding. In many instances, one modality made a wrong prediction while the other modality was correct. Late fusion could correct many of these errors as it fused information from both modalities. This shows that audio and text models committed different types of errors, and thus enabled fusion to be more robust.

Equal weighting fusion was more accurate than accuracy weighted fusion. This is because not all modalities are necessarily equally reliable for each sample. A model that performs well overall may still be uncertain or incorrect for a specific input. Therefore, equal weighting provided a more balanced strategy for this dataset.

6.5 Practical Implications of the Streamlit Demonstration System

The Streamlit application interface showcases the usability of the proposed system in a real-world scenario. In addition to displaying results in the notebook, the interface enables testing of the system with audio uploads, live audio recording and image uploads.

In doing so, this provides added practical value of the project as users can interact directly with the trained models. OCR text extraction and Manual Correction on the interface is also as realistic as it can be. Output from OCR can be incorrect, so that when the user sees the text that it has extracted, he or she can correct it.

The project is not only an experimental, model comparison example, but also a functional prototype of multimodal language identification, as demonstrated by the Streamlit example.

6.6 Threats to Validity and Limitations

While the results are promising, some limitations should be taken into account. First, the text image dataset was generated from Wikipedia text samples. While this provides useful multilingual text data, it may not fully represent real-world handwriting, informal text, screenshots, or low-quality captured documents.

Second, the accuracy of the OCR is very dependent on the quality of the images. The OCR might not perform as well if the image is noisy, blurry or handwritten. The image conditions in the real world may be more complex than those augmented with noise.

Thirdly, the audio data were collected from sources based on VoxLingua. Fusion system may have varying performance in different recording environments, speakers, accents or background noise environments.

Fourth, Decision-level late fusion was used for the fusion system. Although this is effective and simple, it doesn't provide great depth of interaction between the audio and the text features prior to prediction. More advanced fusion techniques are possible to improve the performance.

Lastly, this current system is only in English, Malay and Mandarin. If the system were to be expanded to other languages, there would be additional testing required.

Chapter 7: Conclusion and Recommendations

7.1 Revisiting Research Objectives

The aim of this study was to explore the potential for language identification through multimodal modelling, specifically English, Malay and Mandarin. It was based on three main objectives of the study:

- **To collect, prepare, and generate multimodal datasets, including audio and image-based text, for English, Malay, and Mandarin.**

This was done effectively by creating a special purpose dataset with audio and image text samples. Thus, audio data were collected and segmented into 3-second segments or 10-second segments, and the text data that were needed were produced from text samples from Wikipedia and then converted to image format. The evaluated OCR pipeline was then used for the extraction of OCR. For this reason, EasyOCR is chosen for English, Malay, and Mandarin as it performed better at character level on the languages tested. This OCR engine selection has contributed to enhancing the input of text in all three languages.

- **To develop and evaluate unimodal audio and image-based text language identification models, design multimodal fusion strategies combining speech and OCR text features, and compare performance across English, Malay, and Mandarin.**

The use of two main unimodal classifiers, wav2vec 2.0 for audio and DistilBERT for text, achieved this goal in a comprehensive way. The models were trained and tested separately, and the results showed that text-based classification performed strongly, particularly when OCR quality was high. The

audio-based wav2vec 2.0 model also achieved strong performance and performed better than the CNN audio baseline.

Based on these results, three late fusion strategies were constructed and experimented: Equal Weighting Fusion, Confidence-Adaptive Fusion, and Accuracy-Weighted Fusion. Of these, Equal Weighting Fusion had the largest overall accuracy at **99.18%**, which was superior to the audio-only and OCR text-only results in the fusion experiment and also showed good per-language consistency. The fusion pipeline was also designed to be flexible for real-world deployment situations through the modularity of the pipeline and the ability to integrate OCR and audio modules.

- **To analyse whether multimodal fusion improves recognition accuracy for Malay as a low-resource language.**

This was one of the core objectives of the evaluation. The findings indicated that multimodal fusion was an effective method for increasing recognition accuracy for Malay. In the fusion test set, the Malay audio-only result was **94.32%**, while the Malay text-only result using OCR text was **92.84%**. In the case of Equal Weighting Fusion, Malay accuracy increased to **98.52%**, which was close to the performance of the other two languages. This validates the assumption that the use of OCR-based text features can offset weaknesses in audio-based models and improve recognition performance in low-resource language environments.

Overall, the study was able to achieve all of the mentioned goals. The results show that a lightweight modular multimodal fusion solution is capable of achieving high accuracy in language recognition with limited language-specific data. Another implication of the research is the significance of OCR engine selection and how fusion is beneficial in improving the performance of an under-

resourced language such as Malay. These understandings provide a solid basis for further research on scalable, multilingual, and multimodal language identification systems.

7.2 Key Contributions

This project makes several key contributions to multilingual and multimodal language identification.

The project created a Multimodality dataset preparation pipeline for English, Malay and Mandarin. The data consists of both audio input and image-based text input, well suited to the research purpose mentioned in Chapter 1: to evaluate whether combining speech and OCR-based text can improve multilingual language identification in a Malaysian context.

Second, the project made a comparison between baseline models and transformer-based models in two modalities. The CNN and FastText models gave lightweight baselines, whereas wav2vec 2.0 and DistilBERT models were two stronger transformer-based models. This comparison allowed us to gauge the relative merits and demerits of traditional Deep Learning vs the modern pre-trained models.

Third, the project analyzed the performance of OCR engines for three different languages and found that the selection of OCR engine is crucial for the classification of image-based text. The languages chosen for EasyOCR were English, Malay and Mandarin, as it had better character-level results across the languages evaluated. This result is significant since it has an impact on the text-based language identification pipeline.

Fourth, the project employed multimodal late fusion with the wav2vec 2.0 and DistilBERT prediction probabilities. The results indicated that using combined modality (audio + OCR text predictions) was better than using a single modality. This is consistent with the overall research concept that the multimodal modelling approach will give richer linguistic information than unimodal ones.

Lastly, a demonstration interface was created using Streamlit for an interactive and intuitive representation of the system. Complete deployment was not in the original scope, but the interface was used as a proof of concept for audio upload, live audio recording, image upload, OCR extraction and language prediction.

7.3 Limitations of the Study

The study yielded good outcomes, however, there were some limitations.

The first thing is that the project was limited to three languages: English, Malay and Mandarin. The languages have a high relevance in Malaysia, but the system is yet not tested with other languages, dialects or code-switching situation of Malaysia.

Second, Wikipedia text samples were used to create text image dataset. This creates structured and balanced multilingual text data, but may not be representative of actual handwritten notes, informal text, screenshots, low resolution documents or images with complex backgrounds.

Thirdly, the quality of the OCR is still an issue. This text classifier relies on the extracted output from the OCR, and errors in the OCR extraction could lead to poor predictions. This is particularly relevant to noisy pictures (like photographs) and handwritten text.

Fourth, the audio dataset may not cover all possible speaker accents, background noise conditions, recording devices, or spontaneous speech patterns. Useful data for multilingual speech experiments is available from VoxLingua, but additional tests will be required to assess robustness in real Malaysian environments.

Lastly, the type of fusion for this project was decision-level late fusion. It is successful, but it only merges the probability of the models after the audio and text models have each made their own predictions. This mean it does not acquire more meaningful common representations from audio and text.

7.4 Recommendations for Future Work

There are a few possible enhancements for future work. First, the language support is not limited to English, Malay and Mandarin. Tamil, regional Malaysian language/dialect, or code-switched speech and text can be included, if not already included, in future studies. This would enhance the application of the system in multilingual contexts in Malaysia.

Second, the data set can be expanded with more real-life instances. This may involve various accents, sounds of speech and background noise, audio recording devices, and spontaneous speech. For images, it may be images that contain text, such as screenshots, scanned text, handwritten text, mobile phone images, or low-resolution images.

Thirdly, tuning OCR models or trying new OCR engines can enhance the performance of the OCR. An improvement in OCR extraction will potentially improve the overall multimodal system because the quality of the OCR impacts the text classification pipeline.

Fourth, future studies may be conducted on the more sophisticated fusion strategies. Future work can explore alternative feature-level fusion, attention-based fusion, and multimodal transformer architectures that can learn relationships between audio and text in a more profound manner instead of relying solely on decision-level late fusion.

Fifth, there can be improvement made to the Streamlit demo UI. The following future works would be useful for future versions: more clearly visualizing confidence, batch testing, detailed error analysis, user feedback collection and improved support of noisy or handwritten image input.

7.5 Conclusion

To sum up everything that has been stated so far, this project successfully has created and assessed a multilingual low resource language identification system based on audio, OCR-based text and multimodal late fusion. The study focused on the performance gap that Malay as a relatively under-resourced language encountered when the complementary information of speech and image-based text are combined.

For the audio-language classification, the results indicate that transformer-based models achieve better performance than the CNN baseline, and for the text-language classification, DistilBERT demonstrated good performance. Although FastText was the baseline model in this case, it performed quite well when compared to DistilBERT. The OCR evaluation also showed that OCR engine selection is important when working with multilingual image-based text.

Most importantly, the multimodal late fusion results have shown that the fusion between audio and OCR text predictions enhance recognition performance and robustness. The fusion experiment showed that Equal

Weighting Fusion had the best overall result and the recognition accuracy in the fusion experiment was better than the recognition accuracy of audio-only and OCR text-only prediction on the Malay language.

The results of this work serve as a good foundation for further research around scalable, multilingual, and multimodal language identification. The project also provides valuable insights into the feasibility of integrating audio models, OCR engines, text classifiers, and fusion strategies to create more inclusive multilingual AI systems in Malaysia.

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Appendix A

Project Gantt Chart for **FYP 1**

Task No.	Task Description	Week 1&2	Week 3&4	Week 5&6	Week 7&8	Week 9&10	Week 11&12	Week 13&14
1	Finalize Dataset (Audio + OCR)							
2	Train Unimodal (wav2vec, DistilBERT)							
3	Implement & Test Fusion Strategies							
4	Evaluation (Accuracy, Matrix, etc.)							
5	Chapter 4: Research Methodology							
6	Chapter 5: Data Analysis and Results							
7	Chapter 6: Discussion							
8	Chapter 7: Conclusion and Recommendations							
9	Full Report Compilation							
10	Final Proofreading & Submission							

Project Gantt Chart for **FYP 2**

Task No.	Task Description	Week 1&2	Week 3&4	Week 5&6	Week 7&8	Week 9&10	Week 11&12	Week 13&14
1	Finalize Dataset, OCR Images and Audio Data							
2	Final Model Training and Evaluation							
3	OCR Evaluation and Engine Selection							
4	Multimodal Late Fusion Implementation							
5	Result Analysis and Visualization							
6	Streamlit Demo Development and Testing							
7	Chapter 5: Implementation/Data Analysis and Results							
8	Chapter 6 and Chapter 7 Writing							
9	Poster and Demo Video Preparation							
10	Final Proofreading, Formatting and Submission							

Appendix B



CPT6314 Final Year Project (FYP) 1 Meeting Log Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 4 November 2025	Meeting No.: 1
Meeting Mode: Face-to-Face	
Project ID: 189	Project Type: Research-based
Project Title : Multilingual Low-Resource Speech Recognition	
Student ID : 1211111818	Student Name: Aaron Lim Cjun Shien
Student Programme and Specialisation: Bachelor's in Computer Science (Data Science)	
Supervisor Name: Dr. S Prabha Kumaresan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / ~~Background Study or Literature Review /~~
~~Requirement Analysis or Theoretical Framework /~~ Design or Research Methodology /
~~Prototype Development or Proof of Concept /~~ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Finalising what project title I'm going to choose from the titles I had prepared

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

Identify and collect at least 10 key research papers related to multilingual speech recognition, language identification, and multimodal fusion (audio + text).

Focus Week 1 on foundational works:

- Attention Is All You Need (2017) → Transformer architecture.
- wav2vec 2.0 (2020) → self- supervised speech representation.
- Conformer (2020) → convolution- augmented Transformer for speech.

Summarize each paper (at least 20 lines) highlighting:

- Problem addressed.
- Method/architecture proposed.
- Key results.
- Relevance to low- resource or multilingual tasks.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

Problem: I have limited prior experience and knowledge in multilingual speech recognition and multimodal language identification.

Solution: Began with extensive background study and literature review to build foundational understanding. Focused on summarizing key research papers (Transformer, wav2vec 2.0, Conformer, Whisper, TrOCR) to gradually develop technical familiarity.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Overall progress met expectations.

.....
Supervisor's Signature.....
Student's Signature.....
Co-Supervisor's Signature
(if applicable).....
Company Supervisor's Signature
(if applicable)



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**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 20 November 2025	Meeting No.: 2
Meeting Mode: Face-to-Face	
Project ID: 189	Project Type: Research-based
Project Title : Multilingual Low-Resource Speech Recognition	
Student ID : 1211111818	Student Name: Aaron Lim Cjun Shien
Student Programme and Specialisation: Bachelor's in Computer Science (Data Science)	
Supervisor Name: Dr. S Prabha Kumaresan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identified and collected more key research papers related to multilingual speech recognition, language identification, and multimodal fusion (audio + text).
- Started searching for suitable datasets to use.
- Started searching for suitable models to implement and different model fusion techniques

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

~~Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

- Identify and collect the remaining key research papers related to multilingual speech recognition, language identification, and multimodal fusion (audio + text).
- Began searching for suitable datasets (speech corpora and OCR text sources)
- Started exploring candidate models (CNN, CRNN, wav2vec 2.0, Whisper, FastText, DistilBERT, XLM- R) and different fusion techniques (early fusion, late fusion, attention- based).

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

Problem: : Difficulty in locating suitable datasets, especially for Malay as a low- resource language.

Solution: Began exploring public resources such as Common Voice, VoxLingua107, OpenSLR, and Project Gutenberg. Planned to generate synthetic text images (OCR datasets) from Wikipedia.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Progress was in line with the set objectives.

.....
Supervisor's Signature.....
Student's Signature.....
Co-Supervisor's Signature
(if applicable).....
Company Supervisor's Signature
(if applicable)



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CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 10 December 2025	Meeting No.: 3
Meeting Mode: Face-to-Face	
Project ID: 189	Project Type: Research-based
Project Title : Multilingual Low-Resource Speech Recognition	
Student ID : 1211111818	Student Name: Aaron Lim Cjun Shien
Student Programme and Specialisation: Bachelor's in Computer Science (Data Science)	
Supervisor Name: Dr. S Prabha Kumaresan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: ~~Problem Formulation and Project Planning /~~ Background Study or Literature Review /
~~Requirement Analysis or Theoretical Framework /~~ Design or Research Methodology /
~~Prototype Development or Proof of Concept /~~ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Identified and summarised all key research papers related to multilingual speech recognition, language identification, and multimodal fusion (audio + text).
- Finalised the list of models to be implemented (wav2vec 2.0, DistilBERT, TrOCR, and selected fusion strategies).
- Narrowed down potential datasets (e.g., Common Voice, VoxLingua107, OpenSLR, Wikipedia/Project Gutenberg for text) for English, Malay, and Mandarin.
- Finished Chapter 1: Introduction

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: ~~Problem Formulation and Project Planning~~ / Background Study or Literature-Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / ~~Prototype Development or Proof of Concept~~ / ~~Draft Report Completion~~

(Please strike out the tasks, which are not applicable)

- Discuss with my supervisor about the shortlisted models i have prepared (wav2vec 2.0, DistilBERT, TrOCR, fusion strategies) to confirm suitability.
- Review dataset options with supervisor guidance (e.g., Common Voice, VoxLingua107, OpenSLR, Wikipedia/Project Gutenberg for text).
- Finalise dataset selection and scope (decide number of samples per language, modalities to include).

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

Problem: I have limited prior experience in deciding which models and what kind of effective dataset are most suitable for multimodal language identification.

Solution: Completed the literature review and shortlisted models (wav2vec 2.0, DistilBERT, TrOCR, fusion strategies). Planned to confirm choices with supervisor to ensure alignment with project scope.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

The progress achieved was deemed satisfactory.

.....
Supervisor's Signature.....
Student's Signature.....
Co-Supervisor's Signature
(if applicable).....
Company Supervisor's Signature
(if applicable)



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**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 17 December 2025	Meeting No.: 4
Meeting Mode: Online / Face-to-Face	
Project ID: 189	Project Type: Research-based
Project Title : Multilingual Low-Resource Speech Recognition	
Student ID : 1211111818	Student Name: Aaron Lim Cjun Shien
Student Programme and Specialisation: Bachelor's in Computer Science (Data Science)	
Supervisor Name: Dr. S Prabha Kumaresan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Finalized dataset selection: VoxLingua107 (audio), Wikipedia (text), and custom screenshots for OCR.
- Implemented OCR pipeline using TrOCR (English) and EasyOCR (Malay, Mandarin).
- Preprocessed and aligned audio-text pairs for multimodal input.
- Completed Chapter 2: Literature Review.

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Begin training unimodal models: wav2vec 2.0 (audio) and DistilBERT (text).
- Evaluate OCR output quality and refine preprocessing if needed.
- Start drafting Chapter 3: Methodology.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

Problem: OCR accuracy for Mandarin text was inconsistent using TrOCR.

Solution: Switched to EasyOCR for Mandarin and Malay, retaining TrOCR for English due to better performance.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

The progress achieved was deemed satisfactory.



.....
Supervisor's Signature



.....
Student's Signature

7 January 2026

**FACULTY OF
COMPUTING
& INFORMATICS****CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 7 January 2026	Meeting No.: 5
Meeting Mode: Online / Face-to-Face	
Project ID: 189	Project Type: Research-based
Project Title : Multilingual Low-Resource Speech Recognition	
Student ID : 1211111818	Student Name: Aaron Lim Cjun Shien
Student Programme and Specialisation: Bachelor's in Computer Science (Data Science)	
Supervisor Name: Dr. S Prabha Kumaresan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- Trained unimodal models: wav2vec 2.0 and DistilBERT.
- Implemented three late fusion strategies: Equal Weighting, Confidence-Adaptive, and Accuracy-Weighted.
- Conducted initial evaluation of unimodal and multimodal models.
- Completed Chapter 3: Methodology.

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- ☐ Analyze evaluation results and refine fusion logic if needed.
- ☐ Begin writing Chapter 4: Data Analysis and Results.
- ☐ Prepare visualizations (confusion matrix, accuracy charts) for report.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

Problem: Fusion strategy outputs were inconsistent due to misaligned prediction formats.

Solution: Standardized prediction outputs from both models before fusion; added normalization step.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Progress was in line with the set objectives.



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Supervisor's Signature



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Student's Signature

.....
Co-Supervisor's Signature
(if applicable)

.....
Company Supervisor's Signature
(if applicable)



**FACULTY OF
COMPUTING
& INFORMATICS**

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 21 January 2026	Meeting No.: 6
Meeting Mode: Online / Face-to-Face	
Project ID: 189	Project Type: Research-based
Project Title : Multilingual Low-Resource Speech Recognition	
Student ID : 1211111818	Student Name: Aaron Lim Cjun Shien
Student Programme and Specialisation: Bachelor's in Computer Science (Data Science)	
Supervisor Name: Dr. S Prabha Kumaresan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

[Please write the details of the work done, after the last meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept / Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- ☐ Finalized evaluation results: Equal Weighting Fusion achieved 98.72% accuracy.
- ☐ Completed Chapters 4, 5, and 6 of the report.
- ☐ Compiled full report draft and included meeting logs.
- ☐ Discussed and approved project title change with supervisor.

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]

Tasks: Problem Formulation and Project Planning / Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology / Prototype Development or Proof of Concept/ Draft Report Completion

(Please strike out the tasks, which are not applicable)

Details (in point form):

- ☐ Final proofreading and formatting of the full report.
- ☐ Prepare presentation slides and rehearse for FYP2 proposal defense.
- ☐ Submit final report and meeting logs to eBwise.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

Problem: Original project title did not accurately reflect the project's scope.

Solution: After supervisor consultation, updated title to "Multimodal Low-Resource Language Identification" to better represent the focus on language classification using audio and OCR-based text.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

The progress achieved was deemed satisfactory.



.....
Supervisor's Signature



.....
Student's Signature

.....
Co-Supervisor's Signature
(if applicable)

.....
Company Supervisor's Signature
(if applicable)



FACULTY OF
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CPT6324 Project (FYP2) Meeting Log

Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	21-4-2026	Meeting No.:	1	Meeting Mode:	☒ In-Person	☐ Online
Project ID:	189	Project Type:	☒ Research-based	☐ Application-based		
Project Title:	Multilingual Low-Resource Language Identification					
Student ID:	121111818	Student Name:	Aaron Lim Cjun Shien			
Student Programme and Specialisation:	Bachelor's in Computer Science (DataScience)					
Supervisor Name:	Dr. S Prabha Kumaresan					
Co-Supervisor Name: (if applicable)						
Collaborating Company: (if applicable)						
Company Supervisor Name: (if applicable)						

1. WORK DONE
[Please write the details of the work done, after the last meeting]. Tasks (tick):
☐ Background study ☒ Implementation ☒ Testing ☐ Evaluation
☐ Commercialisation Proposal ☐ Research Paper ☒ Draft Report ☐ Final Report
Details (max 3-5 bullet points):

- Training of the language identification models using the full dataset instead of the initial 60% subset.

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]. Tasks (tick):
☐ Background study ☐ Implementation ☒ Testing ☐ Evaluation
☐ Commercialisation Proposal ☐ Research Paper ☐ Draft Report ☐ Final Report
Details (max 3-5 bullet points):

- Perform detailed error analysis on Malay classification results.
- Update report with evaluation tables and confusion matrices.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Training time increased with full dataset → optimized batch size and
- learning rate to reduce runtime.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)
☐ Not Satisfactory ☒ Satisfactory

Overall progress met expectations.

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature
(if applicable)

Company Supervisor's
Signature (if applicable)

IMPORTANT NOTES TO STUDENTS:

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report. Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee. This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



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CPT6324 Project (FYP2) Meeting Log
Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	28-4-2026	Meeting No.:	2	Meeting Mode:	☒ In-Person	☐ Online
Project ID:	189	Project Type:	☒ Research-based	☐ Application-based		
Project Title:	Multilingual Low-Resource Language Identification					
Student ID:	121111818	Student Name:	Aaron Lim Cjun Shien			
Student Programme and Specialisation:	Bachelor's in Computer Science (DataScience)					
Supervisor Name:	Dr. S Prabha Kumaresan					
Co-Supervisor Name: (if applicable)						
Collaborating Company: (if applicable)						
Company Supervisor Name: (if applicable)						

1. WORK DONE <i>[Please write the details of the work done, after the last meeting]. Tasks (tick):</i>	
☐ Background study ☐ Commercialisation Proposal	☐ Implementation ☐ Research Paper
☒ Testing ☒ Draft Report	
☐ Evaluation ☐ Final Report	
Details (max 3-5 bullet points): - Completed comparative analysis of unimodal vs. multimodal models. - Integrated OCR evaluation results (TrOCR vs. EasyOCR) into the report.	

2. WORK TO BE DONE <i>[Please write the details of the work to be done, before the next meeting]. Tasks (tick):</i>	
☐ Background study ☐ Commercialisation Proposal	☐ Implementation ☐ Research Paper
☒ Testing ☒ Draft Report	
☐ Evaluation ☐ Final Report	
Details (max 3-5 bullet points): - Perform detailed error analysis on Malay classification results. - Update report with evaluation tables and confusion matrices.	

3. PROBLEMS ENCOUNTERED AND SOLUTIONS <i>[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]</i>	
OCR accuracy for English remained low → documented limitation and justified tool choice.	

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)	
☐ Not Satisfactory	☒ Satisfactory
Progress was in line with the set objectives.	

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature
(if applicable)Company Supervisor's
Signature (if applicable)**IMPORTANT NOTES TO STUDENTS:**

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
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CPT6324 Project (FYP2) Meeting Log

Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	4-5-2026	Meeting No.:	3	Meeting Mode:	☒ In-Person	☐ Online
Project ID:	189	Project Type:	☒ Research-based	☐ Application-based		
Project Title:	Multilingual Low-Resource Language Identification					
Student ID:	121111818	Student Name:	Aaron Lim Cjun Shien			
Student Programme and Specialisation:	Bachelor's in Computer Science (DataScience)					
Supervisor Name:	Dr. S Prabha Kumaresan					
Co-Supervisor Name: (if applicable)						
Collaborating Company: (if applicable)						
Company Supervisor Name: (if applicable)						

1. WORK DONE <i>[Please write the details of the work done, after the last meeting]. Tasks (tick):</i>	
☐ Background study ☐ Commercialisation Proposal	☒ Implementation ☐ Research Paper
☒ Testing ☒ Draft Report	
☐ Evaluation ☐ Final Report	
Details (max 3-5 bullet points): - Ran further fusion experiments (Accuracy-Weighted). - Added evaluation tables and bar chart to report. - Incorporated supervisor feedback into Methods section.	

2. WORK TO BE DONE <i>[Please write the details of the work to be done, before the next meeting]. Tasks (tick):</i>	
☐ Background study ☐ Commercialisation Proposal	☐ Implementation ☐ Research Paper
☒ Testing ☒ Draft Report	
☐ Evaluation ☐ Final Report	
Details (max 3-5 bullet points): - Complete Confidence-Adaptive fusion experiments. - Draft Conclusion and Future Work sections.	

3. PROBLEMS ENCOUNTERED AND SOLUTIONS <i>[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]</i>	
Minor inconsistencies in figure captions → standardized across report.	

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)	
☐ Not Satisfactory	☒ Satisfactory
The progress achieved was deemed satisfactory.	

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature
(if applicable)

Company Supervisor's
Signature (if applicable)

IMPORTANT NOTES TO STUDENTS:

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CPT6324 Project (FYP2) Meeting Log
Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	14-5-2026	Meeting No.:	4	Meeting Mode:	☒ In-Person	☐ Online
Project ID:	189	Project Type:	☒ Research-based	☐ Application-based		
Project Title:	Multilingual Low-Resource Language Identification					
Student ID:	1211111818	Student Name:	Aaron Lim Cjun Shien			
Student Programme and Specialisation:	Bachelor's in Computer Science (DataScience)					
Supervisor Name:	Dr. S Prabha Kumaresan					
Co-Supervisor Name: (if applicable)						
Collaborating Company: (if applicable)						
Company Supervisor Name: (if applicable)						

1. WORK DONE <i>[Please write the details of the work done, after the last meeting]. Tasks (tick):</i> ☐ Background study ☒ Implementation ☒ Testing ☐ Evaluation ☐ Commercialisation Proposal ☐ Research Paper ☒ Draft Report ☐ Final Report Details (max 3-5 bullet points):	
Complete Future Work and Conclusion part	

2. WORK TO BE DONE <i>[Please write the details of the work to be done, before the next meeting]. Tasks (tick):</i> ☐ Background study ☐ Implementation ☒ Testing ☐ Evaluation ☐ Commercialisation Proposal ☐ Research Paper ☒ Draft Report ☐ Final Report Details (max 3-5 bullet points):	
Further refine report	

3. PROBLEMS ENCOUNTERED AND SOLUTIONS <i>[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]</i>	
Minor inconsistencies in figure captions → standardized across report.	

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor) ☐ Not Satisfactory ☒ Satisfactory	
The progress achieved was deemed satisfactory.	

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature
(if applicable)Company Supervisor's
Signature (if applicable)**IMPORTANT NOTES TO STUDENTS:**

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
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CPT6324 Project (FYP2) Meeting Log
Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	26-5-2026	Meeting No.:	5	Meeting Mode:	☒ In-Person	☐ Online
Project ID:	189	Project Type:	☒ Research-based	☐ Application-based		
Project Title:	Multilingual Low-Resource Language Identification					
Student ID:	1211111818	Student Name:	Aaron Lim Cjun Shien			
Student Programme and Specialisation:	Bachelor's in Computer Science (DataScience)					
Supervisor Name:	Dr. S Prabha Kumaresan					
Co-Supervisor Name: (if applicable)						
Collaborating Company: (if applicable)						
Company Supervisor Name: (if applicable)						

1. WORK DONE

[Please write the details of the work done, after the last meeting]. Tasks (tick):

☐ Background study ☐ Implementation ☒ Testing ☒ Evaluation
☐ Commercialisation Proposal ☐ Research Paper ☒ Draft Report ☒ Final Report

Details (max 3-5 bullet points):

- Started with demo video for fyp

2. WORK TO BE DONE

[Please write the details of the work to be done, before the next meeting]. Tasks (tick):

☐ Background study ☐ Implementation ☒ Testing ☒ Evaluation
☐ Commercialisation Proposal ☐ Research Paper ☒ Draft Report ☒ Final Report

Details (max 3-5 bullet points):

- Draft for script and prepare slides for demo video

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

- Alignment and work flow for report to be amended

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

☐ Not Satisfactory ☒ Satisfactory

Good, Consistent

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature
(if applicable)

Company Supervisor's
Signature (if applicable)

IMPORTANT NOTES TO STUDENTS:

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
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- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



FACULTY OF
COMPUTING
& INFORMATICS

CPT6324 Project (FYP2) Meeting Log
Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	10-6-2026	Meeting No.:	6	Meeting Mode:	☒ In-Person	☐ Online
Project ID:	189	Project Type:	☒ Research-based	☐ Application-based		
Project Title:	Multilingual Low-Resource Language Identification					
Student ID:	1211111818	Student Name:	Aaron Lim Cjun Shien			
Student Programme and Specialisation:	Bachelor's in Computer Science (DataScience)					
Supervisor Name:	Dr. S Prabha Kumaresan					
Co-Supervisor Name: (if applicable)						
Collaborating Company: (if applicable)						
Company Supervisor Name: (if applicable)						

1. WORK DONE <i>[Please write the details of the work done, after the last meeting]. Tasks (tick):</i> ☐ Background study ☐ Implementation ☒ Testing ☒ Evaluation ☐ Commercialisation Proposal ☐ Research Paper ☒ Draft Report ☒ Final Report Details (max 3-5 bullet points):	
Prepared demo video for report and webapp	

2. WORK TO BE DONE <i>[Please write the details of the work to be done, before the next meeting]. Tasks (tick):</i> ☐ Background study ☐ Implementation ☐ Testing ☐ Evaluation ☒ Commercialisation Proposal ☐ Research Paper ☐ Draft Report ☐ Final Report Details (max 3-5 bullet points):	
Complete Commercialisation Proposal	

3. PROBLEMS ENCOUNTERED AND SOLUTIONS <i>[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]</i>	
Demo video exceed time, redesign slides to save time	

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor) ☐ Not Satisfactory ☒ Satisfactory	
Satisfact, continue good work	

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature
(if applicable)Company Supervisor's
Signature (if applicable)**IMPORTANT NOTES TO STUDENTS:**

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
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- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

Appendix C

FYP Report Draft.pdf

ORIGINALITY REPORT

8%	4%	5%	2%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	arxiv.org Internet Source	1%
2	www.mdpi.com Internet Source	<1%
3	aman.ai Internet Source	<1%
4	K. V. Sambasivarao, Anasuya Sesha Roopa Devi Bhima. "Artificial Intelligence, Computational Intelligence, and Inclusive Technologies - Proceedings of International Conference on Artificial Intelligence, Computational Intelligence, and Inclusive Technologies (ICRAIC2IT – 2025)", CRC Press, 2026 Publication	<1%
5	epochai.org Internet Source	<1%
6	preview.aclanthology.org Internet Source	<1%
7	"Neural Information Processing", Springer Science and Business Media LLC, 2017 Publication	<1%
8	ar5iv.labs.arxiv.org Internet Source	<1%
9	Shiqing Zhang, Yijiao Yang, Chen Chen, Ruixin Liu, Xin Tao, Wenping Guo, Yicheng Xu, Xiaoming Zhao. "Multimodal emotion	<1%

recognition based on audio and text by using hybrid attention networks", Biomedical Signal Processing and Control, 2023

Publication

- 10 Abigail Albuquerque, Samuel Chibuoyim Uche, Emmanuel Agu. "Intoxication detection from speech using representations learned from self-supervised pre-training", Smart Health, 2025

Publication

- 11 H.L. Gururaj, Francesco Flammini, S. Srividhya, M.L. Chayadevi, Sheba Selvam. "Computer Science Engineering", CRC Press, 2024

Publication

- 12 Pushpa Choudhary, Sambit Satpathy, Arvind Dagur, Dharendra Kumar Shukla. "Recent Trends in Intelligent Computing and Communication", CRC Press, 2025

Publication

- 13 Siddharth Bhangale, Pradeep Kumar Roy. "Is It Genuine or Fake? Analyzing E-Commerce Reviews Using Large Language Models", Knowledge-Based Systems, 2025

Publication

- 14 Travis M. Bartley, Fei Jia, Krishna C. Puwada, Samuel Krizan, Boris Ginsburg. "Accidental Learners: Spoken Language Identification in Multilingual Self-Supervised Models", ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023

Publication

- 15 ora.ox.ac.uk

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16	Submitted to Asia Pacific University College of Technology and Innovation (UCTI) Student Paper	<1 %
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25	Nhut Minh Nguyen, Trung Minh Nguyen, Thanh Trung Nguyen, Phuong-Nam Tran et al. "Enhancing multimodal emotion recognition with dynamic fuzzy membership and	<1 %

attention fusion", Engineering Applications of Artificial Intelligence, 2026

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26	aclanthology.org Internet Source	<1 %
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29	"Text, Speech, and Dialogue", Springer Science and Business Media LLC, 2026 Publication	<1 %
30	Ambuj Mehrish, Navonil Majumder, Rishabh Bharadwaj, Rada Mihalcea, Soujanya Poria. "A review of deep learning techniques for speech processing", Information Fusion, 2023 Publication	<1 %
31	Merav Allouch, Noa Mansbach, Amos Azaria, Rina Azoulay. "Utilizing Machine Learning for Detecting Harmful Situations by Audio and Text", Applied Sciences, 2023 Publication	<1 %
32	Samira Hazmoune, Fateh Bougamouza. "Using transformers for multimodal emotion recognition: Taxonomies and state of the art review", Engineering Applications of Artificial Intelligence, 2024 Publication	<1 %
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36	harvest.usask.ca Internet Source	<1 %
37	open.uct.ac.za Internet Source	<1 %
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44	ir.unisa.ac.za Internet Source	<1 %
45	www.frontiersin.org Internet Source	<1 %
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